

Efficient Fuzzy PSI under One-Sided Assumptions

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Abstract

Fuzzy private set intersection (PSI) enables two parties to identify approximately matching elements between their input sets, where two elements are considered a match if their distance is at most a threshold δ under a given metric. Although substantial progress has been made, existing constructions for *general Minkowski distances* either rely on strong two-sided geometric separation assumptions or incur substantial overhead under one-sided assumptions.

In this work, we present the first concretely efficient fuzzy PSI protocols for general $L_{p \in [1, \infty]}$ distances under *one-sided assumptions*, relying solely on lightweight symmetric-key primitives. Our constructions support both sender-sided and receiver-sided settings. We further study sparser input distributions and present more efficient protocols tailored to this case. To reduce the overhead scaling with δ , we non-trivially incorporate prefix trie techniques into our protocols, achieving $O(\log \delta)$ complexity for general $L_{p \in [1, \infty]}$ distances for the first time, improving upon $O((\log \delta)^d)$ or $O(\delta)$ complexities of prior works.

Extensive experiments, across a wide range of parameter settings, show that our protocols significantly outperform prior works under the same assumptions. Specifically, against van Baarsen and Pu (EUROCRYPT'24), our protocols achieve up to 239× faster computation and up to 20× lower communication. Against Dang et al. (CCS'25), we achieve up to 518× speedup and up to 63× communication reduction. Against Bui et al. (ASIACRYPT'25), we achieve up to 4818× faster computation and up to 282× lower communication.

CCS Concepts

• Security and privacy → Cryptography.

Keywords

Fuzzy Private Set Intersection; MPC

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1 Introduction

Standard private set intersection (PSI) [8, 11, 12, 16, 18, 19, 22, 25, 26, 28, 33] has been studied for decades, with significant progress made in both efficiency and scalability. It enables two parties to compute their common elements without revealing any information about non-matching ones. PSI has found applications in contact tracing [19, 33], ad conversion measurement [18], and others [1, 20]. However, its exact-matching requirement limits applicability in settings where data are inherently noisy or imprecise, such as biometric identification or location services.

Fuzzy PSI [6, 10, 13–15, 27, 30–32, 34, 35] generalizes this functionality to support *approximate matching*. Specifically, in a d -dimensional space, let the sender hold a set $Q = \{q_1, \dots, q_m\}$ and the receiver hold a set $W = \{w_1, \dots, w_n\}$. Fuzzy PSI outputs to the receiver all elements $q \in Q$ for which there exists some $w \in W$ satisfying $\text{dist}(q, w) \leq \delta$, where δ is the matching threshold and $\text{dist}(\cdot, \cdot)$ denotes a predefined distance metric, e.g., general $L_{p \in [1, \infty]}$ distances.

To ensure practical efficiency, most existing fuzzy PSI protocols rely on *geometric separation assumptions* over the input sets. These works can be broadly divided into two categories based on the underlying assumptions: *one-sided* and *two-sided*. Protocols based on two-sided assumptions [10, 13, 31, 32, 34, 35] generally achieve better efficiency. For example, under the assumption that points held by both the sender and receiver are pairwise separated by at least 2δ or 4δ , the state-of-the-art construction [32] realizes concretely efficient fuzzy PSI using lightweight OPRF. However, such two-sided assumptions may be unrealistic in practice and can limit its applicability to real-world scenarios.

To relax these constraints, recent works [6, 10, 14, 31] have proposed fuzzy PSI protocols under *one-sided assumptions*, where only one party's inputs are required to satisfy certain assumptions while the other party may hold arbitrary inputs. However, existing constructions either rely on heavyweight cryptographic primitives (e.g., AHE [23, 24]) or do not support general $L_{p \in [1, \infty]}$ distances. Specifically, van Baarsen and Pu [31], later improved by Dang et al. [10], proposed the first fuzzy PSI protocols for general $L_{p \in [1, \infty]}$

distances using AHE. Concurrently, another line of work [6, 14] achieves fuzzy PSI using symmetric-key techniques (e.g., function secret sharing [4, 5]), but supports only the L_∞ distance.

Moreover, these works incur undesirable asymptotic communication and computation overhead in the distance threshold δ . The protocol of [31] scales linearly as $O(\delta)$, while the constructions in [6, 10, 14] incur computation or communication complexity of $O((\log \delta)^d)$. Consequently, the efficiency of these protocols deteriorates with larger δ , particularly in real-world applications where δ represents a normalized integer of some high-precision floating-point value and may thus be large.

In summary, existing works for general $L_{p \in [1, \infty]}$ distances either (1) rely on strong geometric separation assumptions on both parties' inputs, or (2) incur substantial computation and communication overhead under one-sided assumptions. Motivated by these limitations, we raise the following question:

Can we construct efficient fuzzy PSI protocols for general $L_{p \in [1, \infty]}$ distances under one-sided assumptions?

1.1 Our Contributions

To address these limitations, we propose a fuzzy PSI for general $L_{p \in [1, \infty]}$ distances with two key features: (1) it operates under weaker one-sided assumptions (for either sender or receiver), and (2) it relies solely on lightweight symmetric-key primitives, thus achieving significant improvements in both communication and computation efficiency. We summarize our contributions as follows.

- (1) **General fuzzy PSI under one-sided assumptions.** We propose concretely efficient fuzzy PSI protocols for general $L_{p \in [1, \infty]}$ distances under one-sided assumptions, relying solely on lightweight symmetric-key primitives. The core intermediate techniques are efficient protocols for multi-point fuzzy matching, which support one party holding multiple inputs without imposing any assumption. Building on this, we present new constructions of fuzzy PSI supporting both the sender-sided and receiver-sided assumptions.
- (2) **Logarithmic complexity in threshold δ .** We non-trivially incorporate prefix-trie techniques into our protocols. Unlike prior works that incur superlinear complexity $O((\log \delta)^d)$, we design an efficient dimension-by-dimension filtering mechanism to prune the exponential search space. As a result, our optimized protocol achieves logarithmic complexity $O(\log \delta)$ in the distance threshold δ .
- (3) **Extension for sparser distributions.** We further consider sparser input distributions under one-sided assumptions and design more efficient fuzzy PSI protocols tailored to this setting. The key idea is to reverse the roles of the sender and receiver in the spatial hashing procedure, which balances their respective workloads and yields improved concrete performance.
- (4) **Implementation and evaluation.** We implement all proposed protocols and conduct extensive experiments across a wide range of parameter settings. Our protocols significantly outperform all prior works under assumptions of comparable strength, achieving one to three orders of magnitude improvement in efficiency.

2 Related Work

We introduce related works of fuzzy PSI under one-sided assumptions and defer other works to Appendix D. Table 1 compares our protocols with prior works in terms of techniques, assumptions, and asymptotic complexity.

Specifically, Garimella et al. [15] proposed the first fuzzy PSI protocol for L_∞ distance, using a customized scheme of Function Secret Sharing (FSS) [4, 5] to test whether the sender's points fall within the receiver's δ -radius L_∞ balls. Their construction requires the receiver's balls to be disjoint, while the sender's points could be arbitrary. Depending on the distances between the ball centers, they consider different settings, but the computational complexity remains $O(\delta^d)$ in all cases. Following works [6, 14] improved upon this via prefix trie optimizations, reducing the complexity to $O((\log \delta)^d)$ for both communication and computation. These two constructions rely on the *mini-universe* assumption, under which each ball maps uniquely to a distinct vertex of the space and the sender's points remain unconstrained. A common limitation of this line of work [6, 14, 15] is that it supports only L_1 and L_∞ distances, with no known efficient instantiation for general L_p distances.

Subsequently, van Baarsen and Pu [31] studied fuzzy PSI under the one-sided 2δ -*apart* assumption, where only the receiver's points are required to be at least 2δ apart from each other. Their construction is built on a DDH-based set membership test, achieving $O(\delta)$ complexity in threshold δ . To support arbitrary L_p distances, they require a stronger 4δ -*apart* assumption, and the overhead scales as $O(\delta^p)$ since the sender must enumerate all possible p -th power distances. Similarly, Dang et al. [10] further optimized this construction via prefix trie techniques under the same 4δ -*apart* assumption, reducing the complexity to $O((\log \delta)^d)$.

More recently, Richardson et al. [30] and Piske et al. [27] proposed fuzzy PSI protocols based on symmetric-key techniques under the *disjoint hash* assumption, which requires each point to map to a unique representation. In particular, Richardson et al. [30] map elements into bins via spatial hashing and, assuming that each bin of both parties contains at most one element, compare items within corresponding bins. However, as illustrated by Piske et al. [27], their constructions based on Garbled Circuits incur substantial communication and computational overhead. Subsequently, Piske et al. [27] introduced more efficient fuzzy PSI protocols based on a new primitive called distance-aware OT, which transfers values conditioned on the distance between inputs. However, their theoretical construction under one-sided assumptions additionally leaks the density of the arbitrarily distributed receiver set and incurs complexity with an additional multiplicative dependence on this density. Moreover, their concrete instantiation ultimately relies on a two-sided assumption, requiring both the sender's and receiver's points to map to unique blocks or cells.

3 Preliminary

3.1 Notation

We use κ and λ to denote the computational and statistical security parameters, respectively. We use $[a, b]$ to denote the set $\{a, \dots, b\}$ and $[a]$ to denote the set $\{1, \dots, a\}$. For a set S , $|S|$ denotes its cardinality. We write $r \xleftarrow{\$} \mathbb{F}$ to denote that r is sampled uniformly

Table 1: Comparisons of existing fuzzy PSI protocols under one-sided assumptions. The sender holds m points and the receiver holds n points in a d -dimensional space, and δ is the distance threshold. Note that our assumptions are either equivalent to or strictly weaker than existing ones; a detailed analysis is provided in Appendix F.

Metric	Protocol	Technique	Assumption	Communication	Computation	
					Sender	Receiver
L_∞	[6]	FSS	$\mathcal{R}$, mini-universe	$O(dn \log \delta + m2^d (\log \delta)^d)$	$O(m2^d (\log \delta)^d)$	$O((\log \delta)^d n + dm \log \delta)$
	[31]	AHE	$\mathcal{R}$, 2δ -apart	$O(\delta dn + 2^d m)$	$O(2^d dm)$	$O(\delta dn + 2^d m)$
		AHE	$\mathcal{R}$, 4δ -apart	$O(\delta^2 dn + m)$	$O(dm)$	$O(\delta^2 dn + m)$
	[10]	AHE	$\mathcal{R}$, 4δ -apart	$O(d \log \delta (n2^d + m))$	$O(2^d dn \log \delta + m(\log \delta)^d)$	$O(dm \log \delta)$
	[30]	GC	S^* , disj. hash	$O(d \log \delta (n2^s + m2^{d-s}))$	$O(2^{d-s} dm \log \delta)$	$O(2^s dn \log \delta)$
	[27]	OPRF	S^* , disj. hash	$O(d(\delta m + 2^d n))$	$O(\delta dm)$	$O(2^d n + m)$
	Ours	so-OPPRF	$\mathcal{R}/S$, unique cell	$O(d(\delta n + 2^d m))$	$O(2^d dm)$	$O(d(\delta n + 2^d m))$
	Ours-Px	so-OPPRF	$\mathcal{R}/S$, unique cell	$O(\log \delta d(n + 2^d m))$	$O(\log \delta 2^d dm)$	$O(\log \delta d(n + 2^d m))$
	Ours	so-OPPRF	$\mathcal{R}/S$, unique block	$O(d(m + \delta^2 n))$	$O(dm)$	$O(d(m + \delta^2 n))$
L_p	Ours-Px	so-OPPRF	$\mathcal{R}/S$, unique block	$O(\log \delta d(2^d n + m))$	$O(\log \delta dm)$	$O(\log \delta d(2^d n + m))$
	[31]	AHE	$\mathcal{R}$, $2\delta(d^{1/p} + 1)$ -apart	$O(\delta^2 dn + \delta^p m)$	$O((d + \delta^p)m)$	$O(\delta^2 dn + m)$
	[10]	AHE	$\mathcal{R}$, $2\delta(d^{1/p} + 1)$ -apart	$O(2^d p(dn \log \delta + m(\log \delta)^d) + dm \log \delta)$	$O(2^d (dpn \log \delta + m(\log \delta)^d))$	$O(m(dp \log \delta + 2^d (\log \delta)^d))$
	[30]	GC	S^* , disj. hash	$O(d \log(d\delta)(n2^d + m2^{d-s}))$	$O(dm2^{d-s} \log(d\delta))$	$O(dn2^s \log(d\delta))$
	[27]	OPRF	S^* , disj. hash	$O(d(\delta m + 2^d n))$	$O(\delta dm)$	$O(2^d n + m)$
	Ours	so-OPPRF	$\mathcal{R}/S$, unique cell	$O(d(\delta n + 2^d m) + p \log \delta 2^d m)$	$O((d + p \log \delta)2^d m)$	$O(d(\delta n + 2^d m) + p \log \delta 2^d m)$
	Ours-Px	so-OPPRF	$\mathcal{R}/S$, unique cell	$O(\log \delta (dn + 2^d dm + 2^d pm))$	$O(\log \delta (2^d dm + 2^d pm))$	$O(\log \delta (dn + 2^d dm + 2^d pm))$
	Ours	so-OPPRF	$\mathcal{R}/S$, unique block	$O(d(m + \delta^2 n) + p \log \delta m)$	$O((d + p \log \delta)m)$	$O(d(m + \delta^2 n) + p \log \delta m)$
	Ours-Px	so-OPPRF	$\mathcal{R}/S$, unique block	$O(\log \delta (2^d dn + dm + pm))$	$O(\log \delta (dm + pm))$	$O(\log \delta (2^d dn + dm + pm))$

- $\mathcal{R}/S$ denotes that the set of receiver or sender satisfies the assumption.
- S^* denotes that the protocols of [27, 30] require an additional assumption on the receiver's inputs to instantiate the concrete protocol.
- Our protocols for sender-sided and receiver-sided assumptions have the same asymptotic complexity when $m = n$, we report one complexity.
- Mini-universe assumption requires that each L_∞ ball maps uniquely to a distinct mini-universe.
- $2\delta/4\delta$ -apart means the distance between any two points of the set is at least 2δ or 4δ apart. $2\delta(d^{1/p} + 1)$ -apart is the generalization of the 4δ -apart for L_p distances.
- Disj. hash is short for disjoint hash assumption, which means the spatial hashing scheme maps at most one point to the same cell.
- Unique block/unique cell means in spatial hashing, each cell intersects with at most one L_∞ ball and each cell contains at most one point, respectively.

at random from space $\mathbb{F}$. We use $1\{\text{event}\}$ to denote the indicator function that equals 1 if event occurs and 0 otherwise. For a value v , we write $v^{\mathcal{R}}$ to indicate that it is held by the receiver, and $v^{\mathcal{S}}$ for the sender. All protocols in this work are proven secure in the semi-honest model and the formal definition is deferred to Appendix G.1.

3.2 Fuzzy PSI

We formally define the ideal functionality of fuzzy PSI in Figure 1, following prior works [10, 31, 32]. Given a threshold δ and a distance metric $\text{dist}(\cdot, \cdot)$, the functionality takes as input a set from each party and outputs to the receiver all sender elements within distance δ of some receiver element.

Functionality $\mathcal{F}_{\text{FPSI}}$
Parameters: Sender $\mathcal{S}$ and receiver $\mathcal{R}$ with input sizes m, n , respectively. Distance metric $\text{dist}(\cdot, \cdot)$ and threshold δ . Functionality: (1) Wait for input $Q = \{q_j\}_{j \in [m]} \in \mathbb{U}^{d \times m}$ from $\mathcal{S}$ and $W = \{w_i\}_{i \in [n]} \in \mathbb{U}^{d \times n}$ from $\mathcal{R}$. (2) Output $\{q_j \mid \exists i \in [n], \text{dist}(q_j, w_i) \leq \delta\}$ to $\mathcal{R}$.

Figure 1: Functionality of fuzzy PSI.

3.3 Spatial Hashing and Assumptions

Spatial Hashing. To avoid the quadratic complexity of naively comparing every pair of inputs, similar to prior fuzzy PSI under

one-sided assumptions [10, 15, 31, 32], we adopt the spatial hashing technique, which partitions the space $\mathbb{U}^d$ into cells of side length 2δ . For a point x , we define two algorithms:

- (1) $\text{cell}_{2\delta}(x) \rightarrow C$ maps x to the identifier of the cell containing it, where $C = id_1 \| id_2 \| \dots \| id_d$ and $id_k = \lfloor \frac{x_k}{2\delta} \rfloor$ for $k \in [d]$.
- (2) $\text{neigh}_{2\delta}(x) \rightarrow \{C_z\}_{z \in [2^d]}$ maps x to the set of identifiers of all 2^d cells intersecting the L_∞ ball of radius δ centered at x .

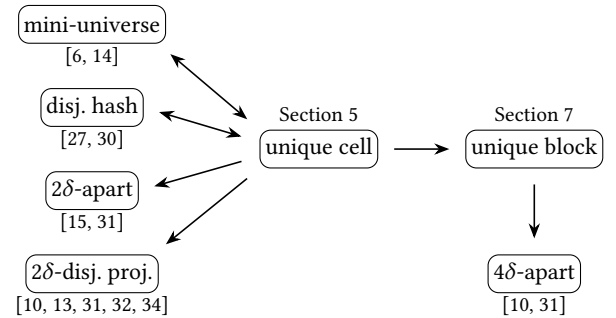


Figure 2: Relationships of existing assumptions. An arrow indicates the direction from the weaker to the stronger assumption, and a double-headed arrow indicates equivalence.

Assumptions. Our protocols operate under two different one-sided assumptions: *unique cell* and *unique block*. The *unique cell* assumption requires that each cell contains at most one point, i.e., every point occupies a distinct cell.

DEFINITION 3.1 (UNIQUE CELL). A set $W \in \mathbb{U}^{n \times d}$ satisfies the *unique cell assumption* if every cell in spatial hashing contains at most one point $\mathbf{w} \in W$. In other words, for any two distinct points $\mathbf{w}, \mathbf{w}' \in W$, $\text{cell}_{2\delta}(\mathbf{w}) \neq \text{cell}_{2\delta}(\mathbf{w}')$.

The *unique block* assumption is sparser, requiring that each cell intersects with at most one L_∞ ball, i.e., every point's L_∞ ball occupies a distinct block of side length 4δ comprising 2^d cells.

DEFINITION 3.2 (UNIQUE BLOCK). A set $W \in \mathbb{U}^{n \times d}$ satisfies the *unique block assumption* if each cell intersects with at most one L_∞ ball centered at $\mathbf{w} \in W$. In other words, for any two distinct points $\mathbf{w}, \mathbf{w}' \in W$, $\text{neigh}_{2\delta}(\mathbf{w}) \cap \text{neigh}_{2\delta}(\mathbf{w}') = \emptyset$.

We provide a comparison of existing assumptions in Figure 2. For clarity, all assumptions are stated with respect to the L_∞ distance; their relationships extend analogously to general L_p distances. Notably, all existing assumptions in the literature are either equivalent to or strictly stronger than the *unique cell* assumption. We defer the formal definitions and detailed analysis of these assumptions to Appendix F.

3.4 Prefix Trie

Chakraborti et al. [7] first introduced the prefix trie technique, which was subsequently studied and formalized by [6, 10, 14, 32]. Let $x = x_\ell x_{\ell-1} \dots x_1 \in \{0, 1\}^\ell$ be a binary string. We adopt the notation of [10]:

- $\text{UpBound}(x_\ell x_{\ell-1} \dots x_k) = x_\ell x_{\ell-1} \dots x_k \| 11 \dots 1$
- $\text{LowBound}(x_\ell x_{\ell-1} \dots x_k) = x_\ell x_{\ell-1} \dots x_k \| 00 \dots 0$
- $\text{Interval}(x_\ell x_{\ell-1} \dots x_k) = \{x_\ell \dots x_k \| x^*\}_{x^* \in \{0, 1\}^{k-\ell}}$

van Baarsen and Pu [32] further summarized the technique and gave the following theorem.

THEOREM 3.1 ([32]). Given an integer interval $[q - \delta, q + \delta] \in \mathbb{Z}_{2^\ell}^{2\delta+1}$ and a point $w \in \mathbb{Z}_{2^\ell}$, there are two algorithms with $\mu, \mu' = O(\log \delta)$:

- (1) $\text{PxTrie}(q - \delta, q + \delta) \rightarrow \{\tilde{q}_1, \dots, \tilde{q}_\mu\}$: The PxTrie algorithm succinctly encodes the interval $[q - \delta, q + \delta]$ into μ prefix nodes $\{\tilde{q}_1, \dots, \tilde{q}_\mu\} \in \mathbb{Z}_{2^\ell}^\mu$ that cover the entire interval without overlap. Specifically, $\mu \leq 2 + \log \delta$ when δ is a power of 2.
- (2) $\text{PxPath}(w, \delta) \rightarrow \{\tilde{w}_1, \dots, \tilde{w}_{\mu'}\}$: The PxPath algorithm expands each query point $w \in \mathbb{Z}_{2^\ell}$ into a path of prefix nodes $\{\tilde{w}_1, \dots, \tilde{w}_{\mu'}\} \in \mathbb{Z}_{2^\ell}^{\mu'}$, where $\mu' := 2 + \log \delta$.

Then, it holds that $\tilde{w}_t \in \{\tilde{q}_1, \dots, \tilde{q}_\mu\}$ for some unique $t \in [\mu']$ if and only if $w \in [q - \delta, q + \delta]$.

For clarity, we assume throughout this paper that δ is a power of 2. We set μ to its maximum value, giving $\mu = \mu' = 2 + \log \delta$. The output of $\text{PxTrie}(q - \delta, q + \delta)$ is always padded to μ prefixes using the default random values. Additionally, we give the following property: the outputs of $\text{PxTrie}(\cdot, \cdot)$ and $\text{PxPath}(\cdot)$ share at most one common element. A formal proof is provided in Appendix G.10.

LEMMA 3.1. For any point $w \in \{0, 1\}^\ell$ and interval $I = [q - \delta, q + \delta]$, the following holds:

$$|\text{PxPath}(w, \delta) \cap \text{PxTrie}(q - \delta, q + \delta)| = \begin{cases} 1, & \text{if } w \in [q - \delta, q + \delta], \\ 0, & \text{otherwise.} \end{cases}$$

3.5 Oblivious Programmable PRF (with Secret-shared Outputs)

An oblivious programmable pseudorandom function (OPPRF) [21] allows the sender to program a pseudorandom function with a set of key-value pairs. When evaluated at a programmed point, the receiver obtains the corresponding programmed value directly; otherwise, it obtains a uniformly random value.

Functionality $\mathcal{F}_{\text{so-OPPRF}}$	
Parameters:	Sender $\mathcal{S}$ and receiver $\mathcal{R}$ with input sizes m, n , respectively.
Functionality:	(1) Wait for input $L = \{(q_j, z_j)\}_{j \in [m]} \subseteq \mathbb{U} \times \mathbb{F}$ from $\mathcal{S}$ and $X = \{x_i\}_{i \in [n]} \subseteq \mathbb{U}$ from $\mathcal{R}$. (2) Sample a random function $F : \mathbb{U} \rightarrow \mathbb{F}$ such that $F(q) = z$ for $(q, z) \in L$. (3) For $i \in [n]$, compute $y_i := F(x_i)$ and sample $y_i^{\mathcal{S}}, y_i^{\mathcal{R}} \leftarrow \mathbb{F}$ such that $y_i^{\mathcal{S}} + y_i^{\mathcal{R}} = y_i \in \mathbb{F}$. (4) Output $\{y_i^{\mathcal{S}}\}_{i \in [n]}$ and O^F to $\mathcal{S}$ and $\{y_i^{\mathcal{R}}\}_{i \in [n]}$ to $\mathcal{R}$.

Figure 3: Functionality of oblivious programmable PRF with secret-shared outputs.

Yang et al. [34] extended the standard OPPRF to support secret-shared outputs, yielding the so-OPPRF. Rather than revealing the outputs directly to the receiver, the so-OPPRF splits it into additive secret shares: when evaluated at a programmed point, both parties obtain shares of the corresponding programmed value; otherwise, they obtain uniformly random shares. The so-OPPRF can be efficiently instantiated from a secret-shared oblivious pseudorandom function (ss-OPRF) [2] and an OKVS. We give the ideal functionality of so-OPPRF in Figure 3.

3.6 Other Building Blocks

Our protocols additionally rely on several standard functionalities, including interval test $\mathcal{F}_{\text{Interval}}$, private equality test $\mathcal{F}_{\text{Eq}}$, secure comparison $\mathcal{F}_{\text{ssCMP}}$, multiplexer $\mathcal{F}_{\text{MUX}}$, and oblivious transfer $\mathcal{F}_{\text{OT}}$. We defer the formal functionalities to Appendix E.

4 Technical Overview

We first introduce the notation used throughout this overview. Let $\mathcal{S}$ (the sender) hold a set $Q = \{q_j\}_{j \in [m]}$ of size m , and let $\mathcal{R}$ (the receiver) hold a set $W = \{w_i\}_{i \in [n]}$ of size n . Each element is a vector in a d -dimensional space, i.e., $q_j, w_i \in \mathbb{U}^d$. We denote by $q_{j,k}$ the k -th coordinate of vector q_j . For clarity, we focus on the L_∞ distance in this technical overview and detail how to extend to general L_p distances in Section 5.

As shown in Figure 4, our starting point of the following technical overview is fuzzy matching (Section 4.1), which can be viewed as a special case of fuzzy PSI in which each party holds a single element. Then, we show how to extend fuzzy matching to multi-point fuzzy matching (Section 4.2), where one party holds multiple points without any assumptions. We identify two possible methods of multi-point fuzzy matching to achieve fuzzy PSI: 1-to- n fuzzy matching (the upper part of Figure 4), where the sender holds a

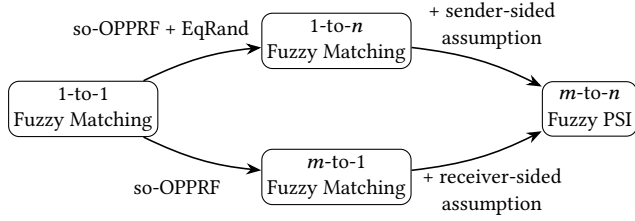


Figure 4: Two approaches to fuzzy PSI, where m is the sender input size and n is the receiver input size. Both approaches only impose an assumption on one of the parties.

single point and the receiver holds n arbitrary points, and m -to-1 fuzzy matching (the lower part of Figure 4), where the receiver holds a single point and the sender holds m arbitrary points. Finally, building on multi-point fuzzy matching, we realize fully fledged fuzzy PSI protocols under sender-sided or receiver-sided assumptions (Section 4.3). In addition, we further optimize our fuzzy PSI protocols using prefix-trie techniques (Section 4.4).

4.1 Revisiting OPPrF-Based Fuzzy Matching

In fuzzy matching, each party holds a single element, denoted by q and w , respectively. The receiver learns q if $\text{dist}(q, w) \leq \delta$, and learns nothing otherwise.

Recently, van Baarsen and Pu [32] presented the state-of-the-art fuzzy matching protocol based on OPPrF (see Section 3.5). At a high level, their key idea is to perform an interval membership test via OPPrF for each dimension and then aggregate the results across all dimensions. Concretely, for each dimension k , the sender samples a uniformly random value s_k and programs all points in the interval $[q_k - \delta, q_k + \delta]$ to the same output s_k . Accordingly, the receiver then evaluates the programmed PRF at input w_k to obtain r_k for each $k \in [d]$. Finally, the sender sends $q \oplus H(\sum_{k \in [d]} s_k)$ to the receiver for an appropriate hash function H , and the receiver recovers q if and only if $\sum_{k \in [d]} r_k = \sum_{k \in [d]} s_k$. This follows because if $\text{dist}(q, w) \leq \delta$, then for every $k \in [d]$, we have $w_k \in [q_k - \delta, q_k + \delta]$, implying $r_k = s_k$ by correctness of OPPrF and thus $\sum_{k \in [d]} r_k = \sum_{k \in [d]} s_k$. Otherwise, if there exists some $k^* \in [d]$ such that $w_{k^*} \notin [q_{k^*} - \delta, q_{k^*} + \delta]$, then $r_{k^*} \neq s_{k^*}$ except with negligible probability and thus $\sum_{k \in [d]} r_k \neq \sum_{k \in [d]} s_k$. Their scheme relies exclusively on symmetric-key operations and achieves substantial efficiency improvements over prior AHE-based approaches [13, 31].

However, the above construction cannot be extended to the setting where the receiver holds multiple points. For example, suppose the receiver holds two points w, w' such that in some dimension k , both w_k, w'_k fall within the same interval $[q_k - \delta, q_k + \delta]$ programmed by the sender. By the correctness of OPPrF, the receiver will obtain the same r_k on these two evaluations. This leads to leakage when there is no match with q : the receiver can infer that the sender has encoded an interval covering both w_k and w'_k in dimension k . Such a collision leaks the distribution of the sender's input. To extend this OPPrF-based fuzzy matching to fuzzy PSI, where both the sender and receiver hold multiple points, [32] prevents this leakage by imposing two-sided assumptions on both parties.

Alternative constructions [10, 31] address the above issue with expensive AHE, thereby keeping the OPPrF outputs hidden from

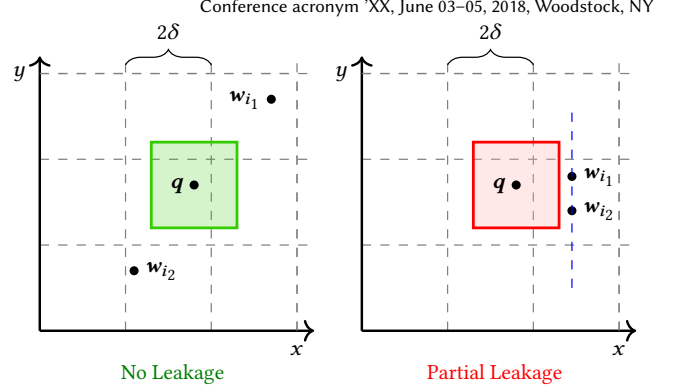


Figure 5: A counterexample for the strawman construction. In the right case, w_{i1} and w_{i2} share the same x -coordinate and both fall within the programmed interval of q along the y -dimension, and are therefore mapped to the same value.

the receiver. In the following, we show how to achieve multi-point fuzzy matching without relying on heavy public-key primitives.

4.2 Multi-point Fuzzy Matching without Assumptions

We propose two variants of multi-point fuzzy matching: 1-to- n fuzzy matching, where the receiver holds n arbitrarily distributed points, and m -to-1 fuzzy matching, where the sender holds m arbitrarily distributed points.

4.2.1 1-to- n Fuzzy Matching. We first introduce our design for the 1-to- n fuzzy matching protocol, in which the receiver holds multiple arbitrary points without any additional assumptions. To address the partial-matching leakage arising from individual dimensions in [32], our key insight is to secret-share the OPPrF output for each dimension so that neither party learns the OPPrF outputs in the clear. We instantiate this approach using shared-output OPPrF (so-OPPrF) [34] (see Section 3.5). Below, we first present a strawman construction that exhibits subtle privacy leakage, and then show how to eliminate it using additional techniques.

Strawman Construction. Let the sender hold a single point q and the receiver hold multiple points $W = \{w_i\}_{i \in [n]}$. Similar to [32], for each dimension k , the sender samples a random s_k and programs the so-OPPrF with key-value pairs $\{(q_k + t, s_k)\}_{t \in [-\delta, \delta]}$. Then, for each $w_i \in W$ and $k \in [d]$, the receiver evaluates the so-OPPrF on input $w_{i,k}$, which outputs shares $r_{i,k}^S$ and $r_{i,k}^R$ to the two parties, respectively. The secret-sharing form avoids the receiver inferring the sender's input from OPPrF outputs of some dimensions. After that, both parties locally aggregate their shares across all dimensions by computing $r_i^S = \sum_{k \in [d]} r_{i,k}^S$ and $r_i^R = \sum_{k \in [d]} r_{i,k}^R$, respectively. Finally, the sender sends all r_i^S and $q \oplus H(\sum_{k \in [d]} s_k)$ to the receiver for an appropriate hash function H . The receiver subsequently reconstructs $r_i = r_i^S + r_i^R$ and recovers q if and only if there exists some $r_i = \sum_{k \in [d]} s_k$.

Partial-matching Leakage. While the above construction appears correct, it introduces subtle privacy leakage. Specifically, although the so-OPPrF outputs remain secret-shared in each dimension, additional information is leaked once the aggregated values r_i

are revealed to the receiver. As illustrated in Figure 5 (right), consider a two-dimensional space in which two distinct receiver points $\mathbf{w}_{i_1}$ and $\mathbf{w}_{i_2}$ share the same x -coordinate and both lie within the programmed interval of $\mathbf{q}$ along the y -coordinate. During the so-OPPRF evaluation, the two points are mapped to the same random value for the x -coordinate. Since $\mathbf{w}_{i_1}$ and $\mathbf{w}_{i_2}$ fall within the same programmed interval along the y -dimension, they are mapped to the same programmed value as well. Consequently, after reconstructing from the shares $r_{i_1}^S$ and $r_{i_2}^S$, the receiver obtains two identical aggregate values, i.e., $r_{i_1} = r_{i_2}$. This collision reveals that there exists a sender point $\mathbf{q}$ that is close to both $\mathbf{w}_{i_1}$ and $\mathbf{w}_{i_2}$ in certain dimensions, even though neither receiver point actually matches $\mathbf{q}$.

The root cause of this leakage is that the aggregated values r_i for non-matching receiver points may coincide, revealing partial information about the sender's input. To eliminate this leakage, we observe that it suffices to re-randomize the secret shares of r_i for non-matching inputs, while leaving those of matching inputs unchanged to preserve correctness.

Enhance Security. To realize this, we introduce a new building block called equality-conditional randomization (EqRand), which randomizes values conditioned on an equality check. Concretely, it takes two pairs (c_1, x_1) from the sender and (c_2, x_2) from the receiver as input, and outputs random shares (y_1, y_2) to the respective parties such that $y_1 + y_2 = x_1 + x_2$ if $c_1 = c_2$; otherwise, $y_1 + y_2$ is uniformly random.

It remains to define an appropriate matching condition. To determine whether a receiver point $\mathbf{w}_i$ matches the sender point $\mathbf{q}$, we additionally program a zero indicator $\mathbf{0}$ together with the target value s_k , encoding each key-value pair as $\mathbf{0} \| s_k$. If $\mathbf{w}_i$ matches the sender point $\mathbf{q}$, it is mapped to $\mathbf{0}$ in every dimension, so the aggregated indicator remains $\mathbf{0}$. Otherwise, the aggregated indicator is uniformly random.

Specifically, the sender programs the so-OPPRF with key-value pairs¹ $\{(k \| \mathbf{q}_k + t, \mathbf{0} \| s_k)\}_{k \in [d], t \in [-\delta, \delta]}$, and the receiver evaluates it on each input $k \| \mathbf{w}_{i,k}$ as before. The two parties obtain packed shares $e_{i,k}^R \| r_{i,k}^R$ and $e_{i,k}^S \| r_{i,k}^S$, respectively, where $e_{i,k}^R$ and $e_{i,k}^S$ are shares of the indicator. They then aggregate these shares across all dimensions $k \in [d]$ to obtain (e_i^R, r_i^R) and (e_i^S, r_i^S) . By the correctness of so-OPPRF, if $\mathbf{w}_i$ matches $\mathbf{q}$, then e_i^R and e_i^S are the shares of $\mathbf{0}$, i.e., $e_i^R + e_i^S = \mathbf{0}$. Otherwise, the shares r_i^R and r_i^S must be re-randomized to prevent collisions. The sender and receiver therefore invoke EqRand with inputs $(-e_i^S, r_i^S)$ and (e_i^R, r_i^R) , using the equality $e_i^R = -e_i^S$ as the matching condition, and obtain fresh shares v_i^R and v_i^S , respectively.

Consequently, the sender can then send v_i^S without incurring leakage, and the receiver reconstructs $v_i = v_i^R + v_i^S$. Finally, as described above, the sender masks $\mathbf{q}$ under the key s using a one-time pad, and the receiver attempts to unmask using each v_i , recovering $\mathbf{q}$ only when $\mathbf{w}_i$ matches $\mathbf{q}$.

4.2.2 m -to-1 Fuzzy Matching. This mirrored path leads to a symmetric setting, where the sender holds multiple points $\mathbf{q}_j \in Q$ and the receiver holds a single point $\mathbf{w}$. Since our multi-point fuzzy matching primitive aims to be assumption-free, directly letting the

sender program all its points into the so-OPPRF, as in the 1-to- n setting, may cause collisions among the key-value pairs. Our key insight is to switch the roles in the so-OPPRF routine: the sender evaluates the so-OPPRF instead of programming it.

This role switching yields a more natural construction. A key difference from the 1-to- n setting is that, for each sender point $\mathbf{q}_j$, the two parties obtain secret shares e_j^R and e_j^S , which are sufficient to jointly determine whether $\mathbf{q}_j$ matches the receiver point $\mathbf{w}$. As a result, the random target value s_k is no longer needed, and the receiver only programs the indicator $\mathbf{0}$.

Concretely, the receiver programs the so-OPPRF with key-value pairs $\{(k \| \mathbf{w}_k + t, \mathbf{0})\}_{k \in [d], t \in [-\delta, \delta]}$, and the sender evaluates it on inputs $\{k \| \mathbf{q}_{j,k}\}_{j \in [m], k \in [d]}$. The two parties obtain shares $e_{j,k}^R$ and $e_{j,k}^S$, and aggregate them across dimensions to obtain $e_j^R = \sum_{k \in [d]} e_{j,k}^R$ and $e_j^S = \sum_{k \in [d]} e_{j,k}^S$. By the correctness of so-OPPRF, the sender point $\mathbf{q}_j$ matches the receiver point if and only if $e_j^R + e_j^S = \mathbf{0}$, equivalently, $e_j^R = -e_j^S$. Then, transferring matching elements becomes considerably simpler. The two parties invoke the private equality test $\mathcal{F}_{\text{Eq}}$ to compare e_j^R and $-e_j^S$, from which the receiver obtains a bit b_j . If $b_j = 0$, then $\mathbf{q}_j$ has no match; otherwise, $\mathbf{q}_j$ matches the receiver point $\mathbf{w}$. They then invoke $\mathcal{F}_{\text{OT}}$ with b_j as the selection bit to transfer $\mathbf{q}_j$ to the receiver.

A notable advantage of this construction is that, since the target value s_k is eliminated and the shares e_j^R and e_j^S remain private throughout, no secret value is ever revealed to either party. Consequently, invoking EqRand is unnecessary. This stands in contrast to the 1-to- n setting in Section 4.2.1, where the aggregated shares are revealed to the receiver and EqRand is required to prevent the receiver from observing identical outputs across multiple inputs.

4.3 Fuzzy PSI under One-sided Assumptions

We now turn to the full m -to- n fuzzy PSI construction. Having established two assumption-free multi-point fuzzy matching protocols in the preceding section, extending each to the m -to- n setting follows the standard paradigm: imposing a separation assumption on the single-input party to allow it to program additional points without collision. To satisfy this requirement, we leverage spatial hashing techniques [10, 31, 32], partitioning the space into cells of side length 2δ and assuming that each cell contains at most one point from the single-input party, while the other party's inputs remain unconstrained. We refer to this as the **unique cell assumption** (see Section 3.3). For each point, we define C_j^S (resp. C_i^R) as the cell identifier of the cell containing $\mathbf{q}_j$ (resp. $\mathbf{w}_i$).

Depending on which party satisfies the unique cell assumption, we distinguish two cases: the *sender-sided assumption* and the *receiver-sided assumption*, each giving rise to a corresponding m -to- n fuzzy PSI construction.

Sender-sided unique cell assumption. In this case, each cell contains at most one sender point. Each sender point $\mathbf{q}_j$ is therefore associated with a unique cell identifier $C_j^S = \text{cell}_{2\delta}(\mathbf{q}_j)$, which the sender concatenates with its programmed keys to build the key-value pairs as $\{(C_j^S \| k \| \mathbf{q}_{j,k} + t, \mathbf{0} \| s_{j,k})\}_{j \in [m], k \in [d], t \in [-\delta, \delta]}$, ensuring that no two key-value pairs collide. Accordingly, as mentioned in [31, 32], the receiver is supposed to evaluate the so-OPPRF on inputs $\{C_{i,z}^R \| k \| \mathbf{w}_{i,k}\}_{i \in [n], k \in [d], z \in [2^d]}$ to maintain correctness, where

¹Here, we encode the dimension index k into each key to allow a single so-OPPRF instance for all dimensions, though it can also be instantiated separately for each dimension as in [32].

$\{C_{i,z}^R\}_{z \in [2^d]} = \text{neigh}_{2\delta}(\mathbf{w}_i)$ (see Section 3.3) denotes the identifiers of all neighboring cells for $\mathbf{w}_i$. This is because two mutually matching points do not necessarily reside in the same cell; they may also fall into neighboring cells. Then, after evaluation, the two parties obtain packed shares $e_{i,k,z}^R \| r_{i,k,z}^R$ and $e_{i,k,z}^S \| r_{i,k,z}^S$, respectively. The remaining steps follow the same process in multi-point fuzzy matching. For the detailed construction, we refer the reader to Section 5.2.

Receiver-sided unique cell assumption. In this case, each cell contains at most one receiver point. Symmetrically, the receiver concatenates the cell identifier C_i^R with its programmed keys to form the key-value pairs $\{(C_i^R \| k \| w_{i,k} + t, \mathbf{0})\}_{i \in [n], k \in [d], t \in [-\delta, \delta]}$, ensuring collision-freeness. The sender then evaluates the so-OPPRF at inputs $\{C_{j,z}^S \| k \| q_{j,k}\}_{j \in [m], k \in [d], z \in [2^d]}$, where $\{C_{j,z}^S\}_{z \in [2^d]}$ denotes all candidate cell identifiers for $\mathbf{q}_j$. Then, after evaluation, the two parties obtain shares $e_{j,k,z}^R$ and $e_{j,k,z}^S$, respectively. The remaining steps also follow the same process in multi-point fuzzy matching. For the detailed construction, we refer the reader to Section 5.3.

4.4 Optimizations from Prefix Trie Techniques without Superlinear Overhead

In this section, we describe how to non-trivially incorporate prefix-trie techniques [10, 32] into our protocol to achieve $O(\log \delta)$ complexity in δ , in contrast to the superlinear $O((\log \delta)^d)$ complexity incurred by prior works [10, 32]. For clarity, we focus on the construction under the receiver-sided assumption; the sender-sided case follows analogously and is deferred to Appendix I.1.

Dang et al. [10] and van Baarsen and Pu [32] introduced an optimization that represents each interval $[w_{i,k} - \delta, w_{i,k} + \delta]$ via $\mu = O(\log \delta)$ prefixes, reducing the number of programmed points from $O(\delta)$ to $O(\log \delta)$. As a trade-off, however, the sender must evaluate the OPPRF along all μ prefix paths of $q_{j,k}$ to maintain correctness, obtaining outputs $\{e_{j,k,h}\}_{h \in [\mu]}$. For each dimension k , since the sender has no knowledge of which prefix is valid, it must exhaustively search all combinations $\sum_{k \in [d]} e_{j,k,h_k}$ for $h_k \in [\mu]$ and compare each against the target value, yielding a complexity² of $O((\log \delta)^d)$ (or $O((2 \log \delta)^d)$ for L_p distance [10]).

Dimension-by-dimension Filtering. Our key observation is that, for each dimension, at most one prefix can match the target interval, rendering all other prefixes unnecessary. This allows both parties to prune the search space on a per-dimension basis by discarding non-matching prefixes, rather than enumerating all prefix combinations across dimensions as in [10, 32]. Since the exponential blowup in prior works arises precisely from taking the Cartesian product of per-dimension prefix sets, retaining only one candidate prefix per dimension collapses the search space from $O((\log \delta)^d)$ combinations to $O(d \cdot \log \delta)$.

Specifically, we denote the prefixes of the interval centered at $w_{i,k}$ as $\{w_{i,k,h}\}_{h \in [\mu]} = \text{PxTrie}(w_{i,k} - \delta, w_{i,k} + \delta)$, and the prefix paths of $q_{j,k}$ as $\{q_{j,k,h}\}_{h \in [\mu]} = \text{PxPath}(q_{j,k}, \delta)$, where $\mu = O(\log \delta)$ denotes the number of prefixes. The sender evaluates the so-OPPRF at inputs $C_{j,z}^S \| k \| q_{j,k,h}$, and the parties obtain shares $e_{j,k,z,h}^R$ and $e_{j,k,z,h}^S$.

²This can be reduced to $O((\log \delta)^{d/2})$ via the meet-in-the-middle technique, but the complexity still grows rapidly for large δ .

According to Lemma 3.1, if $q_{j,k}$ falls within $[w_{i,k} - \delta, w_{i,k} + \delta]$, there exists exactly one $h^* \in [\mu]$ such that $e_{j,k,z,h^*}^R + e_{j,k,z,h^*}^S = \mathbf{0}$.

To extract this unique index h^* and drop out all other non-matching indices, we introduce a new primitive termed conditional selection $\mathcal{F}_{\text{ConSel}}^{f,\mu}$ in Figure 10. Concretely, it takes two tuples of μ secret-shared values $(e_1^R, \dots, e_\mu^R)$ and $(e_1^S, \dots, e_\mu^S)$ as input, and outputs two random shares (y_1, y_2) such that $y_1 + y_2 = e_{h^*}^R + e_{h^*}^S$ if there exists a unique index $h^* \in [\mu]$ with $f(e_{h^*}^R, e_{h^*}^S) = 1$, and $y_1 + y_2$ is uniformly random otherwise. Here, we instantiate the condition function f as an equality check Eq and use the equality $e_{j,k,z,h}^R = -e_{j,k,z,h}^S$ as the condition.

Consequently, for each dimension k , both parties input their per-prefix shares $\{e_{j,k,z,h}^R\}_{h \in [\mu]}$ and $\{-e_{j,k,z,h}^S\}_{h \in [\mu]}$ to $\mathcal{F}_{\text{ConSel}}^{\text{Eq},\mu}$ and receive a single pair of fresh shares $(v_{j,k,z}^R, v_{j,k,z}^S)$. Lemma 3.1 ensures that at most one prefix matches per dimension, so this functionality is well-defined. After this process, each dimension retains exactly one candidate prefix, and therefore yields just one combination across all dimensions, in contrast to the $O((\log \delta)^d)$ combinations in prior works. The complexity of $\mathcal{F}_{\text{ConSel}}$ is $O(\log \delta)$ for each dimension, and $O(d \cdot \log \delta)$ across all dimensions.

Given the shares $(v_{j,k,z}^R, v_{j,k,z}^S)$, with the prefix index eliminated, the remaining steps follow the same procedure as in the non-optimized construction of Section 4.3.

5 Fuzzy PSI under Unique Cell Assumptions

In this section, we present our protocols for general $L_{p \in [1, \infty]}$ distances under one-sided *unique cell* assumptions. That is, only one party (either the sender or the receiver) is required to have at most one point in each cell, while the other party's set may follow an arbitrary distribution.

Functionality $\mathcal{F}_{\text{ConRand}}^f$
Parameters: Sender S and Receiver R . Condition predicate function $f(\cdot, \cdot)$ where f can be an equality or a comparison function. Functionality: (1) Wait for input (e^S, v^S) from S and (e^R, v^R) from R. (2) Compute $b := f(e^S, e^R)$. (3) Generate random shares z^S, z^R such that $z^S + z^R = v^S + v^R$ if $b = 1$, and sample z^S, z^R uniformly at random otherwise. (4) Output z^S to S and z^R to R.

Figure 6: Functionality of conditional randomization.

5.1 Building Block

Before presenting our fuzzy PSI construction, we introduce a building block for preventing information leakage. Specifically, it takes two tuples (e^S, v^S) and (e^R, v^R) as input from the respective parties, and randomizes the values v^S and v^R depending on whether the conditions e^S and e^R satisfy a predefined criterion f . The ideal functionality of Conditional Randomization is given in Figure 6.

We realize this functionality efficiently via a customized secret-shared multiplexer protocol [29]. The conditional function $f(\cdot, \cdot)$ returns a boolean result based on a specified condition, and can be

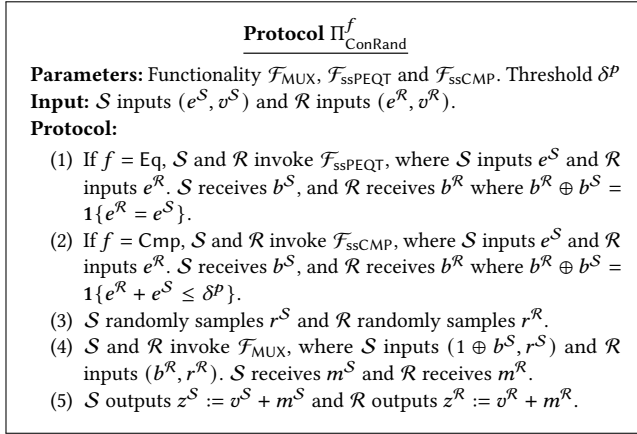


Figure 7: Protocol of conditional randomization.

instantiated with either an equality or a comparison function. Depending on the choice of f , we refer to this functionality as Equality-conditional Randomization (EqRand) or Comparison-conditional Randomization (CmpRand) in the remainder of this paper. The protocol is given in Figure 7 and we defer the proof in Appendix G.8.

5.2 Protocols for Sender-sided Setting

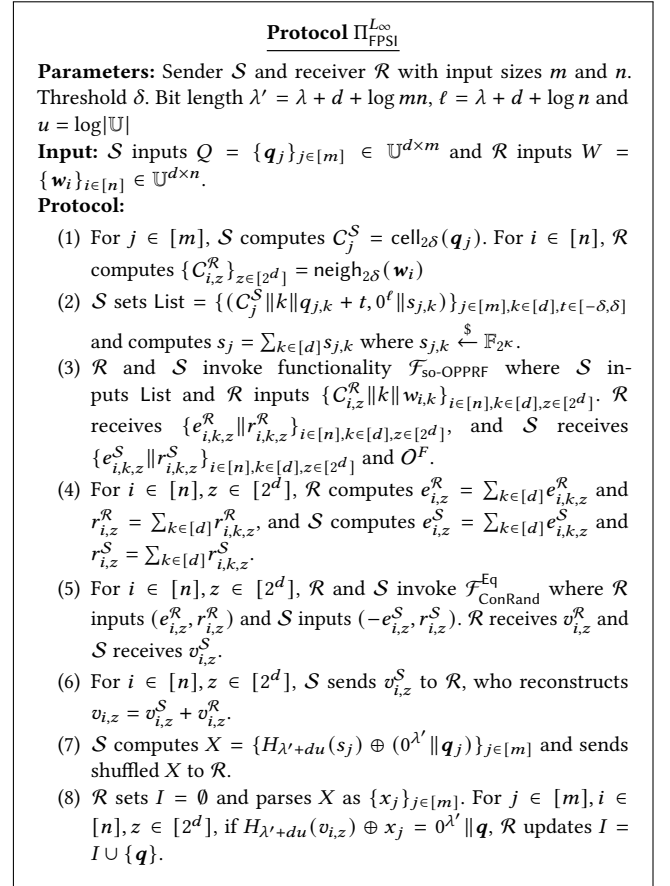
Since we have already presented the main idea for the L_∞ distance in Section 4.2.1 and Section 4.3, we directly provide the detailed protocol in Figure 8. Its security proof is deferred to Appendix G.2.

THEOREM 5.1. *The protocol $\Pi_{\text{FPSI}}^{L_\infty}$ in Figure 8 securely realizes the functionality $\mathcal{F}_{\text{FPSI}}$ for L_∞ distance against semi-honest adversaries in the $(\mathcal{F}_{\text{so-OPPRF}}, \mathcal{F}_{\text{ConRand}}^{\text{Eq}})$ -hybrid model, assuming H is a random oracle.*

L_p distance. We here focus on how to support arbitrary L_p distances. Our key observation is the fact that $\text{dist}_p(\mathbf{w}_i, \mathbf{q}_j) \leq \delta$ implies $\text{dist}_\infty(\mathbf{w}_i, \mathbf{q}_j) \leq \delta$, and we therefore could reuse most parts of the L_∞ construction with only minor modifications. In the L_∞ setting, the sender programs the value $0\|s_{j,k}$, where 0 serves as an indicator of whether $\mathbf{w}_i$ is close to some $\mathbf{q}_j$, as described in Section 4.2.1. For L_p distance, the sender instead encodes $|t|^p\|s_{j,k}$, where $|t|^p$ captures the per-dimension distance contribution when $w_{i,k} - q_{j,k} = t$ for some $t \in [-\delta, \delta]$. The receiver proceeds as before, evaluating the so-OPPRF at $C_{i,z}^R\|k\|\mathbf{w}_{i,k}$. After aggregating across all dimensions, $d_{i,z}^R$ and $d_{i,z}^S$ are secret shares of the p -th power of the distance between $\mathbf{w}_i$ and the candidate $\mathbf{q}_j$, provided they are potentially matched. To avoid the similar leakage mentioned in Section 4.2.1, we invoke the CmpRand to randomize shares $r_{i,z}^R$ and $r_{i,z}^S$ depending on the comparison condition $d_{i,z}^R + d_{i,z}^S \leq \delta^p$. The remaining steps are identical to those in the L_∞ setting, allowing the receiver to obtain all valid matches. We defer the detailed protocol in Figure 21 of Appendix H.1.

5.3 Protocols for Receiver-sided Setting

Since we have presented our idea in Section 4.2.2 and Section 4.3, we directly give the protocol under the receiver-sided assumption

Figure 8: Protocol of fuzzy PSI for L_∞ distance under sender-sided unique cell assumption.

for L_∞ distance in Figure 9. The security proof is deferred to Appendix G.4.

THEOREM 5.2. *The protocol $\Pi_{\text{FPSI}}^{L_\infty}$ in Figure 9 securely realizes the functionality $\mathcal{F}_{\text{FPSI}}$ for L_∞ distance against semi-honest adversaries in the $(\mathcal{F}_{\text{so-OPPRF}}, \mathcal{F}_{\text{Eq}}, \mathcal{F}_{\text{OT}})$ -hybrid model.*

L_p distance. We next show how to extend the protocol to L_p distance. Similarly, the receiver now programs $|t|^p$ in place of 0 as in the L_∞ setting. As noted in Section 4.2.2, invoking conditional randomization is unnecessary under the receiver-sided assumption, since the receiver does not reconstruct $r_{j,z}$. By the correctness of $\mathcal{F}_{\text{so-OPPRF}}$, $r_{j,z}^R$ and $r_{j,z}^S$ are secret shares of the p -th power of the distance between $\mathbf{w}_i$ and the candidate $\mathbf{q}_j$. It therefore suffices to test whether $r_{j,z}^R + r_{j,z}^S$ falls within interval $[0, \delta^p]$. A natural approach would be to invoke the secure comparison $\mathcal{F}_{\text{ssCMP}}$; however, since the output need not be secret-shared, we instead realize this check via an interval test (see Appendix E), following [7, 32].

Specifically, the two parties invoke functionality $\mathcal{F}_{\text{Interval}}^{\delta^p}$, and the receiver obtains outputs $b_{j,z} = 1\{r_{j,z}^R + r_{j,z}^S \leq \delta^p\}$. Since q_j may match multiple receiver elements, the receiver aggregates the per-cell results as $b_j = \bigvee_{z \in [2^d]} b_{j,z}$. Finally, b_j serves as the choice bit, and the sender transfers the matching elements to the receiver

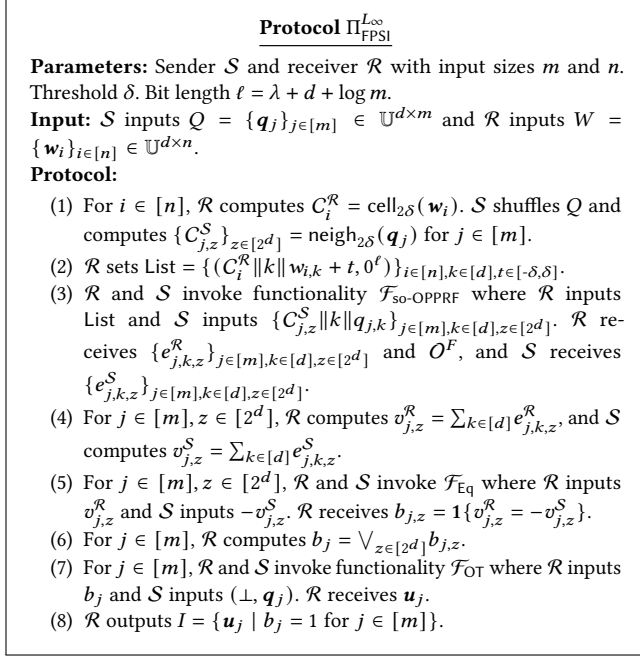


Figure 9: Protocol of fuzzy PSI for L_∞ distance under receiver-sided unique cell assumption.

via an OT protocol. We defer the detailed protocol in Figure 22 of Appendix H.2.

6 Optimized Fuzzy PSI from Prefix Trie

In this section, we optimize our protocols to achieve $O(\log \delta)$ complexity in the threshold δ via prefix trie techniques.

We introduce a core building block that resolves the exponential complexity arising from the incorporation of prefix trie techniques. The ideal functionality is given in Figure 10. Functionality $\mathcal{F}_{\text{ConSel}}^{f,\mu}$ takes as input two sets of μ values and selects a unique pair based on a condition f . Specifically, if there exists a unique index i such that $f(e_i^{\mathcal{S}}, e_i^{\mathcal{R}}) = 1$, it outputs fresh random shares summing to $e_i^{\mathcal{S}} + e_i^{\mathcal{R}}$; otherwise, it outputs uniformly random shares. For each dimension, this functionality acts as a filter, extracting only one prefix and discarding all non-matching prefixes, thereby avoiding exponential complexity. It is worth noting that this functionality is predicated on there being at most one pair of inputs satisfying f , as it produces only a single pair of outputs. The uniqueness of the matching prefix is established in Lemma 3.1, ensuring that this functionality can be safely adopted without compromising correctness when we instantiate the f as an equality test function Eq in our protocol. The concrete construction is presented in Figure 11, and we refer to it as Equality-conditional Selection (EqSel). When $\mu = 1$, this functionality reduces to a special case of EqRand in which the value is the condition itself. Its correctness follows that of $\mathcal{F}_{\text{ConRand}}^f$, and we defer the security proof to Appendix G.9.

We have presented our idea to incorporate prefix trie in Section 4.4, and the prefix-optimized protocol under the receiver-sided assumption for L_∞ distance is illustrated in Figure 12. We defer the

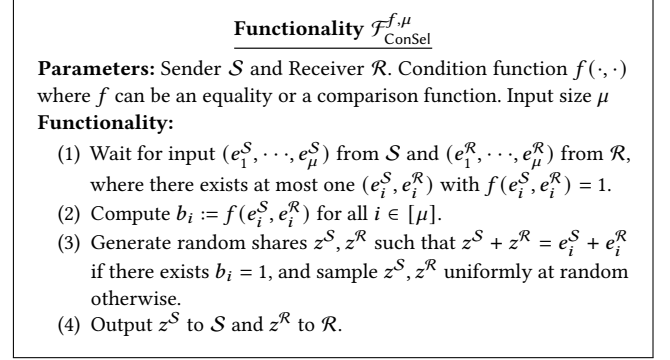


Figure 10: Functionality of conditional selection.

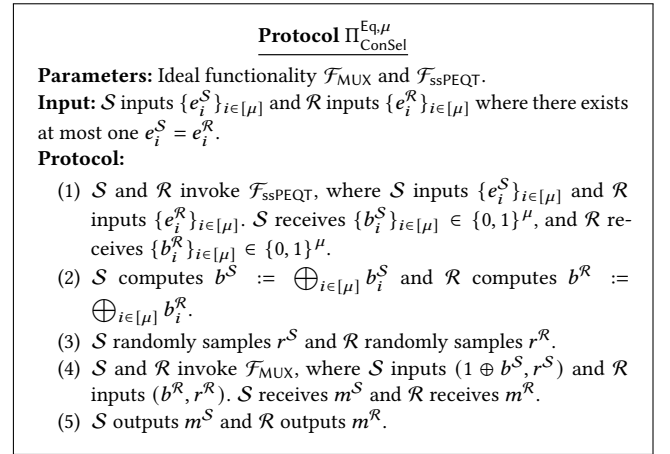


Figure 11: Protocol of equality-conditional selection.

security proof in Appendix G.6. The protocols for L_p distances can be optimized analogously, and we defer the details to Appendix I.

THEOREM 6.1. *The protocol $\Pi_{\text{FPSI-PX}}^{L_\infty}$ in Figure 12 securely realizes the functionality $\mathcal{F}_{\text{FPSI}}$ for L_∞ distance against semi-honest adversaries in the $(\mathcal{F}_{\text{so-OPPRF}}, \mathcal{F}_{\text{ConSel}}^{\text{Eq},\mu}, \mathcal{F}_{\text{Eq}}, \mathcal{F}_{\text{OT}})$ -hybrid model.*

7 Fuzzy PSI for Unique Block Assumptions

We further consider a sparser distribution, termed *unique block* (refer to Appendix F), in which each cell intersects with at most one L_∞ ball centered at a receiver point. This distribution guarantees that each receiver point's L_∞ ball occupies a unique block and is therefore associated with a collision-free set of 2^d neighboring cells.

Unlike the *unique cell* setting, our insight is to reverse the roles of the two parties in the spatial hashing procedure: the receiver programs all 2^d candidate cells without collision, while the sender evaluates only the single cell containing its point. This shifts the 2^d factor from the evaluation side to the programming side. As noted in [34], programming the so-OPPRF primarily involves OKVS [28] encoding operations, which is substantially cheaper than evaluating the so-OPPRF; exploiting this asymmetry therefore yields a more efficient construction.

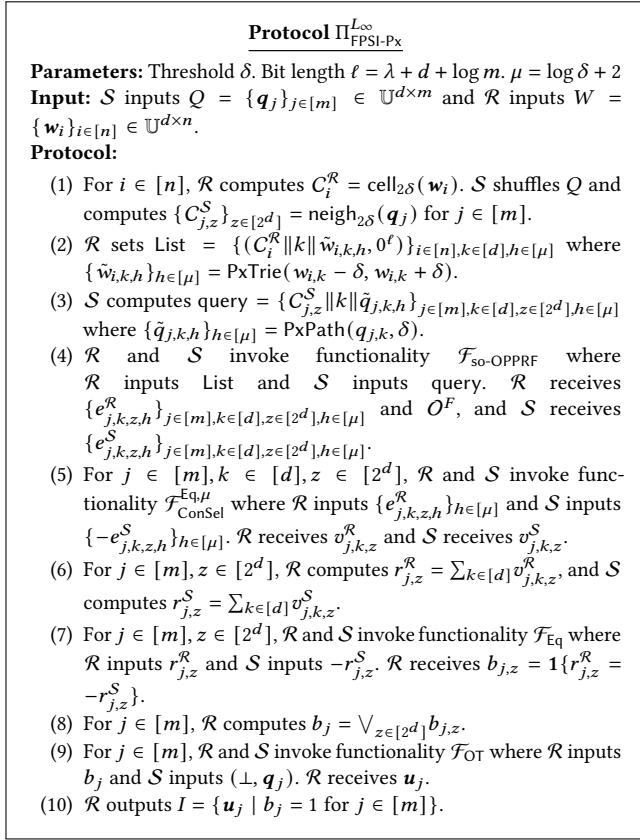


Figure 12: Protocol of prefix-optimized fuzzy PSI for L_∞ distance under receiver-sided unique cell assumption.

The detailed protocol for L_∞ distance is presented in Figure 13 and the security proof is deferred to Appendix G.7. Constructions for general L_p distances and their prefix-optimized variants are provided in Appendix J.

THEOREM 7.1. *The protocol $\Pi_{\text{FPSI}}^{L_\infty}$ in Figure 13 securely realizes the functionality $\mathcal{F}_{\text{FPSI}}$ for L_∞ distance against semi-honest adversaries in the $(\mathcal{F}_{\text{so-OPPRF}}, \mathcal{F}_{\text{Eq}}, \mathcal{F}_{\text{OT}})$ -hybrid model.*

8 Evaluation

We provide a full implementation of our protocols, available at <https://anonymous.4open.science/r/FPSI-One-Sided-1325>. All experiments are conducted on a single server equipped with AMD EPYC 9554 processor and 512 GB of RAM, with the sender and receiver emulated as two separate threads. Each experiment is repeated ten times, and we report the average total running time and communication cost. We set the computational and statistical security parameters to $\kappa = 128$ and $\lambda = 40$, respectively. For implementation, we adopt the so-OPPRF from [2, 34], silent OT from libOTe³, secret-shared comparison from [17], and instantiate hash functions using BLAKE3⁴.

³<https://github.com/osu-crypto/libOTe>

⁴<https://github.com/BLAKE3-team/BLAKE3>

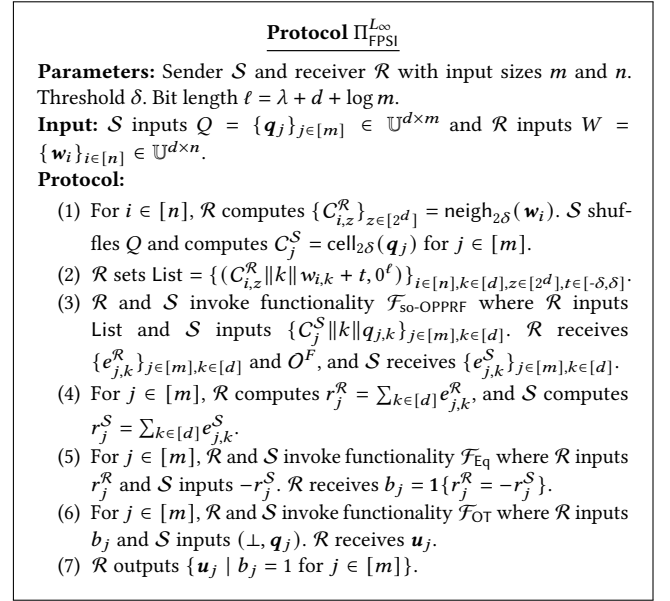


Figure 13: Protocol of fuzzy PSI for L_∞ distance under receiver-sided unique block assumption.

Comparison Baselines. We compare our protocols with state-of-the-art fuzzy PSI constructions under one-sided assumptions [6, 10, 31]. The evaluation is divided according to two assumptions: unique cell and unique block. Under the unique cell assumption, we compare against [6, 31], as our unique cell assumption is strictly weaker than the 2δ -apart assumption of [31] and equivalent to the mini-universe assumption of [6]. Under the unique block assumption, we compare against [10, 31], since our unique block assumption is strictly weaker than the 4δ -apart assumption used in [10, 31]. Please refer to Appendix F for detailed comparisons of different assumptions. For fair comparison with [6], we double its radius parameter, as [6] defines L_∞ balls with side length δ rather than 2δ .

Since these baselines [6, 10, 31] consider only receiver-sided assumptions, we evaluate our protocols under the same receiver-sided setting. Our sender-sided results are deferred to Appendix K.2.

8.1 Performance on Unique Cell Assumptions

We compare our protocols with the state-of-the-art constructions [6, 31] for L_∞ distance under the unique cell assumption in Table 2. The results show that our protocols outperform prior works in both communication and computation across most parameter settings. Since existing works do not support general L_p distance under this assumption, we present our results for L_1 and L_2 distances in Appendix K.1.

Communication. Against van Baarsen and Pu [31], our best protocol reduces communication by up to $20\times$ at $d = 2$, with the advantage growing substantially as δ increases, a direct consequence of our sublinear dependence on δ versus the linear growth of [31]. At higher dimensions ($d = 4, 6$), [31] achieves lower communication at small $\delta = 32$, but our protocols overtake it as δ grows beyond 64 or 128. Against Bui et al. [6], our protocols achieve up to $282\times$ communication reduction at $d = 6$, where the communication of [6]

Table 2: The communication (MB) and running time (s) of fuzzy PSI for L_∞ distance under the unique cell assumption. Results of our protocols for L_p distance are reported in Appendix K.1, as existing works do not support this setting. “—” indicates out-of-memory or execution timeout ($> 2.5h$). The best result is marked in green.

Size $m=n$	Dim. d	Protocol	$\delta = 32$		$\delta = 64$		$\delta = 128$		$\delta = 256$		$\delta = 512$	
			Comm	Time	Comm	Time	Comm	Time	Comm	Time	Comm	Time
2^8	2	van Baarsen et al. [31]	3.11	1.77	6.11	3.47	12.11	6.95	24.11	14.05	48.11	27.98
		Bui et al. [6]	3.56	1.48	4.15	1.53	4.32	1.71	4.98	1.73	5.15	2.07
		Ours	2.59	0.19	3.27	0.19	4.63	0.23	7.34	0.31	12.83	0.37
		Ours-Px	3.26	0.41	3.34	0.39	3.42	0.43	3.77	0.39	3.77	0.40
	4	van Baarsen et al. [31]	6.34	3.72	12.34	7.12	24.34	14.31	48.34	27.77	96.34	55.63
		Bui et al. [6]	52.45	10.53	121.79	14.13	122.19	23.61	248.00	28.92	248.02	42.94
		Ours	8.61	0.36	9.96	0.44	12.68	0.57	18.12	0.62	29.06	0.75
		Ours-Px	15.23	1.23	15.58	1.16	15.58	1.22	15.92	1.27	16.26	1.28
	6	van Baarsen et al. [31]	10.14	6.43	19.14	11.64	37.14	22.08	73.14	43.13	145.14	83.55
		Bui et al. [6]	3070.90	2048.17	11695.09	2755.55	11695.82	7516.48	—	—	—	—
		Ours	39.37	1.38	41.41	1.50	45.50	1.56	53.64	1.76	70.08	2.00
		Ours-Px	82.97	5.25	83.07	5.30	83.31	5.21	83.83	5.29	84.33	5.61
2^{12}	2	van Baarsen et al. [31]	49.75	29.07	97.75	56.37	193.75	112.96	385.75	222.60	769.75	442.27
		Bui et al. [6]	38.18	3.57	47.55	4.64	50.18	6.55	61.06	9.62	63.69	14.78
		Ours	24.57	0.94	35.45	1.18	57.22	1.89	100.78	3.26	188.16	3.94
		Ours-Px	31.45	2.15	32.82	2.13	34.17	2.16	36.90	2.18	39.60	2.28
	4	van Baarsen et al. [31]	101.50	59.81	197.50	115.70	389.50	228.65	773.50	448.70	1541.50	887.83
		Bui et al. [6]	819.28	155.32	1931.30	206.54	1936.54	341.16	3954.58	453.26	3959.78	704.95
		Ours	120.25	4.06	142.02	4.85	185.59	6.30	272.76	8.79	447.53	9.80
		Ours-Px	223.49	13.23	226.21	14.06	228.92	12.95	234.36	13.85	239.79	14.13
	6	van Baarsen et al. [31]	162.25	106.67	306.25	192.80	594.25	354.21	1170.25	690.49	2322.25	1357.23
		Bui et al. [6]	—	—	—	—	—	—	—	—	—	—
		Ours	612.54	21.07	645.25	22.27	710.70	25.39	841.68	27.75	1104.24	29.37
		Ours-Px	1311.37	83.49	1315.45	79.59	1319.52	83.74	1327.71	80.75	1335.89	81.50
	8	van Baarsen et al. [31]	796.00	478.07	1564.00	904.65	3100.00	1836.01	6172.00	3693.64	12316.00	7499.23
		Bui et al. [6]	583.20	39.28	733.25	55.02	775.30	80.48	949.34	122.20	991.39	240.33
		Ours	376.23	12.40	550.66	17.76	899.68	31.78	1598.21	48.09	2997.97	69.62
		Ours-Px	484.52	27.98	506.41	27.71	528.06	28.68	571.86	31.23	615.18	31.37
	10	van Baarsen et al. [31]	1624.00	970.83	3160.00	1877.65	6232.00	3778.05	12376.00	7453.94	—	—
		Bui et al. [6]	13078.68	2522.63	30874.60	3446.77	30958.79	5633.09	—	—	—	—
		Ours	1907.65	63.12	2256.70	75.11	2955.23	95.62	4352.80	132.53	7153.31	168.36
		Ours-Px	3579.83	219.52	3623.63	218.46	3666.95	220.18	3754.61	218.06	3841.31	220.76
2^{16}	12	van Baarsen et al. [31]	2596.00	1818.51	4900.00	3066.45	9508.00	5825.47	18724.00	—	37156.00	—
		Bui et al. [6]	—	—	—	—	—	—	—	—	—	—
		Ours	9787.03	344.87	10311.51	349.66	11361.00	392.36	13467.23	420.58	17669.67	488.08
		Ours-Px	21121.37	1380.11	21186.78	1383.61	21252.35	1437.40	21383.25	1506.43	21514.43	1399.69

blows up due to its $O((\log \delta)^d)$ complexity. At $d = 2$, the gap is smaller (1.2–1.7 $\times$), as [6] benefits from low-dimensional setting.

Computation. The computational advantage is significant across all settings. Against van Baarsen and Pu [31], our protocols achieve speedups of 4–239 $\times$, with the largest gains at large δ and large set sizes, reaching over 200 $\times$ at $n = 2^{16}$ and $\delta = 512$. Against Bui et al. [6], we achieve speedups of 3–4818 $\times$, with the most dramatic gains at $d = 6$, where the running time of [6] becomes prohibitive due to its $O((\log \delta)^d)$ complexity. These gains stem primarily from our exclusive use of symmetric-key primitives, avoiding both the superlinear complexity of [6] and the expensive additive homomorphic operations of [31].

Effect of Prefix Optimization. These two variants expose a clear communication-computation tradeoff. The non-optimized protocol scales linearly in δ and achieves the lowest running time for small δ

(32 and 64). The prefix-optimized protocol scales logarithmically in δ , making it superior in communication at large δ ; for instance, at $\delta = 512$ and $d = 2$, it reduces communication by up to 20 $\times$ over [31] while maintaining a substantial computational advantage.

8.2 Performance on Unique Block Assumptions

We compare our protocols with the state-of-the-art constructions [10, 31] under the unique block assumption. Table 3 reports results for set size 2^{12} as a representative case; other settings exhibit similar trends, since both prior works and our protocols scale linearly with the set size.

Communication. Compared to van Baarsen and Pu [31], the communication reduction grows substantially with the threshold: at $\delta = 32$ we achieve 6–18 $\times$ reduction, increasing to 34–100 $\times$ at $\delta = 256$ and up to 67 $\times$ at $\delta = 512$. This rapid scaling is a direct

Table 3: The communication (MB) and running time (s) of fuzzy PSI under the unique block assumption. The set size is $m = n = 2^{12}$. “—” indicates out-of-memory or execution timeout ($> 2.5h$). The best result is marked in green.

Metric	Dim. d	Protocol	$\delta = 32$		$\delta = 64$		$\delta = 128$		$\delta = 256$		$\delta = 512$	
			Comm	Time	Comm	Time	Comm	Time	Comm	Time	Comm	Time
L_∞	2	van Baarsen et al. [31]	195.25	111.97	387.25	223.08	771.25	446.28	1539.25	893.64	3075.25	1806.35
		Dang et al. [10]	127.56	14.42	145.48	16.16	164.38	19.12	183.55	18.61	201.86	25.58
		Ours	48.64	1.63	92.21	2.70	179.38	5.24	353.82	11.08	702.82	20.15
		Ours-Px	29.82	1.92	33.77	2.07	37.85	2.37	41.89	2.56	45.81	2.82
	4	van Baarsen et al. [31]	1560.25	932.15	3096.25	1836.02	6168.25	3732.99	12312.25	7697.61	—	—
		Dang et al. [10]	2289.54	196.76	1576.67	142.00	2006.78	161.69	2578.55	208.96	2866.33	280.98
		Ours	361.74	10.41	710.78	20.49	1409.32	39.75	2806.88	77.75	5603.26	162.37
		Ours-Px	86.59	3.93	98.69	4.47	110.97	4.89	123.19	5.51	135.36	5.89
	6	van Baarsen et al. [31]	9360.25	5507.11	—	—	—	—	—	—	—	—
		Dang et al. [10]	13731.49	1192.01	9443.74	821.75	12016.79	1013.32	15450.58	1275.52	17170.10	1617.10
		Ours	2141.18	60.29	4241.20	124.89	8443.85	254.36	16852.49	545.97	33544.37	1302.04
		Ours-Px	301.30	8.60	344.08	9.75	386.42	10.76	429.03	12.25	472.16	13.39
L_1	2	van Baarsen et al. [31]	195.77	112.01	388.27	224.03	773.27	454.50	1543.27	893.72	3083.27	1787.95
		Dang et al. [10]	312.84	155.45	381.93	207.62	456.06	279.29	534.66	359.87	618.78	461.89
		Ours	51.89	1.77	95.50	3.00	182.70	5.58	357.28	10.52	706.39	20.70
		Ours-Px	80.24	3.70	93.18	4.56	106.17	5.15	119.33	5.55	132.42	6.21
	4	van Baarsen et al. [31]	1560.77	901.23	3097.27	1823.36	6170.27	3660.91	12316.27	7279.21	—	—
		Dang et al. [10]	3609.97	603.43	4944.93	5443.02	5517.94	5624.82	6376.92	5695.84	—	—
		Ours	367.31	10.78	716.39	20.94	1414.96	42.03	2812.66	83.38	5609.16	168.24
		Ours-Px	207.22	8.90	241.11	10.50	275.20	11.76	309.72	12.68	343.99	14.94
	6	van Baarsen et al. [31]	9360.77	5481.36	—	—	—	—	—	—	—	—
		Dang et al. [10]	22946.21	7243.12	—	—	—	—	—	—	—	—
		Ours	2149.07	64.90	4249.12	129.43	8451.80	258.02	16860.58	562.71	33552.58	1333.52
		Ours-Px	604.28	24.96	704.89	25.36	805.00	28.15	904.95	31.66	1004.74	40.38
L_2	2	van Baarsen et al. [31]	211.27	112.01	451.27	221.86	1027.27	448.16	2563.27	891.59	7171.27	1803.15
		Dang et al. [10]	530.49	151.32	642.30	212.06	766.64	285.79	911.44	383.57	1092.79	566.46
		Ours	52.33	1.74	96.40	3.13	183.63	5.44	358.31	10.79	707.55	20.88
		Ours-Px	108.60	4.68	126.64	5.12	144.30	6.07	162.18	6.58	180.02	7.55
	4	van Baarsen et al. [31]	1576.27	896.00	3160.27	1795.90	6424.27	3604.21	13336.27	7208.44	—	—
		Dang et al. [10]	6995.61	627.84	7493.30	5470.09	8644.48	5484.85	10379.32	5568.10	—	—
		Ours	367.74	10.91	717.29	20.96	1415.90	43.00	2813.70	84.26	5610.32	166.81
		Ours-Px	263.35	10.25	306.95	11.92	350.33	13.68	394.20	14.97	437.85	16.77
	6	van Baarsen et al. [31]	9376.27	5485.92	—	—	—	—	—	—	—	—
		Dang et al. [10]	43227.86	7374.09	—	—	—	—	—	—	—	—
		Ours	2149.51	63.81	4250.02	128.59	8452.74	252.92	16861.62	548.45	33553.74	1311.12
		Ours-Px	688.17	26.87	803.11	29.06	917.13	31.56	1031.06	35.08	1144.85	45.69

consequence of our logarithmic dependence on δ , whereas [31] incurs communication growing linearly in δ or δ^p . Against Dang et al. [10], our prefix-optimized protocol consistently reduces communication by 4–63× across all metrics and parameter settings, with the largest gains at higher dimension ($d = 6$). We note that our protocol achieves better performance at larger parameter settings, owing to the underlying silent OT [3] used in our implementation, which offers superior amortized communication efficiency for large batch generation.

Computation. The computational advantage is even more pronounced. Against van Baarsen and Pu [31], our protocol achieves speedups of 63–1397× across all metrics and dimensions. Against Dang et al. [10], we achieve 8–518× speedups. These dramatic gains stem from two factors. First, our prefix-optimized protocol achieves $O(\log \delta)$ complexity, while [31] scales linearly in δ and [10] incurs $O((\log \delta)^d)$ complexity. Second, our protocols rely exclusively on

symmetric-key primitives, whereas both [10, 31] depend on expensive public-key operations such as DDH and Paillier.

Effect of Prefix Optimization. The prefix-optimized variant attains the best performance in nearly all settings, in contrast to the previous case. This is because, under the *unique block* assumption, the dominant bottleneck is the receiver programming $2^d \cdot d \cdot n \cdot (2\delta + 1)$ key-value pairs, while the sender evaluates at only $d \cdot m$ points. Incorporating prefix trie techniques reduces the receiver’s overhead by a factor of $O(\delta / \log \delta)$, while increasing the sender’s overhead by only $O(\log \delta)$, yielding a clear improvement in overall efficiency. Moreover, this advantage becomes more pronounced as the dimension d and the threshold δ increase, indicating that the prefix-optimized variant is well-suited to large-scale settings.

9 Conclusion

In this work, we presented concretely efficient fuzzy PSI protocols for general $L_{p \in [1, \infty]}$ distances under one-sided assumptions, relying solely on lightweight symmetric-key primitives. The core technique is new multi-point fuzzy matching protocols, upon which we derived protocols flexibly supporting both sender-sided and receiver-sided settings. By incorporating prefix-trie techniques, we further achieved $O(\log \delta)$ complexity in the distance threshold δ , improving upon the $O((\log \delta)^d)$ and $O(\delta)$ complexities of prior works. Extensive experiments confirm that our protocols significantly outperform all prior works under the same assumptions.

References

- [1] Navid Alamati, Pedro Branco, Nico Döttling, Sanjam Garg, Mohammad Hajiabadi, and Sihang Pu. 2021. Laconic private set intersection and applications. In *Theory of Cryptography Conference*. Springer, 94–125.
- [2] Navid Alamati, Guru-Vamsi Policharla, Srinivasan Raghuraman, and Peter Rindal. 2024. Improved alternating-moduli PRFs and post-quantum signatures. In *Annual International Cryptology Conference*. Springer, 274–308.
- [3] Elette Boyle, Geoffroy Couteau, Niv Gilboa, Yuval Ishai, Lisa Kohl, Peter Rindal, and Peter Scholl. 2019. Efficient two-round OT extension and silent non-interactive secure computation. In *Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security*. 291–308.
- [4] Elette Boyle, Niv Gilboa, and Yuval Ishai. 2015. Function secret sharing. In *Annual international conference on the theory and applications of cryptographic techniques*. Springer, 337–367.
- [5] Elette Boyle, Niv Gilboa, and Yuval Ishai. 2016. Function secret sharing: Improvements and extensions. In *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*. 1292–1303.
- [6] Dung Bui, Gayathri Garimella, Peihan Miao, and Phuoc Van Long Pham. 2025. New Framework for Structure-Aware PSI From Distributed Function Secret Sharing. In *International Conference on the Theory and Application of Cryptology and Information Security*. Springer, 294–326.
- [7] Anrin Chakraborti, Giulia Fanti, and Michael K Reiter. 2023. Distance-Aware private set intersection. In *32nd USENIX Security Symposium (USENIX Security 23)*. 319–336.
- [8] Hao Chen, Kim Laine, and Peter Rindal. 2017. Fast private set intersection from homomorphic encryption. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*. 1243–1255.
- [9] Wutichai Chongchitmate, Steve Lu, and Rafail Ostrovsky. 2024. Approximate psi with near-linear communication. *Cryptology ePrint Archive* (2024).
- [10] Chengrui Dang, Xv Zhou, and Bei Liang. 2025. Efficient Fuzzy PSI Based on Prefix Representation. In *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*. 2204–2218.
- [11] Changyu Dong, Liqun Chen, and Zikai Wen. 2013. When private set intersection meets big data: an efficient and scalable protocol. In *Proceedings of the 2013 ACM SIGSAC conference on Computer & communications security*. 789–800.
- [12] Michael J Freedman, Kobbi Nissim, and Benny Pinkas. 2004. Efficient private matching and set intersection. In *International conference on the theory and applications of cryptographic techniques*. Springer, 1–19.
- [13] Ying Gao, Lin Qi, Xiang Liu, Yuanhao Luo, and Longxin Wang. 2024. Efficient fuzzy private set intersection from fuzzy mapping. In *International Conference on the Theory and Application of Cryptology and Information Security*. Springer, 36–68.
- [14] Gayathri Garimella, Benjamin Goff, and Peihan Miao. 2024. Computation efficient structure-aware psi from incremental function secret sharing. In *Annual International Cryptology Conference*. Springer, 309–345.
- [15] Gayathri Garimella, Mike Rosulek, and Jaspal Singh. 2022. Structure-aware private set intersection, with applications to fuzzy matching. In *Annual International Cryptology Conference*. Springer, 323–352.
- [16] Meng Hao, Weiran Liu, Liqiang Peng, Hongwei Li, Cong Zhang, Hanxiao Chen, and Tianwei Zhang. 2024. Unbalanced Circuit-PSI from Oblivious Key-Value Retrieval. In *33rd USENIX Security Symposium (USENIX Security 24)*. 6435–6451.
- [17] Zhicong Huang, Wen-jie Lu, Cheng Hong, and Jiansheng Ding. 2022. Cheetah: Lean and fast secure Two-Party deep neural network inference. In *31st USENIX Security Symposium (USENIX Security 22)*. 809–826.
- [18] Mihaela Ion, Ben Kreuter, Erhan Nergiz, Sarvar Patel, Shobhit Saxena, Karn Seth, David Shanahan, and Moti Yung. 2017. Private intersection-sum protocol with applications to attributing aggregate ad conversions. *Cryptology ePrint Archive* (2017).
- [19] Daniel Kales, Christian Rechberger, Thomas Schneider, Matthias Senker, and Christian Weinert. 2019. Mobile private contact discovery at scale. In *28th USENIX Security Symposium (USENIX Security 19)*. 1447–1464.
- [20] Ágnes Kiss, Jian Liu, Thomas Schneider, N Asokan, and Benny Pinkas. 2017. Private set intersection for unequal set sizes with mobile applications. In *Privacy Enhancing Technologies Symposium*. De Gruyter, 177–197.
- [21] Vladimir Kolesnikov, Naor Matania, Benny Pinkas, Mike Rosulek, and Ni Trieu. 2017. Practical multi-party private set intersection from symmetric-key techniques. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*. 1257–1272.
- [22] Catherine Meadows. 1986. A more efficient cryptographic matchmaking protocol for use in the absence of a continuously available third party. In *1986 IEEE Symposium on Security and Privacy*. IEEE, 134–134.
- [23] M. Naor and O. Reingold. 1997. Number-theoretic constructions of efficient pseudo-random functions. In *Proceedings 38th Annual Symposium on Foundations of Computer Science*. 458–467. doi:10.1109/SFCS.1997.646134
- [24] Pascal Paillier. 1999. Public-key cryptosystems based on composite degree residuosity classes. In *International conference on the theory and applications of cryptographic techniques*. Springer, 223–238.
- [25] Benny Pinkas, Mike Rosulek, Ni Trieu, and Avishay Yanai. 2020. PSI from PaXoS: Fast, malicious private set intersection. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*. Springer, 739–767.
- [26] Benny Pinkas, Thomas Schneider, Gil Segev, and Michael Zohner. 2015. Phasing: Private set intersection using permutation-based hashing. In *24th USENIX Security Symposium (USENIX Security 15)*. 515–530.
- [27] Lucas Piske, Jaspal Singh, Ni Trieu, Vladimir Kolesnikov, and Vassilis Zikas. 2025. Distance-aware OT with application to fuzzy PSI. In *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*. 4679–4691.
- [28] Srinivasan Raghuraman and Peter Rindal. 2022. Blazing fast PSI from improved OKVS and subfield VOLE. In *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security*. 2505–2517.
- [29] Deevashwer Rathee, Mayank Rathee, Nishant Kumar, Nishanth Chandran, Divya Gupta, Aseem Rastogi, and Rahul Sharma. 2020. Cryptflow2: Practical 2-party secure inference. In *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*. 325–342.
- [30] David Richardson, Mike Rosulek, and Jiayu Xu. 2024. Fuzzy PSI via oblivious protocol routing. *Cryptology ePrint Archive* (2024).
- [31] Aron van Baarsen and Sihang Pu. 2024. Fuzzy private set intersection with large hyperballs. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*. Springer, 340–369.
- [32] Aron van Baarsen and Sihang Pu. 2025. Fuzzy private set intersection from VOLE. In *International Conference on the Theory and Application of Cryptology and Information Security*. Springer, 327–360.
- [33] Mingli Wu and Tsz Hon Yuen. 2023. Efficient unbalanced private set intersection cardinality and user-friendly privacy-preserving contact tracing. In *32nd USENIX Security Symposium (USENIX Security 23)*. 283–300.
- [34] Xinpeng Yang, Meng Hao, Chenkai Weng, Robert H Deng, Yonggang Wen, and Tianwei Zhang. 2026. Efficient fuzzy private set intersection from secret-shared OPRF. In *2026 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2442–2461.
- [35] Cong Zhang, Yu Chen, Yang Cao, Yujie Bai, Shuaishuai Li, Juntong Lin, Anyu Wang, and Xiaoyun Wang. 2025. Fast Fuzzy PSI from Symmetric-Key Techniques. *Cryptology ePrint Archive* (2025).

A Open Science

We follow the principles of the Open Science Policy. Our research artifacts, comprising the source code of the proposed protocols, benchmarking scripts, and comprehensive technical documentation, have been consolidated into an open-source repository, which is currently accessible for peer review at <https://anonymous.4open.science/r/FPSI-One-Sided-1325>. Upon acceptance and publication of the camera-ready version of this manuscript, we will formally release all associated artifacts to the scientific community, thereby ensuring unrestricted access for independent verification, reproducibility assessment, and subsequent extension of our research.

B Ethical Considerations

Stakeholder Identification and Impacts. We organize this section as a stakeholder-based analysis, distinguishing direct stakeholders (those who design, implement, or deploy our fuzzy PSI protocols) from indirect stakeholders (people and organizations affected by systems that rely on them). Direct stakeholders include

the research team and organizations that integrate our protocols into privacy-preserving biometric matching systems. Indirect stakeholders include end users whose biometric templates are compared through such systems, regulators responsible for oversight of data-sharing practices, and adversaries who may attempt to abuse or subvert these systems.

Research Team. Our work involves no human subjects, no physical experiments, and no collection of real-world private data. All evaluations were conducted on synthetic datasets generated specifically for benchmarking, which contain no personally identifiable information. We explicitly acknowledge and respect the intellectual contributions of each team member, ensuring that authorship reflects meaningful involvement.

Organizations Deploying Fuzzy PSI. Organizations that operate biometric matching services may adopt our protocols to improve efficiency and security. However, adoption may require non-trivial engineering effort for integration, incur audit costs for cryptographic review, and impose performance constraints in certain settings (e.g., high-dimensional data or large set sizes). We have documented quantitative performance benchmarks and known limitations to help deployers make informed decisions.

Cryptography Community and Society. *Beneficence.* We propose fuzzy PSI protocols that enable parties to identify approximate matches between their data without revealing the underlying records, aligning with data-protection principles *Respect for Persons*. We have carefully cited prior work to ensure accurate attribution and to avoid plagiarism. *Justice.* Our constructions rely only on standard cryptographic assumptions and open primitives, with no proprietary dependencies, keeping the core ideas accessible to researchers and practitioners worldwide. *Respect for Law and Public Interest.* Our research complies with all relevant academic and cryptographic research policies. By advancing the state of the art in fuzzy PSI, this work serves the public interest by enabling privacy-preserving data collaboration that supports compliance with data-protection regulations.

Publication Impacts. Our protocols may influence future research on fuzzy matching under additional distance metrics or security models. To reduce misuse or misinterpretation, we clearly state the security model, assumptions, and parameter settings under which the stated guarantees hold.

C Generative AI Usage

This paper used LLMs (including ChatGPT and Gemini) solely to assist with grammar checking and language polishing. The LLMs did not contribute any conceptual ideas, methodological innovations, experimental designs, or analytical insights to this work. All intellectual contributions originate from the authors. The content provided to the LLMs for linguistic refinement contains no sensitive, personal, or ethically problematic information.

D Other Related Work

van Baarsen and Pu [31] proposed the first general fuzzy PSI construction under the “*existing disjoint*” assumption, requiring that each point in both parties’ sets is at least 2δ apart from all other points in some dimension. The overall complexity scales as $O((d\delta)^2)$, with no additional exponential factor. Gao et al. [13] subsequently

improved this to $O(d\delta)$, achieving fully linear complexity for the first time. Their protocol, however, still operates under the same two-sided “*existing disjoint*” assumption, as the symmetric structure of their construction requires both parties’ inputs to satisfy the condition. Zhang et al. [35] adopted a stronger “*s-separate*” assumption and proposed an efficient fuzzy PSI protocol from symmetric-key primitives, at the cost of $O(\delta^s)$ overhead, which becomes prohibitive for large δ . Dang et al. [10] improved upon Gao et al. [13] under the same assumption via prefix trie techniques, reducing the overhead from $O(d\delta)$ to $O(d \log \delta)$. This was further improved by Yang et al. [34] via secret-shared OPRF [2], eliminating the reliance on public-key primitives. Recently, van Baarsen and Pu [32] revisited their earlier construction [31] using symmetric-key primitives. Their new protocol relies on spatial hashing to map each point to a unique cell, requiring both parties’ points to be at least 2δ or 4δ apart to prevent additional leakage. Chongchitmate et al. [9] proposed a nearly linear fuzzy PSI protocol, but their construction requires a strong gap assumption: every pair of points across the two sets must be either close or far apart. This is a strict assumption, as it imposes a joint constraint across both parties’ inputs.

While these works have achieved substantial efficiency improvements, their advantages rely on two-sided assumptions, which are harder to satisfy in real-world applications. Removing the assumption on either party’s input would render these protocols insecure, inapplicable, or subject to dramatically increased overhead.

Two additional constructions operate under the arbitrary distribution setting. The first [14] incurs $O(mn(\log \delta)^d)$ complexity, which is quadratic in the set sizes m and n . The second [32] achieves $O((m+n)(\log \delta)^d)$ complexity. However, both constructions suffer from the worse $O((\log \delta)^d)$ factor in both communication and computation, making them far from practical. Moreover, they support only L_∞ distance with no known efficient extension to general $L_{p \in [1, \infty]}$ distances; we therefore omit a detailed discussion.

E Other Functionalities

E.1 Interval Test

Chakraborti et al. [7] introduced an efficient interval test protocol, which runs in $O(\log \delta)$ complexity with respect to the interval size δ . We give the ideal functionality in Figure 14.

Functionality $\mathcal{F}_{\text{Interval}}^\delta$	
Parameters:	Sender $\mathcal{S}$ and receiver $\mathcal{R}$. Threshold δ .
Functionality:	(1) Wait for input $x^{\mathcal{S}}$ from $\mathcal{S}$ and $x^{\mathcal{R}}$ from $\mathcal{R}$. (2) Output 1 to $\mathcal{R}$ if $x^{\mathcal{S}} + x^{\mathcal{R}} \leq \delta$, otherwise output 0.

Figure 14: Functionality of private interval test.

E.2 MUX

The ideal functionality of MUX is given in Figure 15.

E.3 Secret-shared Private Equality Test

The ideal functionality of secret-shared private equality test is given in Figure 16.

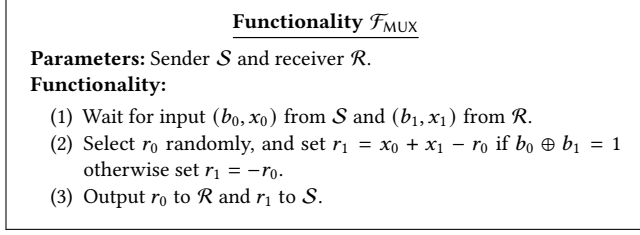


Figure 15: Functionality of MUX.

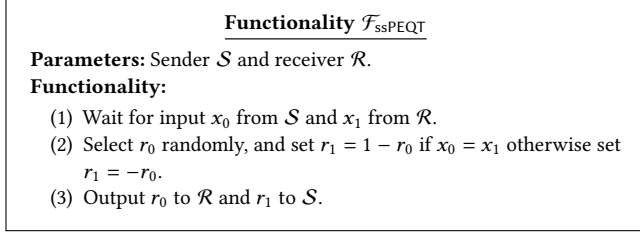


Figure 16: Functionality of secret-shared PEQT.

E.4 Secret-shared Comparison

The ideal functionality of secret-shared comparison is given in Figure 17.

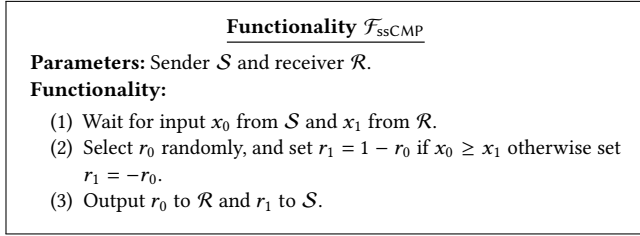


Figure 17: Functionality of secret-shared comparison.

E.5 Private Equality Test

The ideal functionality of private equality test is given in Figure 18.

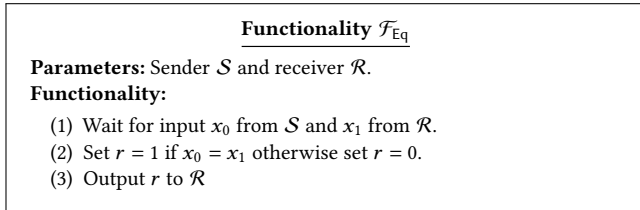


Figure 18: Functionality of private equality test.

F Comparisons of Assumptions

We compare the unique cell assumption of Definition 3.1 with existing assumptions as follows.

F.1 Unique Cell

F.1.1 Comparison with disjoint hash assumption. The disjoint hash assumption [27, 30] (abbreviated as disj. hash in Figure 2) requires the spatial hash to map each point to a distinct cell, which is identical to our *unique cell* assumption. However, while [27, 30] rely on this assumption in their theoretical constructions, their concrete protocols require additional assumptions on the other party.

F.1.2 Comparison with mini-universe assumption. We revisit the *mini-universe* assumption from [6, 14]. Since [6, 14] defines L_∞ balls of side length δ rather than 2δ , we double the radius parameter to align with our setting. Consider a d -dimensional space tiled by cells of side length 2δ . For any grid vertex $\mathbf{o} = (o_1, o_2, \dots, o_d)$, where coordinate o_i is an integer multiple of 2δ , we define the *mini-universe* with origin $\mathbf{o}$ as the d -dimensional hypercube of side length 4δ , namely $\text{Univ}_{\mathbf{o}} := [o_1, o_1 + 4\delta) \times \dots \times [o_d, o_d + 4\delta)$. For any point $\mathbf{x} \in \mathbb{U}^d$, we define $\text{Ball}_{\mathbf{x}}$ as the L_∞ ball of side length 2δ with lower-left corner at $\mathbf{x}$, namely $\text{Ball}_{\mathbf{x}} := [x_1, x_1 + 2\delta) \times \dots \times [x_d, x_d + 2\delta)$.

DEFINITION F.1 (MINI-UNIVERSE [6, 14]). A set $W \in \mathbb{U}^{n \times d}$ satisfies the *mini-universe* assumption if for every $\mathbf{x} \in W$, there exists a unique *mini-universe* $\text{Univ}_{\mathbf{o}}$ containing $\text{Ball}_{\mathbf{x}}$. In other words, for any two distinct points $\mathbf{x}, \mathbf{x}' \in W$ with $\text{Ball}_{\mathbf{x}} \subseteq \text{Univ}_{\mathbf{o}}$ and $\text{Ball}_{\mathbf{x}'} \subseteq \text{Univ}_{\mathbf{o}'}$, it holds that $\text{Univ}_{\mathbf{o}} \neq \text{Univ}_{\mathbf{o}'}$.

Below, we formally prove that the unique cell assumption is equivalent to the mini-universe assumption.

LEMMA F.1. A set $W \in \mathbb{U}^{n \times d}$ satisfies the *unique cell* assumption if and only if it satisfies the *mini-universe* assumption.

PROOF. We first show that the *unique cell* assumption implies the *mini-universe* assumption. Let $\mathbf{x} \in W$ lie in the grid cell with lower-left corner $\mathbf{o}$, i.e., $x_i \in [o_i, o_i + 2\delta)$ for all $i \in [d]$. Then $x_i + 2\delta \leq o_i + 4\delta$, so $[x_i, x_i + 2\delta) \subseteq [o_i, o_i + 4\delta)$ for all $i \in [d]$. Therefore, it holds that $\text{Ball}_{\mathbf{x}} \subseteq \text{Univ}_{\mathbf{o}}$, meaning every point in W has its ball $\text{Ball}_{\mathbf{x}}$ contained in the *mini-universe* $\text{Univ}_{\mathbf{o}}$ determined by its cell's lower-left corner $\mathbf{o}$. For any two distinct points $\mathbf{x}, \mathbf{x}' \in W$, the *unique cell* assumption gives $\text{cell}_{2\delta}(\mathbf{x}) \neq \text{cell}_{2\delta}(\mathbf{x}')$, so $\mathbf{o} \neq \mathbf{o}'$ and hence $\text{Univ}_{\mathbf{o}} \neq \text{Univ}_{\mathbf{o}'}$. Thus the *mini-universe* containing each ball is unique.

We next show that the *mini-universe* assumption implies the *unique cell* assumption. Let $\mathbf{x} \in W$ with $\text{Ball}_{\mathbf{x}} \subseteq \text{Univ}_{\mathbf{o}}$ for some corner $\mathbf{o}$. Since $\text{Ball}_{\mathbf{x}} = [x_1, x_1 + 2\delta) \times \dots \times [x_d, x_d + 2\delta)$ and $\text{Univ}_{\mathbf{o}} = [o_1, o_1 + 4\delta) \times \dots \times [o_d, o_d + 4\delta)$, the containment implies $o_i \leq x_i < o_i + 2\delta$ for all $i \in [d]$. Since each o_i is an integer multiple of 2δ , the region $[o_1, o_1 + 2\delta) \times \dots \times [o_d, o_d + 2\delta)$ is precisely the grid cell with lower-left corner $\mathbf{o}$, and $\mathbf{x}$ lies in this cell. For any two distinct points $\mathbf{x}, \mathbf{x}' \in W$ with $\text{Ball}_{\mathbf{x}} \subseteq \text{Univ}_{\mathbf{o}}$ and $\text{Ball}_{\mathbf{x}'} \subseteq \text{Univ}_{\mathbf{o}'}$, the *mini-universe* assumption gives $\text{Univ}_{\mathbf{o}} \neq \text{Univ}_{\mathbf{o}'}$, hence $\mathbf{o} \neq \mathbf{o}'$. By the above argument, $\mathbf{x}$ and $\mathbf{x}'$ lie in distinct cells, so $\text{cell}_{2\delta}(\mathbf{x}) \neq \text{cell}_{2\delta}(\mathbf{x}')$. $\square$

The difference between the two assumptions stems solely from how the L_∞ ball of a point is defined: our construction centers the ball at the point, whereas the sa-PSI line of work [6, 14] places the ball's lower-left corner at the point. Additionally, $\text{Ball}_{\mathbf{x}}$ spans $[x_i, x_i + 2\delta)$ in each dimension, a half-open interval of length 2δ , while the L_∞ ball in fuzzy PSI uses closed intervals of side length

exactly $2\delta + 1$. Despite these differences in convention, the two assumptions impose the same distributional constraint on the input set.

F.1.3 Comparison with 2δ -apart assumption. We recall the 2δ -apart assumption from [31] as follows.

DEFINITION F.2 (2δ -APART). A set $W \in \mathbb{U}^{n \times d}$ satisfies the 2δ -apart assumption if for any two distinct points $\mathbf{w}, \mathbf{w}' \in W$, $\text{dist}_p(\mathbf{w}, \mathbf{w}') \geq 2\delta d^{1/p}$. Specifically, if $p = \infty$, this reduces to $\text{dist}_\infty(\mathbf{w}, \mathbf{w}') \geq 2\delta$.

The following lemma from [31] illustrates that the 2δ -apart assumption is strictly stronger than the *unique cell* assumption. Intuitively, the *unique cell* assumption permits the L_∞ balls of input points to overlap as long as their centers fall into distinct cells, whereas the 2δ -apart assumption ensures that these L_∞ balls are pairwise disjoint.

LEMMA F.2 ([31]). Suppose there are multiple L_p balls ($p \in [1, \infty]$) with radius δ lying in a d -dimensional space which is tiled by cells with side length 2δ . If these balls' centers are at least $2\delta d^{1/p}$ apart, then for each cell, there is at most one center of the balls lying in this cell. Specifically, if $p = \infty$, then the *unique center* holds for disjoint balls since $2\delta d^{1/p}$ degrades to 2δ in this case.

F.1.4 Comparison with 2δ -disjoint projection assumption. The 2δ -disjoint projection assumption (abbreviated as 2δ -disj. proj. in Figure 2), adopted in [10, 13, 15, 31, 32, 34], requires that for every point $\mathbf{w} \in W$, there exists a dimension k such that $|w_k - w'_k| \geq 2\delta$ for all $\mathbf{w}' \in W \setminus \{\mathbf{w}\}$. It is strictly stronger than the *unique cell* assumption for general L_p distances, since $|w_k - w'_k| \geq 2\delta$ for some dimension k implies $\text{cell}_{2\delta}(\mathbf{w}) \neq \text{cell}_{2\delta}(\mathbf{w}')$.

To the best of our knowledge, all existing constructions relying on the 2δ -disjoint projection assumption require it to hold for both parties' input sets, and no known technique extends it to the one-sided setting.

F.2 Unique Block

We compare the unique block assumption of Definition 3.2 with existing assumptions as follows.

F.2.1 Comparison with *unique cell* assumption. We note that the *unique block* assumption is strictly stronger than the *unique cell* assumption, as it requires the L_∞ balls to have no common cells and therefore their centers fall into different cells.

F.2.2 Comparison with 4δ -apart assumption. We recall the 4δ -apart assumption from [31] as follows.

DEFINITION F.3 (4δ -APART). A set $W \in \mathbb{U}^{n \times d}$ satisfies the 4δ -apart assumption if for any two distinct points $\mathbf{w}, \mathbf{w}' \in W$, $\text{dist}_p(\mathbf{w}, \mathbf{w}') \geq 2\delta(d^{1/p} + 1)$. In particular, for $p = \infty$, this reduces to $\text{dist}_\infty(\mathbf{w}, \mathbf{w}') \geq 4\delta$.

The following lemma from [31] illustrates that the 4δ -apart assumption is strictly stronger than the *unique block* assumption.

LEMMA F.3 ([31]). Suppose there are multiple δ -radius L_p balls ($p \in [1, \infty]$) distributed in a d -dimensional space which is tiled by cells of side length 2δ . If these balls' centers are at least $2\delta(d^{1/p} + 1)$ apart from each other, then there exists at most one ball intersecting

with the same cell. Specifically, if $p = \infty$, this holds for L_∞ balls with 4δ -apart centers.

G Security Proof

G.1 Threat Model

Similar to prior works [6, 10, 13, 31, 32, 34], we consider static semi-honest probabilistic polynomial-time (PPT) adversaries. Namely, a PPT adversary $\mathcal{A}$ passively corrupts either the sender $\mathcal{S}$ or the receiver $\mathcal{R}$ at the beginning of the protocol and honestly follows the protocol specification. We use the standard simulation-based security definition for secure two-party computation. Our construction invokes multiple sub-protocols, and we use the *hybrid model* to describe them. By convention, a protocol invoking a functionality $\mathcal{F}$ is referred to as the $\mathcal{F}$ -hybrid model. We give the formal security definition as follows.

DEFINITION G.1. Let $\text{view}_S^\Pi(x, y)$ and $\text{view}_R^\Pi(x, y)$ be the views of $\mathcal{S}$ and $\mathcal{R}$ in a protocol Π , respectively, where x is the input of $\mathcal{S}$ and y is the input of $\mathcal{R}$. Let $\text{out}(x, y)$ be the protocol's output of both parties and $\mathcal{F}(x, y)$ be the functionality's output. Π is said to securely compute a functionality $\mathcal{F}$ in the semi-honest model if for every PPT adversary $\mathcal{A}$ there exists PPT simulators Sim_S and Sim_R such that for all inputs x and y ,

$$\{\text{view}_S^\Pi(x, y), \text{out}(x, y)\} \approx_c \{\text{Sim}_S(x, \mathcal{F}_S(x, y)), \mathcal{F}(x, y)\},$$

$$\{\text{view}_R^\Pi(x, y), \text{out}(x, y)\} \approx_c \{\text{Sim}_R(y, \mathcal{F}_R(x, y)), \mathcal{F}(x, y)\}.$$

G.2 L_∞ Distance under Sender-sided Unique Cell Assumptions

PROOF. Correctness. We first consider that for some $\mathbf{q}_j \in Q$, there exists $\mathbf{w}_i \in W$ such that $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$. By the correctness of spatial hashing, there exists an index $z^* \in [2^d]$ such that $C_{i,z^*}^\mathcal{R} = C_j^\mathcal{S}$. Since $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$, it holds that $|q_{j,k} - w_{i,k}| \leq \delta$ for all $k \in [d]$. Therefore, for each $k \in [d]$, there exists $t_k \in [-\delta, \delta]$ such that $q_{j,k} + t_k = w_{i,k}$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, $e_{i,k,z^*}^\mathcal{R} \| r_{i,k,z^*}^\mathcal{R}$ and $e_{i,k,z^*}^\mathcal{S} \| r_{i,k,z^*}^\mathcal{S}$ are the shares of $0^\ell \| s_{j,k}$. And then the aggregated shares $e_{i,z^*}^\mathcal{R}$ and $e_{i,z^*}^\mathcal{S}$ are the shares of 0^ℓ , $r_{i,z^*}^\mathcal{R}$ and $r_{i,z^*}^\mathcal{S}$ are the shares of $s_j = \sum_{k \in [d]} s_{j,k}$. Since $e_{i,z^*}^\mathcal{R} + e_{i,z^*}^\mathcal{S} = 0$, by the correctness of functionality $\mathcal{F}_{\text{ConRand}}^{\text{Eq}}$, it holds that $v_{i,z^*}^\mathcal{R} + v_{i,z^*}^\mathcal{S} = s_j$. Then, the $\mathcal{R}$ reconstructs $v_{i,z^*} = s_j$. After shuffling, there must be $x_{j^*} \in X$ and $x_{j^*} = H_{\lambda'+du}(s_j) \oplus (0^{\lambda'} \| \mathbf{q}_j)$. Finally, the $\mathcal{R}$ un.masks x_{j^*} by $H_{\lambda'+du}(v_{i,z^*}) \oplus x_{j^*}$ and correctly gets $\mathbf{q}_j$.

We then consider that for some $\mathbf{q}_j \in Q$, it holds $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) > \delta$ for any $\mathbf{w}_i \in W$. We consider the following two cases.

(1) The first case, there exists an index $z^* \in [2^d]$ such that $C_{i,z^*}^\mathcal{R} = C_j^\mathcal{S}$. Since $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) > \delta$, it holds that there exists an index $k^* \in [d]$ such that $|q_{j,k^*} - w_{i,k^*}| > \delta$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{i,k^*,z^*}^\mathcal{R} \| r_{i,k^*,z^*}^\mathcal{R}$ and $e_{i,k^*,z^*}^\mathcal{S} \| r_{i,k^*,z^*}^\mathcal{S}$ are uniformly random. Then, after aggregation, the shares $e_{i,z^*}^\mathcal{R} \| r_{i,z^*}^\mathcal{R}$ and $e_{i,z^*}^\mathcal{S} \| r_{i,z^*}^\mathcal{S}$ are also uniformly random. By the correctness of functionality $\mathcal{F}_{\text{ConRand}}^{\text{Eq}}$, the new shares $v_{i,z^*}^\mathcal{R}$ and $v_{i,z^*}^\mathcal{S}$ are uniformly random. After reconstruction, v_{i,z^*} is uniformly random.

(2) The second case, for any $z \in [2^d]$, it holds that $C_{i,z}^\mathcal{R} \neq C_j^\mathcal{S}$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{i,k,z}^\mathcal{R} \| r_{i,k,z}^\mathcal{R}$

and $e_{i,k,z}^S \| r_{i,k,z}^S$ are uniformly random for all $z \in [2^d]$, $k \in [d]$. Similarly, $v_{i,z}$ is uniformly random.

In both cases, $v_{i,z}$ is uniformly random for all $i \in [n]$, $z \in [2^d]$. By setting the bit length of s_j as $\kappa > \lambda + d + \log mn$, the probability of $v_{i,z} = s_j$ for some $j \in [m]$ is at most $2^{-\lambda}$, which is negligible. On the other hand, if $v_{i,z} \neq s_j$, the decryption $H_{\lambda'+du}(v_{i,z}) \oplus x_j$ is uniformly random for all $i \in [n]$, $j \in [m]$, $z \in [2^d]$. By setting $\lambda' = \lambda + d + \log mn$, the probability of valid decryption is at most $2^{-\lambda}$, which is negligible. A union bound shows that the receiver can obtain q_j with the probability at most $2^{-\lambda}(2 - 2^{-\lambda}) < 2^{-\lambda+1}$, which is negligible. This completes the correctness proof.

Corrupted sender. The simulator Sim_S receives the sender's input Q and proceeds as follows:

- (1) Sim_S maintains a table for queries of the random oracle H . For each query to H , Sim_S returns a consistent random value.
- (2) Sim_S emulates $\mathcal{F}_{\text{so-OPPRF}}$ by sampling uniformly random shares $e_{i,k,z}^S, r_{i,k,z}^S$ for $i \in [n]$, $k \in [d]$, $z \in [2^d]$, and sampling a random function F such that $F(C_j^S \| k \| q_{j,k} + t) = 0^\ell \| s_{j,k}$ for $j \in [m]$, $k \in [d]$, $t \in [-\delta, \delta]$, and sends $e_{i,k,z}^S, r_{i,k,z}^S$ and O^F to $\mathcal{A}$.
- (3) Sim_S emulates $\mathcal{F}_{\text{ConRand}}^{\text{Eq}}$ by sampling uniformly random shares $v_{i,z}^S$ for $i \in [n]$, $z \in [2^d]$, and sends them to $\mathcal{A}$.

Indistinguishability. The adversary's view in the real execution is identically distributed to its view in the simulated execution. This follows because the values $e_{i,k,z}^S, r_{i,k,z}^S$ received from $\mathcal{F}_{\text{so-OPPRF}}$ and $v_{i,z}^S$ from $\mathcal{F}_{\text{ConRand}}^{\text{Eq}}$ are uniformly random in the real world and are sampled from the same distribution by the simulator in the ideal world. Moreover, the random function oracle O^F subject to $F(C_j^S \| k \| q_{j,k} + t) = 0^\ell \| s_{j,k}$, both of which are independent of S 's actual input Q . Therefore, the view in the real-world execution is identically distributed to its view in the simulation.

Corrupted receiver. We show the simulator $\text{Sim}_R(W, I)$:

- (1) Sim_R maintains a table for queries of the random oracle H . For each query to H , Sim_R returns a consistent random value.
- (2) Sim_R emulates $\mathcal{F}_{\text{so-OPPRF}}$ by sampling uniformly random shares $e_{i,k,z}^R, r_{i,k,z}^R$ for $i \in [n]$, $k \in [d]$, $z \in [2^d]$, and sends $e_{i,k,z}^R, r_{i,k,z}^R$ to $\mathcal{A}$.
- (3) Sim_R emulates $\mathcal{F}_{\text{ConRand}}^{\text{Eq}}$ by sampling uniformly random shares $v_{i,z}^R$ for $i \in [n]$, $z \in [2^d]$, and sends them to $\mathcal{A}$.
- (4) For each $q_j \in I$, Sim_R samples uniformly random s_j and sets $v_{i,z}^S = s_j - v_{i,z}^R$ for all $i \in [n]$ and $z \in [2^d]$ such that $C_{i,z}^R = C_j^S$ and $\text{dist}(q_j, w_i) \leq \delta$. For the remaining $v_{i,z}^S$'s, Sim_R samples uniformly random values. Sim_R sends these $v_{i,z}^S$ to $\mathcal{A}$.
- (5) Sim_R initializes an empty set X . For each $q_j \in I$, where there exists w_i and $z \in [2^d]$ such that $C_{i,z}^R = C_j^S$ and $\text{dist}(q_j, w_i) \leq \delta$, it computes $x = H(s_j) \oplus (0^{\lambda'} \| q_j)$ and adds x to X . For the remaining $m - |I|$ points, Sim_R samples uniformly random strings of length $\lambda' + d \cdot u$ and adds them to X . Sim_R shuffles X and sends them to $\mathcal{A}$.

Indistinguishability. We prove security via a sequence of hybrids.

Hybrid H_0 . This is the real execution of the protocol.

Hybrid H_1 . This hybrid is identical to H_0 , except that Sim_R acts as $\mathcal{F}_{\text{so-OPPRF}}$ and $\mathcal{F}_{\text{ConRand}}^{\text{Eq}}$, but follows the specification of

these functionalities and uses the real sender's input Q . This is only a conceptual change, and hence the adversary's view in Hybrid H_0 and Hybrid H_1 are identically distributed.

Hybrid H_2 . This hybrid is identical to H_1 , except for the following modifications. For each $q_j \in I$, Sim_R sets $v_{i,z}^S = s_j - v_{i,z}^R$ for all $i \in [n]$ and $z \in [2^d]$ such that $C_{i,z}^R = C_j^S$ and $\text{dist}(q_j, w_i) \leq \delta$. For the remaining $v_{i,z}^S$'s, Sim_R samples uniformly random values. Besides, for each $q_j \in I$, it computes $x = H(s_j) \oplus (0^{\lambda'} \| q_j)$ and adds x to X . For the remaining $m - |I|$ points, Sim_R samples uniformly random strings of length $\lambda' + d \cdot u$ and adds them to X .

In H_1 , for each w_i and $z \in [2^d]$ such that there is no $q_j \in Q$ satisfying $C_{i,z}^R = C_j^S$ and $\text{dist}(q_j, w_i) \leq \delta$, there exists at least one k_i^* such that $e_{i,k_i^*,z}^S$ and $e_{i,k_i^*,z}^R$ are uniformly random by the correctness of so-OPPRF. This implies $e_{i,z}^S$ and $e_{i,z}^R$ are uniformly random. By setting the bit length as $\ell = \lambda + d + \log n$, a union bound shows that $e_{i,z}^S \neq -e_{i,z}^R$ with overwhelming probabilities. According to the correctness of $\mathcal{F}_{\text{ConRand}}^{\text{Eq}}$, $v_{i,z}^S$ for the above w_i and $z \in [2^d]$ is uniformly random and hence identical in both worlds. Besides, the above analysis implies that s_j for $q_j \notin I$ is hidden from $\mathcal{A}$. Moreover, due to $|s_j| = \kappa$, the probability that the adversary queries H on s_j is negligible. Then, for each $q_j \notin I$, $H(s_j) \oplus (0^{\lambda'} \| q_j)$ yields a string identically distributed to a uniform random string. Therefore, the adversary's view of Hybrid H_2 is computationally indistinguishable from that of Hybrid H_1 .

Hybrid H_3 . This hybrid is identical to H_2 , except that Sim_R simulates $\mathcal{F}_{\text{so-OPPRF}}$ and $\mathcal{F}_{\text{ConRand}}^{\text{Eq}}$ as follows. In particular, Sim_R samples uniformly random shares $e_{i,k,z}^R, r_{i,k,z}^R, v_{i,z}^R$ for $i \in [n]$, $k \in [d]$, $z \in [2^d]$, and sends them to $\mathcal{A}$.

Since they are uniformly random shares from the adversary's view, the adversary's view in Hybrid H_2 and Hybrid H_3 are identically distributed.

Observe that Hybrid H_3 is exactly the simulated execution produced by Sim_R . Therefore, the real and ideal executions are computationally indistinguishable, completing the proof. $\square$

G.3 L_p Distance under Sender-sided Unique Cell Assumptions

PROOF. Correctness. We first consider that for some $q_j \in Q$, there exists $w_i \in W$ such that $\text{dist}_p(q_j, w_i) \leq \delta$ and therefore $\text{dist}_\infty(q_j, w_i) \leq \delta$. By the correctness of spatial hashing, there exists an index $z^* \in [2^d]$ such that $C_{i,z^*}^R = C_j^S$. Since $\text{dist}_p(q_j, w_i) \leq \delta$, it holds that $|q_{j,k} - w_{i,k}| \leq \delta$ for all $k \in [d]$. Therefore, for each $k \in [d]$, there exists $t_k \in [-\delta, \delta]$ such that $C_j^S \| k \| q_{j,k} + t_k = w_{i,k}$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, $e_{i,k,z^*}^R \| r_{i,k,z^*}^R$ and $e_{i,k,z^*}^S \| r_{i,k,z^*}^S$ are the shares of $|t_k|^p \| s_{j,k}$. And by the correctness of functionality $\mathcal{F}_{\text{B2A}}$, d_{i,z^*}^R and d_{i,z^*}^S are the shares of $\sum_{k \in [d]} |t_k|^p = \sum_{k \in [d]} (|q_{j,k} - w_{i,k}|)^p$. Since $\sum_{k \in [d]} (|q_{j,k} - w_{i,k}|)^p \leq \delta^p$, by the correctness of functionality $\mathcal{F}_{\text{ConRand}}^{\text{Cmp}}$, it holds that $v_{i,z^*}^R + v_{i,z^*}^S = s_j$. Then, the $\mathcal{R}$ reconstructs $v_{i,z^*} = s_j$. After shuffling, there must be

$x_{j^*} \in X$ and $x_{j^*} = H_{\lambda'+du}(s_j) \oplus (0^{\lambda'} \| \mathbf{q}_j)$. Finally, the $\mathcal{R}$ un.masks x_{j^*} by $H_{\lambda'+du}(v_{i,z^*}) \oplus x_{j^*}$ and correctly gets $\mathbf{q}_j$.

We then consider that for some $\mathbf{q}_j \in Q$, it holds $\text{dist}_p(\mathbf{q}_j, \mathbf{w}_i) > \delta$ for any $\mathbf{w}_i \in W$. We consider the following three cases.

(1) The first case: for any $z \in [2^d]$, it holds that $C_{i,z}^{\mathcal{R}} \neq C_j^{\mathcal{S}}$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{i,k,z}^{\mathcal{R}} \| r_{i,k,z}^{\mathcal{R}}$ and $e_{i,k,z}^{\mathcal{S}} \| r_{i,k,z}^{\mathcal{S}}$ are uniformly random for all $z \in [2^d]$, $k \in [d]$. By the correctness of $\mathcal{F}_{\text{B2A}}$, after aggregation, the shares $d_{i,z}^{\mathcal{R}}$ and $d_{i,z}^{\mathcal{S}}$ are also uniformly random. By the correctness of $\mathcal{F}_{\text{ConRand}}^{\text{Cmp}}$, the new shares $v_{i,z}^{\mathcal{R}}$ and $v_{i,z}^{\mathcal{S}}$ are uniformly random. Therefore, after reconstruction, $v_{i,z}$ is uniformly random.

(2) The second case: there exists an index $z^* \in [2^d]$ such that $C_{i,z^*}^{\mathcal{R}} = C_j^{\mathcal{S}}$ but $\text{dist}_{\infty}(\mathbf{q}_j, \mathbf{w}_i) > \delta$. Then there exists an index $k^* \in [d]$ such that $|q_{j,k^*} - w_{i,k^*}| > \delta$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{i,k^*,z^*}^{\mathcal{R}} \| r_{i,k^*,z^*}^{\mathcal{R}}$ and $e_{i,k^*,z^*}^{\mathcal{S}} \| r_{i,k^*,z^*}^{\mathcal{S}}$ are uniformly random. By the correctness of $\mathcal{F}_{\text{B2A}}$, after aggregation, the shares $d_{i,z^*}^{\mathcal{R}}$ and $d_{i,z^*}^{\mathcal{S}}$ are also uniformly random. Similarly, v_{i,z^*} is uniformly random.

(3) The third case: there exists an index $z^* \in [2^d]$ such that $C_{i,z^*}^{\mathcal{R}} = C_j^{\mathcal{S}}$ and $\text{dist}_{\infty}(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$ and $\mathcal{F}_{\text{B2A}}$, it holds that $d_{i,z^*}^{\mathcal{R}} + d_{i,z^*}^{\mathcal{S}} = \sum_{k \in [d]} (|w_{i,k} - q_{j,k}|)^p$. Since $\text{dist}_p(\mathbf{q}_j, \mathbf{w}_i) > \delta$, therefore $d_{i,z^*}^{\mathcal{R}} + d_{i,z^*}^{\mathcal{S}} > \delta^p$. By the correctness of functionality $\mathcal{F}_{\text{ConRand}}^{\text{Cmp}}$, the outputs $v_{i,z^*}^{\mathcal{R}}$ and $v_{i,z^*}^{\mathcal{S}}$ are uniformly random. After reconstruction, v_{i,z^*} is also uniformly random.

In all cases, $v_{i,z}$ is uniformly random for all $i \in [n]$, $z \in [2^d]$. By setting the bit length of s_j as $\kappa > \lambda + d + \log nm$, the probability of $v_{i,z} = s_j$ for some $j \in [m]$ is at most $2^{-\lambda}$, which is negligible. On the other hand, if $v_{i,z} \neq s_j$, the decryption $H_{\lambda'+du}(v_{i,z}) \oplus x_j$ is uniformly random for all $i \in [n]$, $j \in [m]$, $z \in [2^d]$. By setting $\lambda' = \lambda + d + \log mn$, the probability of valid decryption is at most $2^{-\lambda}$, which is negligible. A union bound shows that the receiver can obtain valid $\mathbf{q}_j$ with the probability at most $2^{-\lambda}(2 - 2^{-\lambda}) < 2^{-\lambda+1}$, which is negligible.

This completes the correctness proof. $\square$

G.4 L_{∞} Distance under Receiver-sided Unique Cell Assumptions

PROOF. Correctness. We first consider that for some $\mathbf{q}_j \in Q$, there exists $\mathbf{w}_i \in W$ such that $\text{dist}_{\infty}(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$. By the correctness of spatial hashing, there exists an index $z^* \in [2^d]$ such that $C_{j,z^*}^{\mathcal{S}} = C_i^{\mathcal{R}}$. Since $\text{dist}_{\infty}(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$, it holds that $|q_{j,k} - w_{i,k}| \leq \delta$ for all $k \in [d]$. Therefore, for each $k \in [d]$, there exists $t_k \in [-\delta, \delta]$ such that $C_j^{\mathcal{S}} \| k \| q_{j,k} + t_k = w_{i,k}$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, $e_{j,k,z^*}^{\mathcal{R}}$ and $e_{j,k,z^*}^{\mathcal{S}}$ are the shares of 0^{ℓ} . And then the aggregated shares $r_{j,z^*}^{\mathcal{R}}$ and $r_{j,z^*}^{\mathcal{S}}$ are still the shares of 0^{ℓ} . By the correctness of $\mathcal{F}_{\text{Eq}}$, it holds that $b_{j,z^*} = 1$ and therefore $b_j = 1$. By the correctness of $\mathcal{F}_{\text{OT}}$, $\mathcal{R}$ correctly gets $\mathbf{q}_j$.

We then consider that for some $\mathbf{q}_j \in Q$, it holds $\text{dist}_{\infty}(\mathbf{q}_j, \mathbf{w}_i) > \delta$ for any $\mathbf{w}_i \in W$. We consider the following two cases.

(1) The first case: there exists an index z^* such that $C_{j,z^*}^{\mathcal{S}} = C_i^{\mathcal{R}}$. Since $\text{dist}_{\infty}(\mathbf{q}_j, \mathbf{w}_i) > \delta$, it holds that there exists an index $k^* \in [d]$

such that $|q_{j,k^*} - w_{i,k^*}| > \delta$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{j,k^*,z^*}^{\mathcal{R}}$ and $e_{j,k^*,z^*}^{\mathcal{S}}$ are uniformly random. Then, after aggregation, $r_{j,z^*}^{\mathcal{R}}$ and $r_{j,z^*}^{\mathcal{S}}$ are also uniformly random.

(2) The second case: for any $z \in [2^d]$, it holds that $C_{j,z}^{\mathcal{S}} \neq C_i^{\mathcal{R}}$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{j,k,z}^{\mathcal{R}}$ and $e_{j,k,z}^{\mathcal{S}}$ are uniformly random. Then, $r_{j,z}^{\mathcal{R}}$ and $r_{j,z}^{\mathcal{S}}$ are also uniformly random.

In both cases, $r_{j,z}^{\mathcal{R}}$ and $r_{j,z}^{\mathcal{S}}$ are uniformly random for all $j \in [m]$, $z \in [2^d]$. By the correctness of functionality $\mathcal{F}_{\text{Eq}}$, the probability of $b_{j,z} = 1$ is at most $1/|\mathbb{F}|$. A union bound shows that the probability of there existing $j \in [m]$ such that $b_j = 1$ is at most $m2^d/|\mathbb{F}|$. By setting $\log|\mathbb{F}| = \ell = \lambda + d + \log m$, the probability $m2^d/|\mathbb{F}| = 2^{-\lambda}$ is negligible. By the correctness of $\mathcal{F}_{\text{OT}}$, $\mathcal{R}$ learns nothing about $\mathbf{q}_j$. This completes the correctness proof.

Corrupted sender. The simulator $\text{Sim}_{\mathcal{S}}$ receives the sender's input Q and proceeds as follows:

- (1) $\text{Sim}_{\mathcal{S}}$ emulates $\mathcal{F}_{\text{so-OPPRF}}$ by sampling uniformly random shares $e_{j,k,z}^{\mathcal{S}}$ for all $j \in [m]$, $k \in [d]$, and $z \in [2^d]$, and sends them to $\mathcal{A}$.
- (2) $\text{Sim}_{\mathcal{S}}$ emulates $\mathcal{F}_{\text{Eq}}$ and $\mathcal{F}_{\text{OT}}$, both of which output nothing to $\mathcal{A}$.

Indistinguishability. The adversary's view in the real execution is identically distributed to its view in the simulated execution. This follows because the values $e_{j,k,z}^{\mathcal{S}}$ received from $\mathcal{F}_{\text{so-OPPRF}}$ are uniformly random in the real world and are sampled from the same distribution by the simulator in the ideal world.

Corrupted receiver. The simulator $\text{Sim}_{\mathcal{R}}$ receives the receiver's input W and output I , and proceeds as follows:

- (1) $\text{Sim}_{\mathcal{R}}$ emulates $\mathcal{F}_{\text{so-OPPRF}}$ by sampling uniformly random shares $e_{j,k,z}^{\mathcal{R}}$ for all $j \in [m]$, $k \in [d]$, and $z \in [2^d]$, and by sampling a random function F such that $F(C_i^{\mathcal{R}} \| k \| w_{i,k} + t) = 0$ for all $i \in [n]$, $k \in [d]$, and $t \in [-\delta, \delta]$. It then sends these shares $e_{j,k,z}^{\mathcal{R}}$ and O^F to $\mathcal{A}$.
- (2) $\text{Sim}_{\mathcal{R}}$ emulates $\mathcal{F}_{\text{Eq}}$ by padding I with $\perp$ to size m and randomly permuting the padded I . For each $j \in [m]$ and $z \in [2^d]$, it sets

$$b_{j,z} = \begin{cases} 1, & \text{if } \mathbf{q}_j = I[j] \neq \perp \text{ and there exists } \mathbf{w}_i \text{ such that} \\ & C_i^{\mathcal{R}} = C_{j,z}^{\mathcal{S}} \text{ and } \text{dist}(\mathbf{q}_j, \mathbf{w}_i) \leq \delta, \\ 0, & \text{otherwise.} \end{cases}$$

$\text{Sim}_{\mathcal{R}}$ sends all values $b_{j,z}$ to $\mathcal{A}$.

- (3) $\text{Sim}_{\mathcal{R}}$ emulates $\mathcal{F}_{\text{OT}}$ and sends the padded and permuted I to $\mathcal{A}$.

Indistinguishability. We prove security via a sequence of hybrids.

Hybrid H_0 . This is the real execution of the protocol.

Hybrid H_1 . This hybrid is identical to H_0 , except that $\text{Sim}_{\mathcal{R}}$ acts as $\mathcal{F}_{\text{so-OPPRF}}$, $\mathcal{F}_{\text{Eq}}$, and $\mathcal{F}_{\text{OT}}$, but follows the specification of these functionalities and uses the real sender's input Q . This is only a conceptual change, and hence the adversary's view in Hybrid H_0 and Hybrid H_1 are identically distributed.

Hybrid H_2 . This hybrid is identical to H_1 , except that instead of computing $b_{j,z}$'s (resp. u_j 's) according to $\mathcal{F}_{\text{Eq}}$ (resp. $\mathcal{F}_{\text{OT}}$), $\text{Sim}_{\mathcal{R}}$ derives them utilizing the receiver's input W and output I . Specifically, it pads I with $\perp$ to size m , applies a random

permutation, and for every $j \in [m], z \in [2^d]$ sets

$$b_{j,z} = \begin{cases} 1, & \text{if } \mathbf{q}_j = I[j] \neq \perp \text{ and there exists } \mathbf{w}_i \text{ such that} \\ & C_i^R = C_{j,z}^S \text{ and } \text{dist}(\mathbf{q}_j, \mathbf{w}_i) \leq \delta, \\ 0, & \text{otherwise.} \end{cases}$$

The values $b_{j,z}$'s (resp. $\mathbf{u}_j$'s) are returned to $\mathcal{A}$ as the outputs of $\mathcal{F}_{\text{Eq}}$ (resp. $\mathcal{F}_{\text{OT}}$).

The adversary's views in Hybrid H_1 and Hybrid H_2 are identically distributed, because functionality Eq outputs exactly these bits $b_{j,z}$ and the outputs of functionality OT are exactly determined by $b_{j,z}$ and I .

Hybrid H_3 . This hybrid is identical to H_2 , except that we replace the sender's interaction in $\mathcal{F}_{\text{so-OPPRF}}$ with a simulation as follows: sample random shares $e_{j,k,z}^R$ for all $j \in [m], k \in [d], z \in [2^d]$, and sample a random function F conditioned on $F(C_i^R \| k \| w_{i,k} + t) = 0$ for all $i \in [n], k \in [d]$, and $t \in [-\delta, \delta]$. The simulator sends $e_{j,k,z}^R$ and O^F to $\mathcal{A}$.

By the security of $\mathcal{F}_{\text{so-OPPRF}}$, the adversarial views of Hybrid H_2 and Hybrid H_3 are identically distributed.

Observe that Hybrid H_3 is exactly the simulated execution produced by $\text{Sim}_{\mathcal{R}}$. Therefore, the real and ideal executions are perfectly indistinguishable, completing the proof. $\square$

G.5 L_p Distance under Receiver-sided Unique Cell Assumptions

PROOF. Correctness. We first consider that for some $\mathbf{q}_j \in Q$, there exists $\mathbf{w}_i \in W$ such that $\text{dist}_p(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$. By the correctness of spatial hashing, there exists an index $z^* \in [2^d]$ such that $C_{j,z^*}^S = C_i^R$. Since $\text{dist}_p(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$, it holds that $|q_{j,k} - w_{i,k}| \leq \delta$ for all $k \in [d]$. Therefore, for each $k \in [d]$, there exists $t_k \in [-\delta, \delta]$ such that $q_{j,k} + t_k = w_{i,k}$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, e_{j,k,z^*}^R and e_{j,k,z^*}^S are the shares of $|t_k|^p$. By the correctness of $\mathcal{F}_{\text{B2A}}$, $d_{j,k,z^*}^R + d_{j,k,z^*}^S = |t_k|^p$. Then, after aggregation, $r_{j,z^*}^R + r_{j,z^*}^S = \sum_{k \in [d]} |t_k|^p = \sum_{k \in [d]} |q_{j,k} - w_{i,k}|^p$. Since $\text{dist}_p(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$, by the correctness of $\mathcal{F}_{\text{Interval}}^{\delta p}$, it holds that $b_{j,z^*} = 1$ and therefore $b_j = 1$. By the correctness of $\mathcal{F}_{\text{OT}}$, $\mathcal{R}$ correctly gets $\mathbf{q}_j$.

We then consider that for some $\mathbf{q}_j \in Q$, it holds $\text{dist}_p(\mathbf{q}_j, \mathbf{w}_i) > \delta$ for any $\mathbf{w}_i \in W$. We consider the following three cases.

(1) The first case, for any $z \in [2^d]$, it holds that $C_{j,z}^S \neq C_i^R$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{j,k,z}^R$ and $e_{j,k,z}^S$ are uniformly random for all $k \in [d], z \in [2^d]$. By the correctness of $\mathcal{F}_{\text{B2A}}$, the shares $d_{j,k,z}^R$ and $d_{j,k,z}^S$ are still uniformly random. Then, after aggregation, $r_{j,z}^R$ and $r_{j,z}^S$ are also uniformly random.

(2) The second case, there exists an index z^* such that $C_{j,z^*}^S = C_i^R$ but $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) > \delta$. Therefore, it holds that there exists an index $k^* \in [d]$ such that $|q_{j,k^*} - w_{i,k^*}| > \delta$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs e_{j,k^*,z^*}^R and e_{j,k^*,z^*}^S are uniformly random. By the correctness of functionality $\mathcal{F}_{\text{B2A}}$, d_{j,k^*,z^*}^R and d_{j,k^*,z^*}^S are also uniformly random. Then, after aggregation, r_{j,z^*}^R and r_{j,z^*}^S are also uniformly random.

In the first two cases, $r_{j,z}^R$ and $r_{j,z}^S$ are uniformly random for all $j \in [m], z \in [2^d]$. By the correctness of functionality $\mathcal{F}_{\text{Interval}}^{\delta p}$, the probability of $b_{j,z} = 1$ is at most $\delta^p / |\mathbb{F}|$. A union bound shows that the probability of there existing $j \in [m]$ such that $b_j = 1$ is at most $m 2^d \delta^p / |\mathbb{F}|$. By setting $\log |\mathbb{F}| = \ell = \lambda + d + \log m + p \log \delta$, the probability $m 2^d \delta^p / |\mathbb{F}| = 2^{-\lambda}$ is negligible. By the correctness of $\mathcal{F}_{\text{OT}}$, $\mathcal{R}$ learns nothing about $\mathbf{q}_j$.

(3) The third case, there exists an index z^* such that $C_{j,z^*}^S = C_i^R$ and $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$ and $\mathcal{F}_{\text{B2A}}$, $r_{j,z^*}^R + r_{j,z^*}^S = \text{dist}_p(\mathbf{q}_j, \mathbf{w}_i)^p > \delta^p$. By the correctness of functionality $\mathcal{F}_{\text{Interval}}^{\delta p}$ and $\mathcal{F}_{\text{OT}}$, $\mathcal{R}$ learns nothing about $\mathbf{q}_j$.

This completes the correctness proof. $\square$

G.6 L_∞ Distance under Receiver-sided Unique Cell Assumptions with Prefix Optimization

PROOF. Correctness. We first consider that for some $\mathbf{q}_j \in Q$, there exists $\mathbf{w}_i \in W$ such that $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$. By the correctness of spatial hashing, there exists an index $z^* \in [2^d]$ such that $C_{j,z^*}^S = C_i^R$. Since $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$, it holds that $|q_{j,k} - w_{i,k}| \leq \delta$ for all $k \in [d]$.

Therefore, for each $k \in [d]$, $q_{j,k} \in [w_{i,k} - \delta, w_{i,k} + \delta]$. By Lemma 3.1, there exists exactly one index $h^* \in [\mu]$ such that $\tilde{q}_{j,k,h^*} \in \{\tilde{w}_{i,k,h}\}_{h \in [\mu]}$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, e_{j,k,z^*,h^*}^R and e_{j,k,z^*,h^*}^S are the shares of 0^ℓ . By the correctness of functionality $\mathcal{F}_{\text{ConSel}}^{\text{Eq}, \mu}$, $v_{j,k,z^*}^R + v_{j,k,z^*}^S = e_{j,k,z^*,h^*}^R + e_{j,k,z^*,h^*}^S = 0$. Then, after aggregation, $r_{j,z^*}^R + r_{j,z^*}^S = 0$. By the correctness of functionality $\mathcal{F}_{\text{Eq}}$, $b_{j,z^*} = 1$ and therefore $b_j = 1$. By the correctness of functionality $\mathcal{F}_{\text{OT}}$, $\mathcal{R}$ correctly obtains $\mathbf{q}_j$.

We then consider that for some $\mathbf{q}_j \in Q$, it holds $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) > \delta$ for any $\mathbf{w}_i \in W$. We consider the following two cases.

(1) The first case, there exists an index z^* such that $C_{j,z^*}^S = C_i^R$. Since $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) > \delta$, it holds that there exists an index $k^* \in [d]$ such that $|q_{j,k^*} - w_{i,k^*}| > \delta$, therefore $q_{j,k^*} \notin [w_{i,k^*} - \delta, w_{i,k^*} + \delta]$. By Lemma 3.1, it holds that $\{\tilde{q}_{j,k,h}\}_{h \in [\mu]} \cap \{\tilde{w}_{i,k,h}\}_{h \in [\mu]} = \emptyset$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{j,k^*,z^*,h}^R$ and $e_{j,k^*,z^*,h}^S$ are uniformly random for all $h \in [\mu]$. Then, by the correctness of functionality $\mathcal{F}_{\text{ConSel}}^{\text{Eq}, \mu}$, v_{j,k^*,z^*}^R and v_{j,k^*,z^*}^S are uniformly random. After aggregation, r_{j,z^*}^R and r_{j,z^*}^S are also uniformly random.

(2) The second case, for any $z \in [2^d]$, it holds that $C_{j,z}^S \neq C_i^R$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{j,k,z}^R$ and $e_{j,k,z}^S$ are uniformly random. Then, $r_{j,z}^R$ and $r_{j,z}^S$ are also uniformly random.

In both cases, $r_{j,z}^R$ and $r_{j,z}^S$ are uniformly random for all $j \in [m], z \in [2^d]$. By the correctness of functionality $\mathcal{F}_{\text{Eq}}$, the probability of $b_{j,z} = 1$ is at most $1/|\mathbb{F}|$. A union bound shows that the probability of there existing $j \in [m]$ such that $b_j = 1$ is at most $m 2^d / |\mathbb{F}|$. By setting $\log |\mathbb{F}| = \ell = \lambda + d + \log m$, the probability $m 2^d / |\mathbb{F}| = 2^{-\lambda}$ is negligible. By the correctness of $\mathcal{F}_{\text{OT}}$, $\mathcal{R}$ learns nothing about $\mathbf{q}_j$. This completes the correctness proof.

Corrupted sender. The simulator $\text{Sim}_{\mathcal{S}}$ receives the sender's input Q and proceeds as follows:

- (1) Sim_S emulates $\mathcal{F}_{\text{so-OPPRF}}$ by sampling uniformly random shares $e_{j,k,z,h}^S$ for all $j \in [m]$, $k \in [d]$, $z \in [2^d]$, and $h \in [\mu]$ and sends them to $\mathcal{A}$.
- (2) Sim_S emulates $\mathcal{F}_{\text{ConSel}}^{\text{Eq},\mu}$ by sampling uniformly random shares $v_{j,k,z}^S$ for all $j \in [m]$, $k \in [d]$, and $z \in [2^d]$, and sends them to $\mathcal{A}$.
- (3) Sim_S emulates $\mathcal{F}_{\text{Eq}}$ and $\mathcal{F}_{\text{OT}}$, both of which output nothing to $\mathcal{A}$.

Indistinguishability. The adversary's view in the real execution is identically distributed to its view in the simulated execution. This follows because the values $e_{j,k,z,h}^S$ received from $\mathcal{F}_{\text{so-OPPRF}}$ and $v_{j,k,z}^S$ from $\mathcal{F}_{\text{ConSel}}^{\text{Eq},\mu}$ are uniformly random in the real world and are sampled from the same distribution by the simulator in the ideal world.

Corrupted receiver. The simulator Sim_R receives the receiver's input W and output I , and proceeds as follows:

- (1) Sim_R emulates $\mathcal{F}_{\text{so-OPPRF}}$ by sampling uniformly random shares $e_{j,k,z,h}^R$ for all $j \in [m]$, $k \in [d]$, $z \in [2^d]$, $h \in [\mu]$, and by sampling a random function F such that $F(C_i^R \| k \| \tilde{w}_{i,k,h}) = 0$ for all $i \in [n]$, $k \in [d]$, and $h \in [\mu]$. It then sends these shares $e_{j,k,z,h}^R$ and O^F to $\mathcal{A}$.
- (2) Sim_R emulates $\mathcal{F}_{\text{ConSel}}^{\text{Eq},\mu}$ by sampling uniformly random shares $v_{j,k,z}^R$ for all $j \in [m]$, $k \in [d]$, and $z \in [2^d]$, and sends them to $\mathcal{A}$.
- (3) Sim_R emulates $\mathcal{F}_{\text{Eq}}$ by padding I with $\perp$ to size m and randomly permuting the padded I . For each $j \in [m]$ and $z \in [2^d]$, it sets

$$b_{j,z} = \begin{cases} 1, & \text{if } \mathbf{q}_j = I[j] \neq \perp \text{ and there exists } \mathbf{w}_i \text{ such that} \\ & C_i^R = C_{j,z}^S \text{ and } \text{dist}(\mathbf{q}_j, \mathbf{w}_i) \leq \delta, \\ 0, & \text{otherwise.} \end{cases}$$

Sim_R sends all values $b_{j,z}$ to $\mathcal{A}$.

- (4) Sim_R emulates $\mathcal{F}_{\text{OT}}$ and sends the padded and permuted I to $\mathcal{A}$.

Indistinguishability. We prove security via a sequence of hybrids.

Hybrid H_0 . This is the real execution of the protocol.

Hybrid H_1 . This hybrid is identical to H_0 , except that Sim_R acts as $\mathcal{F}_{\text{so-OPPRF}}$, $\mathcal{F}_{\text{ConSel}}^{\text{Eq},\mu}$, $\mathcal{F}_{\text{Eq}}$, and $\mathcal{F}_{\text{OT}}$, but follows the specification of these functionalities and uses the real sender's input Q . This is only a conceptual change, and hence the adversary's view in Hybrid H_0 and Hybrid H_1 are identically distributed.

Hybrid H_2 . This hybrid is identical to H_1 , except that instead of computing $b_{j,z}$'s (resp. $\mathbf{u}_j$'s) according to $\mathcal{F}_{\text{Eq}}$ (resp. $\mathcal{F}_{\text{OT}}$), Sim_R derives them utilizing the receiver's input W and output I . Specifically, it pads I with $\perp$ to size m , applies a random permutation, and for every $j \in [m]$, $z \in [2^d]$ sets

$$b_{j,z} = \begin{cases} 1, & \text{if } \mathbf{q}_j = I[j] \neq \perp \text{ and there exists } \mathbf{w}_i \text{ such that} \\ & C_i^R = C_{j,z}^S \text{ and } \text{dist}(\mathbf{q}_j, \mathbf{w}_i) \leq \delta, \\ 0, & \text{otherwise.} \end{cases}$$

The values $b_{j,z}$'s (resp. $\mathbf{u}_j$'s) are returned to $\mathcal{A}$ as the outputs of $\mathcal{F}_{\text{Eq}}$ (resp. $\mathcal{F}_{\text{OT}}$).

The adversary's views in Hybrid H_1 and Hybrid H_2 are identically distributed, because functionality Eq outputs exactly

these bits $b_{j,z}$ and the outputs of functionality OT are exactly determined by $b_{j,z}$ and I .

Hybrid H_3 . This hybrid is identical to H_2 , except that we replace the sender's interaction in $\mathcal{F}_{\text{so-OPPRF}}$ and $\mathcal{F}_{\text{ConSel}}^{\text{Eq},\mu}$ with a simulation as follows. Sim_R emulates $\mathcal{F}_{\text{so-OPPRF}}$ by sampling uniformly random shares $e_{j,k,z,h}^R$ for all $j \in [m]$, $k \in [d]$, $z \in [2^d]$, $h \in [\mu]$, and by sampling a random function F such that $F(C_i^R \| k \| \tilde{w}_{i,k,h}) = 0$ for all $i \in [n]$, $k \in [d]$, and $h \in [\mu]$. It then sends these shares $e_{j,k,z,h}^R$ and O^F to $\mathcal{A}$. Then, Sim_R emulates $\mathcal{F}_{\text{ConSel}}^{\text{Eq},\mu}$ by sampling uniformly random shares $v_{j,k,z}^R$ for all $j \in [m]$, $k \in [d]$, and $z \in [2^d]$, and sends them to $\mathcal{A}$. By the security of $\mathcal{F}_{\text{so-OPPRF}}$ and $\mathcal{F}_{\text{ConSel}}^{\text{Eq},\mu}$, the adversarial views of Hybrid H_2 and Hybrid H_3 are identically distributed.

Observe that Hybrid H_3 is exactly the simulated execution produced by Sim_R . Therefore, the real and ideal executions are perfectly indistinguishable, completing the proof. $\square$

G.7 L_∞ Distance under Receiver-sided Unique Block Assumptions

PROOF. Correctness. We first consider that for some $\mathbf{q}_j \in Q$, there exists $\mathbf{w}_i \in W$ such that $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$. By the correctness of spatial hashing, there exists an index $z^* \in [2^d]$ such that $C_j^S = C_{i,z^*}^R$. Since $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) \leq \delta$, it holds that $|q_{j,k} - w_{i,k}| \leq \delta$ for all $k \in [d]$. Therefore, for each $k \in [d]$, there exists $t_k \in [-\delta, \delta]$ such that $q_{j,k} + t_k = w_{i,k}$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, $e_{j,k}^R$ and $e_{j,k}^S$ are the shares of 0^ℓ . And then the aggregated shares r_j^R and r_j^S are still the shares of 0^ℓ . By the correctness of $\mathcal{F}_{\text{Eq}}$, it holds that $b_j = 1$. By the correctness of $\mathcal{F}_{\text{OT}}$, $\mathcal{R}$ correctly obtains $\mathbf{q}_j$.

We then consider that for some $\mathbf{q}_j \in Q$, it holds $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) > \delta$ for any $\mathbf{w}_i \in W$. We consider the following two cases.

(1) The first case: there exists an index z^* such that $C_j^S = C_{i,z^*}^R$. Since $\text{dist}_\infty(\mathbf{q}_j, \mathbf{w}_i) > \delta$, it holds that there exists an index $k^* \in [d]$ such that $|q_{j,k^*} - w_{i,k^*}| > \delta$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs e_{j,k^*}^R and e_{j,k^*}^S are uniformly random. Then, after aggregation, r_j^R and r_j^S are also uniformly random.

(2) The second case: for any $z \in [2^d]$, it holds that $C_j^S \neq C_{i,z}^R$. By the correctness of functionality $\mathcal{F}_{\text{so-OPPRF}}$, the outputs $e_{j,k}^R$ and $e_{j,k}^S$ are uniformly random. Then, r_j^R and r_j^S are also uniformly random.

In both cases, r_j^R and r_j^S are uniformly random for all $j \in [m]$. By the correctness of functionality $\mathcal{F}_{\text{Eq}}$, the probability of $b_j = 1$ is at most $1/|\mathbb{F}|$. A union bound shows that the probability of there existing $j \in [m]$ such that $b_j = 1$ is at most $m/|\mathbb{F}|$. By setting $\log|\mathbb{F}| = \ell = \lambda + \log m$, the probability $m/|\mathbb{F}| = 2^{-\lambda}$ is negligible. By the correctness of $\mathcal{F}_{\text{OT}}$, $\mathcal{R}$ learns nothing about $\mathbf{q}_j$. This completes the correctness proof.

Corrupted sender. The simulator Sim_S receives the sender's input Q and proceeds as follows:

- (1) Sim_S emulates $\mathcal{F}_{\text{so-OPPRF}}$ by sampling uniformly random shares $e_{j,k}^S$ for all $j \in [m]$ and $k \in [d]$, and sends them to $\mathcal{A}$.
- (2) Sim_S emulates $\mathcal{F}_{\text{Eq}}$ and $\mathcal{F}_{\text{OT}}$, both of which output nothing to $\mathcal{A}$.

Indistinguishability. The adversary's view in the real execution is identically distributed to its view in the simulated execution. This follows because the values $e_{j,k}^S$ received from $\mathcal{F}_{\text{so-OPPRF}}$ are uniformly random in the real world and are sampled from the same distribution by the simulator in the ideal world.

Corrupted receiver. The simulator $\text{Sim}_{\mathcal{R}}$ receives the receiver's input W and output I , and proceeds as follows:

- (1) $\text{Sim}_{\mathcal{R}}$ emulates $\mathcal{F}_{\text{so-OPPRF}}$ by sampling uniformly random shares $e_{j,k}^R$ for all $j \in [m]$ and $k \in [d]$, and by sampling a random function F such that $F(C_{i,z}^R \| k \| w_{i,k} + t) = 0$ for all $i \in [n]$, $k \in [d]$, $z \in [2^d]$, and $t \in [-\delta, \delta]$. It then sends these shares $e_{j,k}^R$ and O^F to $\mathcal{A}$.
- (2) $\text{Sim}_{\mathcal{R}}$ emulates $\mathcal{F}_{\text{Eq}}$ by padding I with $\perp$ to size m and randomly permuting the padded I . For each $j \in [m]$, it sets

$$b_j = \begin{cases} 1, & \text{if } q_j = I[j] \neq \perp \text{ and there exists } w_i \text{ and } z \in [2^d] \\ & \text{such that } C_{i,z}^R = C_j^S \text{ and } \text{dist}(q_j, w_i) \leq \delta, \\ 0, & \text{otherwise.} \end{cases}$$

$\text{Sim}_{\mathcal{R}}$ sends all values b_j to $\mathcal{A}$.

- (3) $\text{Sim}_{\mathcal{R}}$ emulates $\mathcal{F}_{\text{OT}}$ and sends the padded and permuted I to $\mathcal{A}$.

Indistinguishability. We prove security via a sequence of hybrids.

Hybrid H_0 . This is the real execution of the protocol.

Hybrid H_1 . This hybrid is identical to H_0 , except that $\text{Sim}_{\mathcal{R}}$ acts as $\mathcal{F}_{\text{so-OPPRF}}$, $\mathcal{F}_{\text{Eq}}$, and $\mathcal{F}_{\text{OT}}$, but follows the specification of these functionalities and uses the real sender's input Q . This is only a conceptual change, and hence the adversary's view in Hybrid H_0 and Hybrid H_1 are identically distributed.

Hybrid H_2 . This hybrid is identical to H_1 , except that instead of computing b_j 's (resp. u_j 's) according to $\mathcal{F}_{\text{Eq}}$ (resp. $\mathcal{F}_{\text{OT}}$), $\text{Sim}_{\mathcal{R}}$ derives them utilizing the receiver's input W and output I . Specifically, it pads I with $\perp$ to size m , applies a random permutation, and for every $j \in [m]$ sets

$$b_j = \begin{cases} 1, & \text{if } q_j = I[j] \neq \perp \text{ and there exists } w_i \text{ and } z \in [2^d] \\ & \text{such that } C_{i,z}^R = C_j^S \text{ and } \text{dist}(q_j, w_i) \leq \delta, \\ 0, & \text{otherwise.} \end{cases}$$

The values b_j 's (resp. u_j 's) are returned to $\mathcal{A}$ as the outputs of $\mathcal{F}_{\text{Eq}}$ (resp. $\mathcal{F}_{\text{OT}}$).

The adversary's views in Hybrid H_1 and Hybrid H_2 are identically distributed, because functionality Eq outputs exactly these bits b_j and the outputs u_j of functionality OT are exactly determined by b_j and I .

Hybrid H_3 . This hybrid is identical to H_2 , except that we replace the sender's interaction in $\mathcal{F}_{\text{so-OPPRF}}$ with a simulation as follows: sample random shares $e_{j,k}^R$ for all $j \in [m]$, $k \in [d]$, and sample a random function F conditioned on $F(C_{i,z}^R \| k \| w_{i,k} + t) = 0$ for all $i \in [n]$, $k \in [d]$, $z \in [2^d]$, and $t \in [-\delta, \delta]$. The simulator sends $\{e_{j,k}^R\}$ and O^F to $\mathcal{A}$.

By the security of $\mathcal{F}_{\text{so-OPPRF}}$, the adversary's view of Hybrid H_2 and Hybrid H_3 are identically distributed.

Observe that Hybrid H_3 is exactly the simulated execution produced by $\text{Sim}_{\mathcal{R}}$. Therefore, the real and ideal executions are perfectly indistinguishable, completing the proof.

□

G.8 Security Proof of $\mathcal{F}_{\text{ConRand}}$

PROOF. Correctness. For any e^S and e^R such that $f(e^S, e^R) = 1$, the outputs b^S and b^R are secret shares of 1. By the correctness of $\mathcal{F}_{\text{MUX}}$, the MUX outputs m^S and m^R are secret shares of 0, and hence $z^S + z^R = v^S + v^R$.

Conversely, for any e^S and e^R such that $f(e^S, e^R) = 0$, the outputs b^S and b^R are secret shares of 0. Since r^S and r^R are sampled uniformly at random, by the correctness of $\mathcal{F}_{\text{MUX}}$, m^S and m^R are uniformly random, and consequently $v^S + m^S$ and $v^R + m^R$ are each uniformly random. This completes the correctness proof.

Corrupted sender. The simulator Sim_S receives the sender's input (e^S, v^S) and output z^S , and proceeds as follows:

- (1) Sim_S emulates $\mathcal{F}_{\text{ssPEQT}}/\mathcal{F}_{\text{ssCMP}}$ by sampling uniformly random shares b^S and sends it to $\mathcal{A}$.
- (2) Sim_S emulates $\mathcal{F}_{\text{MUX}}$ by computing $m^S = z^S - v^S$ and sends it to $\mathcal{A}$.

Indistinguishability. The adversary's view in the real execution is identically distributed to its view in the simulated execution. This follows because the values b^S received from $\mathcal{F}_{\text{ssPEQT}}/\mathcal{F}_{\text{ssCMP}}$ are uniformly random in the real execution and are sampled from the same distribution by the simulator in the ideal execution. Moreover, the value m^S is identically distributed in both worlds conditioned on the relation $z^S = v^S + m^S$. Therefore, the joint distribution of the adversary's view is identical in the real and ideal executions.

Corrupted receiver. Since the protocol execution is symmetric, the security of the corrupted receiver follows similarly. This completes the proof. □

G.9 Security Proof of $\mathcal{F}_{\text{ConSel}}$

PROOF. Corrupted sender. The simulator Sim_S receives the sender's input $\{e_i^S\}_{i \in [\mu]}$ and output m^S , and proceeds as follows:

- (1) Sim_S emulates $\mathcal{F}_{\text{ssPEQT}}/\mathcal{F}_{\text{ssCMP}}$ by sampling uniformly random shares b_i^S for $i \in [\mu]$ and sends them to $\mathcal{A}$.
- (2) Sim_S emulates $\mathcal{F}_{\text{MUX}}$ and sends m^S to $\mathcal{A}$.

Indistinguishability. The adversary's view in the real execution is identically distributed to its view in the simulated execution. This follows because the values b_i^S received from $\mathcal{F}_{\text{ssPEQT}}/\mathcal{F}_{\text{ssCMP}}$ are uniformly random in the real execution and are sampled from the same distribution by the simulator in the ideal execution. Moreover, the output m^S is identically distributed in both worlds. Therefore, the joint distribution of the adversary's view is identical in the real and ideal executions.

Corrupted receiver. Since the protocol execution is symmetric, the security of the corrupted receiver follows similarly. This completes the proof. □

G.10 Proof of Lemma 3.1

PROOF. By definition 3.1 of [10], let $\{b_j\}_{j \in [\mu]} = \text{PxTrie}(q - \delta, q + \delta)$. Then:

$$[q - \delta, q + \delta] = \bigcup_{j \in [\mu]} \text{Interval}(b_j)$$

and for any $b \in \{b_j\}_{j \in [\mu]}$, if a binary string b' is a prefix of b , then:

$$\text{Interval}(b') \not\subseteq [q - \delta, q + \delta]$$

Therefore, for any b_j , we have $\text{Interval}(b_j) \subseteq [q - \delta, q + \delta]$. Let k_j be the number of wildcards in b_j . Since the length of the interval $|\text{Interval}(b_j)| = 2^{k_j}$ cannot exceed the total length of the range $2\delta + 1$, it holds that $2^{k_j} \leq 2\delta + 1$, which implies $k_j \leq \lfloor \log_2(2\delta + 1) \rfloor$.

Case 1: $w \in [q - \delta, q + \delta]$. First, we show **existence**. Since $w \in [q - \delta, q + \delta]$, the completeness property of definition 3.1 of [10] ensures there exists an index $j^* \in [\mu]$ such that $w \in \text{Interval}(b_{j^*})$. By the definition of $\text{Interval}(\cdot)$, b_{j^*} must be a prefix of w . As derived above, the wildcard count of b_{j^*} satisfies $k_{j^*} \leq \lfloor \log_2(2\delta + 1) \rfloor$. By the definition of $\text{PxPath}(w, \delta)$ provided in section 3.4, b_{j^*} is a valid prefix on the path of root-to- w within this wildcard bound. Thus, $b_{j^*} \in \text{PxPath}(w, \delta) \cap \text{PxTrie}(q - \delta, q + \delta)$, and the intersection is non-empty.

Next, we show **uniqueness** by contradiction. Suppose there exist two distinct nodes x, y in the intersection. Since $x, y \in \text{PxPath}(w, \delta)$, both are prefixes of w , implying one must be a prefix of the other. However, definition 3.1 of [10] states that for any $b \in \text{PxTrie}$, its proper prefixes b' must satisfy $\text{Interval}(b') \not\subseteq [q - \delta, q + \delta]$. This prefix-free property ensures that no node in PxTrie can be a prefix of another, contradicting the existence of both x and y . Thus, the intersection size is exactly 1.

Case 2: $w \notin [q - \delta, q + \delta]$. We show the intersection is empty by contradiction. Suppose there exists some $b_i \in \text{PxPath}(w, \delta) \cap \text{PxTrie}(q - \delta, q + \delta)$. The membership $b_i \in \text{PxPath}(w, \delta)$ implies b_i is a prefix of w , so $w \in \text{Interval}(b_i)$. The membership $b_i \in \text{PxTrie}(q - \delta, q + \delta)$ implies $\text{Interval}(b_i) \subseteq [q - \delta, q + \delta]$ per definition 3.1 of [10]. It follows that $w \in [q - \delta, q + \delta]$, which contradicts the assumption. Hence, the intersection must be empty.

Combining the above two cases completes the proof. $\square$

Functionality $\mathcal{F}_{\text{ExtConSel}}^{f, \mu}$

Parameters: Sender $\mathcal{S}$ and Receiver $\mathcal{R}$. Condition function $f(\cdot, \cdot)$ where f can be an equality or a comparison function. Input size μ

Functionality:

- (1) Wait for input $\{(e_1^S, r_1^S), \dots, (e_\mu^S, r_\mu^S)\}$ from $\mathcal{S}$ and $\{(e_1^R, r_1^R), \dots, (e_\mu^R, r_\mu^R)\}$ from $\mathcal{R}$, where there exists at most one (e_i^S, e_i^R) with $f(e_i^S, e_i^R) = 1$.
- (2) Compute $b_i := f(e_i^S, e_i^R)$ for all $i \in [\mu]$.
- (3) If exist $b_i = 1$, generate two random shares such that $z^S + z^R = r_i^S + r_i^R$.
Otherwise, generate two random shares (z^S, z^R) .
- (4) Output z^S to $\mathcal{S}$ and z^R to $\mathcal{R}$.

Figure 19: Functionality of Extended Conditional Selection.

H Additional Protocols under Unique Cell Assumptions

H.1 Sender-sided Setting

The protocol for L_p distances under sender-sided unique cell assumption is given in Figure 21. The proof is given in Appendix G.3.

Protocol $\Pi_{\text{ExtConSel}}^{\text{Eq}, \mu}$

Parameters: Ideal functionality $\mathcal{F}_{\text{MUX}}$ and $\mathcal{F}_{\text{ssPEQT}}$.

Input: $\mathcal{S}$ inputs $\{(e_i^S, r_i^S)\}_{i \in [\mu]}$ and $\mathcal{R}$ inputs $\{(e_i^R, r_i^R)\}_{i \in [\mu]}$.

Protocol:

- (1) $\mathcal{S}$ and $\mathcal{R}$ invoke $\mathcal{F}_{\text{ssPEQT}}$, where $\mathcal{S}$ inputs $\{e_i^S\}_{i \in [\mu]}$ and $\mathcal{R}$ inputs $\{e_i^R\}_{i \in [\mu]}$. $\mathcal{S}$ receives $\{b_i^S\}_{i \in [\mu]} \in \{0, 1\}^\mu$, and $\mathcal{R}$ receives $\{b_i^R\}_{i \in [\mu]} \in \{0, 1\}^\mu$.
- (2) $\mathcal{S}$ and $\mathcal{R}$ invoke $\mathcal{F}_{\text{MUX}}$, where $\mathcal{S}$ inputs $\{(b_i^S, r_i^S)\}_{i \in [\mu]}$ and $\mathcal{R}$ inputs $\{(b_i^R, r_i^R)\}_{i \in [\mu]}$. $\mathcal{S}$ receives $\{t_i^S\}_{i \in [\mu]}$ and $\mathcal{R}$ receives $\{t_i^R\}_{i \in [\mu]}$.
- (3) $\mathcal{S}$ computes $t^S := \sum_{i \in [\mu]} t_i^S$ and $\mathcal{R}$ computes $t^R := \sum_{i \in [\mu]} t_i^R$.
- (4) $\mathcal{S}$ computes $b^S := \bigoplus_{i \in [\mu]} b_i^S$ and $\mathcal{R}$ computes $b^R := \bigoplus_{i \in [\mu]} b_i^R$.
- (5) $\mathcal{S}$ randomly samples r^S and $\mathcal{R}$ randomly samples r^R .
- (6) $\mathcal{S}$ and $\mathcal{R}$ invoke $\mathcal{F}_{\text{MUX}}$, where $\mathcal{S}$ inputs $(1 - b^S, r^S)$ and $\mathcal{R}$ inputs (b^R, r^R) . $\mathcal{S}$ receives m^S and $\mathcal{R}$ receives m^R .
- (7) $\mathcal{S}$ outputs $z^S = m^S + t^S$ and $\mathcal{R}$ outputs $z^R = m^R + t^R$.

Figure 20: Protocol of Extended Equality-conditional Selection.

H.2 Receiver-sided Setting

The protocol for L_p distances under the receiver-sided unique cell assumption is given in Figure 22. The proof is given in Appendix G.5.

I Additional Prefix-optimized Protocols

I.1 Sender-sided Setting

The prefix-optimized protocol under the sender-sided *unique cell* assumption for L_∞ distances is given in Figure 23. The protocol for L_p distances can be optimized analogously, and we omit the details for brevity. The core idea follows Section 4.4. Here, we extend $\mathcal{F}_{\text{ConSel}}^{f, \mu}$ to a more general functionality $\mathcal{F}_{\text{ExtConSel}}^{f, \mu}$ that handles additional inputs. Specifically, $\mathcal{F}_{\text{ExtConSel}}^{f, \mu}$ takes as input a set of tuples, each consisting of a condition and an associated value to be randomized, and randomizes each value depending on whether its condition is satisfied. $\mathcal{F}_{\text{ConSel}}^{f, \mu}$ can be viewed as a special case where the condition itself serves as the value. The ideal functionality of $\mathcal{F}_{\text{ExtConSel}}^{f, \mu}$ is given in Figure 19, and its instantiation, which follows the same structure as $\mathcal{F}_{\text{ConSel}}^{f, \mu}$, is presented in Figure 20.

We note that when invoking $\mathcal{F}_{\text{ExtConSel}}^{f, \mu}$, there is no need to additionally invoke EqRand , as the output shares $v_{i,z}^R$ and $v_{i,z}^S$ are already re-randomized for non-matching points w_i .

I.2 Receiver-sided Setting

The prefix-optimized protocol under the receiver-sided *unique cell* assumption for L_p distances is given in Figure 26. Since each prefix covers a range of points rather than a single value, the receiver cannot program the exact p -th power distance $|t|^p$ as in the L_∞ construction. We therefore adopt the approach of [10], where the receiver programs the distance from $w_{i,k}$ to the endpoint of the

Protocol Π_{FPSI}^{Lp}

Parameters: Sender $\mathcal{S}$ and receiver $\mathcal{R}$ with input sizes m and n . Threshold δ . Bit length $\lambda' = \lambda + d + \log mn$, $\ell = \lambda + d + \log m + p \log \delta$, and $u = \log|\mathbb{U}|$.

Input: $\mathcal{S}$ inputs $Q = \{q_j\}_{j \in [m]} \in \mathbb{U}^{d \times m}$ and $\mathcal{R}$ inputs $W = \{w_i\}_{i \in [n]} \in \mathbb{U}^{d \times n}$.

Protocol:

- (1) For $j \in [m]$, $\mathcal{S}$ computes $C_j^S = \text{cell}_{2\delta}(q_j)$. For $i \in [n]$, $\mathcal{R}$ computes $\{C_{i,z}^R\}_{z \in [2^d]} = \text{neigh}_{2\delta}(w_i)$.
- (2) $\mathcal{S}$ sets $\text{List} = \left\{ \left(C_j^S \| k \| q_{j,k} + t, |t|^p \| s_{j,k} \right) \right\}_{j \in [m], k \in [d], t \in [-\delta, \delta]}$ and computes $s_j = \sum_{k \in [d]} s_{j,k}$ where $s_{j,k} \xleftarrow{\$} \mathbb{F}_{2^\kappa}$.
- (3) $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{so-OPPRF}}$ where $\mathcal{S}$ inputs List and $\mathcal{R}$ inputs $\{C_{i,z}^R \| k \| w_{i,k}\}_{i \in [n], k \in [d], z \in [2^d]}$. $\mathcal{R}$ receives $\{e_{i,k,z}^R \| r_{i,k,z}^R\}_{i \in [n], k \in [d], z \in [2^d]}$ and $\mathcal{S}$ receives $\{e_{i,k,z}^S \| r_{i,k,z}^S\}_{i \in [n], k \in [d], z \in [2^d]}$ and $\mathcal{O}^F$.
- (4) For $i \in [n]$, $k \in [d]$, $z \in [2^d]$, $\mathcal{S}$ and $\mathcal{R}$ invoke $\mathcal{F}_{\text{B2A}}$, where $\mathcal{S}$ inputs $e_{i,k,z}^S$ and $\mathcal{R}$ inputs $e_{i,k,z}^R$. $\mathcal{S}$ receives $d_{i,k,z}^S$ and $\mathcal{R}$ receives $d_{i,k,z}^R$.
- (5) For $i \in [n]$, $z \in [2^d]$, $\mathcal{R}$ computes $d_{i,z}^R = \sum_{k \in [d]} d_{i,k,z}^R$ and $r_{i,z}^R = \sum_{k \in [d]} r_{i,k,z}^R$, and $\mathcal{S}$ computes $d_{i,z}^S = \sum_{k \in [d]} d_{i,k,z}^S$ and $r_{i,z}^S = \sum_{k \in [d]} r_{i,k,z}^S$.
- (6) For $i \in [n]$, $z \in [2^d]$, $\mathcal{R}$ and $\mathcal{S}$ invoke $\mathcal{F}_{\text{ConRand}}^{\text{Cmp}}$ where $\mathcal{R}$ inputs $(d_{i,z}^R, r_{i,z}^R)$ and $\mathcal{S}$ inputs $(d_{i,z}^S, r_{i,z}^S)$. $\mathcal{R}$ receives $v_{i,z}^R$ and $\mathcal{S}$ receives $v_{i,z}^S$.
- (7) For $i \in [n]$, $z \in [2^d]$, $\mathcal{S}$ sends $v_{i,z}^S$ and $\mathcal{R}$ reconstructs $v_{i,z} = v_{i,z}^S + v_{i,z}^R$.
- (8) $\mathcal{S}$ computes $X = \{H_{\lambda'+du}(s_j) \oplus (0^{\lambda'} \| q_j)\}_{j \in [m]}$ and sends shuffled X to $\mathcal{R}$.
- (9) $\mathcal{R}$ sets $I = \emptyset$ and parses X as $\{x_j\}_{j \in [m]}$. For $j \in [m]$, $i \in [n]$, $z \in [2^d]$, if $H_{\lambda'+du}(v_{i,z}) \oplus x_j = 0^{\lambda'} \| q_j$, update $I = I \cup \{q_j\}$.

Figure 21: Protocol of fuzzy PSI for L_p distance under sender-sided unique cell assumption.

interval covered by its prefix. The sender then evaluates the so-OPPRF to obtain shares of this distance and computes the total distance via MPC multiplication and addition.

Specifically, let $\{\tilde{w}_{i,k,h'}\}_{h' \in [\mu]}$ and $\{\tilde{q}_{j,k,h}\}_{h \in [\mu]}$ denote the prefixes of $w_{i,k}$ and $q_{j,k}$, respectively, with upper endpoints w^* and q^* . The receiver programs $|w_{i,k} - w^*|$ via the so-OPPRF, and the sender evaluates at $\tilde{q}_{j,k,h}$. In addition, the sender knows $q_{j,k}$ and q^* locally, therefore it can compute $|q^* - q_{j,k}|$ directly. If $\tilde{q}_{j,k,h}$ matches some $\tilde{w}_{i,k,h'}$, then $q^* = w^*$, and the distance decomposes as $|w_{i,k} - q_{j,k}| = |w_{i,k} - w^*| + |q^* - q_{j,k}|$. Since $|w_{i,k} - w^*|$ is secret-shared between the two parties and $|q^* - q_{j,k}|$ is known to the sender, the p -th power can be expanded via the binomial theorem:

$$|w_{i,k} - q_{j,k}|^p = \sum_{c=0}^p \binom{p}{c} |w_{i,k} - w^*|^c \cdot |q^* - q_{j,k}|^{p-c}.$$

This decomposition holds when $q_{j,k} \leq w_{i,k}$; the symmetric case $q_{j,k} > w_{i,k}$ is handled using the lower endpoint instead. To cover both cases, we split $[w_{i,k} - \delta, w_{i,k} + \delta]$ into $[w_{i,k} - \delta, w_{i,k}]$ and

Protocol Π_{FPSI}^{Lp}

Parameters: Sender $\mathcal{S}$ and receiver $\mathcal{R}$ with input sizes m and n . Threshold δ . Bit length $\ell = \lambda + d + \log m + p \log \delta$.

Input: $\mathcal{S}$ inputs $Q = \{q_j\}_{j \in [m]} \in \mathbb{U}^{d \times m}$ and $\mathcal{R}$ inputs $W = \{w_i\}_{i \in [n]} \in \mathbb{U}^{d \times n}$.

Protocol:

- (1) For $i \in [n]$, $\mathcal{R}$ computes $C_i^R = \text{cell}_{2\delta}(w_i)$. For $j \in [m]$, $\mathcal{S}$ computes $\{C_{j,z}^S\}_{z \in [2^d]} = \text{neigh}_{2\delta}(q_j)$.
- (2) $\mathcal{R}$ sets $\text{List} = \{(C_i^R \| k \| w_{i,k} + t, |t|^p)\}_{i \in [n], k \in [d], t \in [-\delta, \delta]}$.
- (3) $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{so-OPPRF}}$ where $\mathcal{R}$ inputs List and $\mathcal{S}$ inputs $\{C_{j,z}^S \| k \| q_{j,k}\}_{j \in [m], k \in [d], z \in [2^d]}$. $\mathcal{R}$ receives $\{e_{j,k,z}^R\}_{j \in [m], k \in [d], z \in [2^d]}$ and $\mathcal{O}^F$ and $\mathcal{S}$ receives $\{e_{j,k,z}^S\}_{j \in [m], k \in [d], z \in [2^d]}$.
- (4) For $j \in [m]$, $k \in [d]$, $z \in [2^d]$, $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{B2A}}$, where $\mathcal{R}$ inputs $e_{j,k,z}^R$ and $\mathcal{S}$ inputs $e_{j,k,z}^S$. $\mathcal{R}$ receives $d_{j,k,z}^R$ and $\mathcal{S}$ receives $d_{j,k,z}^S$.
- (5) For $j \in [m]$, $z \in [2^d]$, $\mathcal{R}$ computes $r_{j,z}^R = \sum_{k \in [d]} d_{j,k,z}^R$ and $\mathcal{S}$ computes $r_{j,z}^S = \sum_{k \in [d]} d_{j,k,z}^S$.
- (6) For $j \in [m]$, $z \in [2^d]$, $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{Interval}}^{\delta p}$ where $\mathcal{R}$ inputs $r_{j,z}^R$ and $\mathcal{S}$ inputs $r_{j,z}^S$. $\mathcal{R}$ receives $b_{j,z} = 1\{r_{j,z}^R + r_{j,z}^S \leq \delta^p\}$.
- (7) For $j \in [m]$, $\mathcal{R}$ computes $b_j = \bigvee_{z \in [2^d]} b_{j,z}$.
- (8) For $j \in [m]$, $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{OT}}$ where $\mathcal{R}$ inputs b_j and $\mathcal{S}$ inputs $(\perp, q_j)$. $\mathcal{R}$ receives u_j .
- (9) $\mathcal{R}$ outputs $I = \{u_j \mid b_j = 1 \text{ for } j \in [m]\}$.

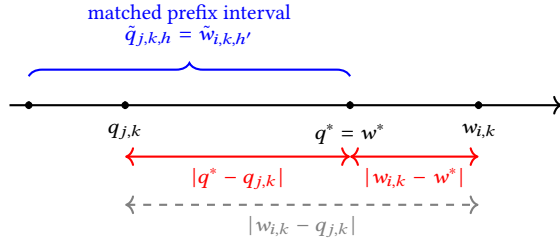
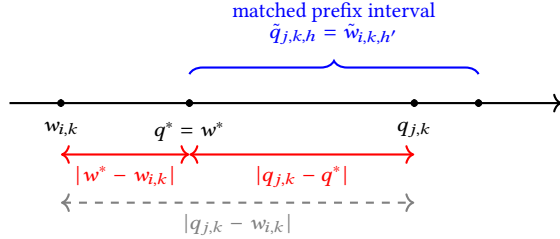
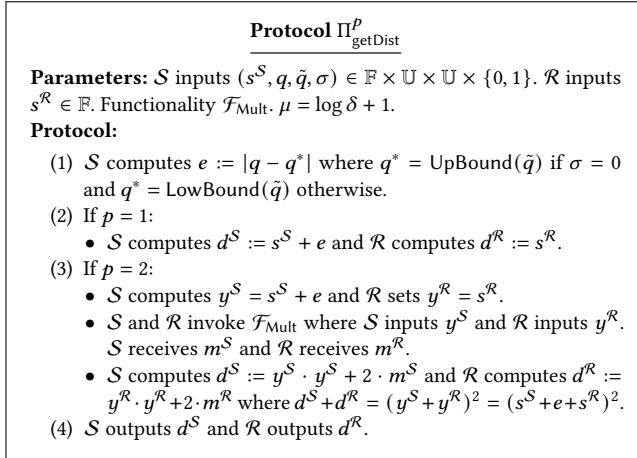
Figure 22: Protocol of fuzzy PSI for L_p distance under receiver-sided unique cell assumption.

$[w_{i,k} + 1, w_{i,k} + \delta]$, applying the upper endpoint to the left sub-interval and the lower endpoint to the right. We use $\sigma \in \{0, 1\}$ to index the two sub-intervals. The two cases are illustrated in Figure 24, and the detailed sub-protocol for computing the distance is given in Figure 25.

J Additional Protocols under Unique Block Assumptions

The protocol under the receiver-sided *unique block* assumption for L_p distances is given in Figure 27. The key difference from the L_∞ construction is that the receiver programs the p -th power of the per-dimension distance rather than 0, and invokes $\mathcal{F}_{\text{Interval}}^{\delta p}$ to test whether the aggregated distance falls within the threshold δ^p . The prefix-optimized variant follows the approach of Section 4.4, and the detailed protocol is given in Figure 28.

The construction for general L_p distances follows the same idea as in Appendix I.2, with the prefix-optimized protocol given in Figure 29. The protocols under the sender-sided unique block assumption follow analogously, and we omit the details for brevity.

(a) Left sub-interval: $q_{j,k} < w_{i,k}$, upper endpoint $q^* = w^*$.(b) Right sub-interval: $q_{j,k} > w_{i,k}$, lower endpoint $q^* = w^*$.**Figure 24: An example to compute the distances with prefix trie techniques****Figure 25: Protocol for computing distance for $p = 1$ and $p = 2$. The protocol can be extended to arbitrary p .****Protocol $\Pi_{\text{FPSI-PX}}^{L_\infty}$**

Parameters: Sender $\mathcal{S}$ and receiver $\mathcal{R}$ with input sizes m and n . Threshold δ . Bit length $\lambda' = \lambda + d + \log mn$, $\ell = \lambda + d + \log mn$ and $u = \log|\mathbb{U}|$.

Input: $\mathcal{S}$ inputs $Q = \{q_j\}_{j \in [m]} \in \mathbb{U}^{d \times m}$ and $\mathcal{R}$ inputs $W = \{w_i\}_{i \in [n]} \in \mathbb{U}^{d \times n}$.

Protocol:

- (1) For $j \in [m]$, $\mathcal{S}$ computes $C_j^S = \text{cell}_{2\delta}(q_j)$. For $i \in [n]$, $\mathcal{R}$ computes $\{C_{i,z}^R\}_{z \in [2^d]} = \text{neigh}_{2\delta}(w_i)$.
- (2) $\mathcal{S}$ encodes $\text{List} = \{(C_j^S \| k \| \tilde{q}_{j,k,h}, 0^\ell \| s_{j,k})\}_{j \in [m], k \in [d], h \in [\mu]}$ where $s_{j,k} \xleftarrow{\$} \mathbb{F}_{2^\kappa}$ and $\{\tilde{q}_{j,k,h}\}_{h \in [\mu]} = \text{PxTrie}(q_{j,k} - \delta, q_{j,k} + \delta)$.
- (3) $\mathcal{R}$ computes $\text{query} = \{(C_j^S \| k \| \tilde{w}_{i,k,h}\}_{i \in [n], k \in [d], z \in [2^d], h \in [\mu]}$ where $\{\tilde{w}_{i,k,h}\}_{h \in [\mu]} = \text{PxPath}(w_{i,k}, \delta)$.
- (4) $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{so-OPPRF}}$ where $\mathcal{S}$ inputs List and $\mathcal{R}$ inputs query . $\mathcal{R}$ receives $\{e_{i,k,z,h}^R \| r_{i,k,z,h}^R\}_{i \in [n], k \in [d], z \in [2^d], h \in [\mu]}$ and $\mathcal{S}$ receives $\{e_{i,k,z,h}^S \| r_{i,k,z,h}^S\}_{i \in [n], k \in [d], z \in [2^d], h \in [\mu]}$.
- (5) For $i \in [n], k \in [d], z \in [2^d]$, $\mathcal{R}$ and $\mathcal{S}$ invoke $\mathcal{F}_{\text{ExtConSel}}^{\text{Eq}, \mu}$ where $\mathcal{R}$ inputs $\{(e_{i,k,z,h}^R \| r_{i,k,z,h}^R)\}_{h \in [\mu]}$ and $\mathcal{S}$ inputs $\{(-e_{i,k,z,h}^S \| r_{i,k,z,h}^S)\}_{h \in [\mu]}$. $\mathcal{R}$ receives $v_{i,k,z}^R$ and $\mathcal{S}$ receives $v_{i,k,z}^S$.
- (6) For $i \in [n], z \in [2^d]$, $\mathcal{R}$ computes $v_{i,z}^R = \sum_{k \in [d]} v_{i,k,z}^R$. $\mathcal{S}$ computes $v_{i,z}^S = \sum_{k \in [d]} v_{i,k,z}^S$.
- (7) For $i \in [n], z \in [2^d]$, $\mathcal{S}$ sends $v_{i,z}^S$ and $\mathcal{R}$ reconstructs $v_{i,z} = v_{i,z}^S + v_{i,z}^R$.
- (8) For $j \in [m]$, $\mathcal{S}$ computes $s_j = \sum_{k \in [d]} s_{j,k}$.
- (9) $\mathcal{S}$ computes $X = \{H_{\lambda' + du}(s_j) \oplus (0^{\lambda'} \| q_j)\}_{j \in [m]}$ and sends shuffled X to $\mathcal{R}$.
- (10) $\mathcal{R}$ sets $I = \emptyset$ and parses X as $\{x_j\}_{j \in [m]}$. For $j \in [m], i \in [n], z \in [2^d]$, if $H_{\lambda' + du}(v_{i,z}) \oplus x_j = 0^{\lambda'} \| q_j$, update $I = I \cup \{q_j\}$.

Figure 23: Protocol of prefix-optimized fuzzy PSI for L_∞ distance under sender-sided unique cell assumption.**K Additional Experimental Results****K.1 Performance of our protocols for L_p distances under unique cell assumption**

We present the results of our protocols for L_p distances under the receiver-sided unique cell assumption in Table 4. As no existing work supports L_p distances under this assumption, we omit a direct comparison.

K.2 Performance of our protocols under the sender-sided unique cell assumption

We present the results of our protocols for L_∞ , L_1 , and L_2 distances under the sender-sided unique cell assumption in Table 5. As no existing work operates under this assumption, we omit a direct comparison.

Protocol $\Pi_{\text{FPSI-Px}}^{Lp}$

Parameters: Sender S and receiver R with input sizes m and n . Threshold δ . Bit length $\ell = \lambda + d + \log m + p \log \delta$. $\mu = 1 + \log \delta$

Input: S inputs $Q = \{q_j\}_{j \in [m]} \in \mathbb{U}^{d \times m}$ and R inputs $W = \{w_i\}_{i \in [n]} \in \mathbb{U}^{d \times n}$.

Protocol:

- (1) For $i \in [n]$, R computes $C_i^R = \text{cell}_{2\delta}(w_i)$. For $j \in [m]$, S computes $\{C_{j,z}^S\}_{z \in [2^d]} = \text{neigh}_{2\delta}(q_j)$.
- (2) R sets $\text{List} = \{(C_i^R \| k \| \sigma \| \tilde{w}_{i,k,h,\sigma}, |w^* - w_{i,k}|)\}_{i \in [n], k \in [d], h \in [\mu], \sigma \in [0,1]}$ where $\{\tilde{w}_{i,k,h,0}\}_{h \in [\mu]} = \text{PxTrie}(w_{i,k} - \delta, w_{i,k})$ and $\{\tilde{w}_{i,k,h,1}\}_{h \in [\mu]} = \text{PxTrie}(w_{i,k} + 1, w_{i,k} + \delta)$ and $w^* = \text{UpBound}(\tilde{w}_{\sigma,i,k,h})$ if $\sigma = 0$ otherwise $w^* = \text{LowBound}(\tilde{w}_{\sigma,i,k,h})$.
- (3) S computes query $= \{C_{j,z}^S \| k \| \sigma \| \tilde{q}_{j,k,h}\}_{j \in [m], k \in [d], z \in [2^d], h \in [\mu], \sigma \in [0,1]}$ where $\{\tilde{q}_{j,k,h}\}_{h \in [\mu]} = \text{PxPath}(q_{j,k}, \delta/2)$.
- (4) R and S invoke functionality $\mathcal{F}_{\text{SO-OPPRF}}$ where R inputs List and S inputs query. R receives $\{e_{j,k,z,h,\sigma}^R\}_{j \in [m], k \in [d], z \in [2^d], h \in [\mu], \sigma \in [0,1]}$ and S receives $\{e_{j,k,z,h,\sigma}^S\}_{j \in [m], k \in [d], z \in [2^d], h \in [\mu], \sigma \in [0,1]}$.
- (5) For $j \in [m]$, $k \in [d]$, $z \in [2^d]$, $h \in [\mu]$, $\sigma \in [0,1]$, R and S invoke functionality $\mathcal{F}_{\text{B2A}}$, where R inputs $e_{j,k,z,h,\sigma}^R$ and S inputs $e_{j,k,z,h,\sigma}^S$. R receives $s_{j,k,z,h,\sigma}^R$ and S receives $s_{j,k,z,h,\sigma}^S$.
- (6) For $j \in [m]$, $k \in [d]$, $z \in [2^d]$, $h \in [\mu]$, $\sigma \in [0,1]$, R and S invoke Π_{getDist}^P where R inputs $s_{j,k,z,h,\sigma}^R$ and S inputs $(s_{j,k,z,h,\sigma}^S, q_{j,k}, \tilde{q}_{j,k,h}, \sigma)$. R receives $v_{j,k,z,h,\sigma}^R$ and S receives $v_{j,k,z,h,\sigma}^S$.
- (7) For $j \in [m]$, $k \in [d]$, $z \in [2^d]$, R and S invoke functionality $\mathcal{F}_{\text{ConSel}}^{\text{Cmp}, 2 \cdot \mu}$ where R inputs $\{v_{j,k,z,h,\sigma}^R\}_{h \in [\mu], \sigma \in [0,1]}$ and S inputs $\{v_{j,k,z,h,\sigma}^S\}_{h \in [\mu], \sigma \in [0,1]}$. R receives $d_{j,k,z}^R$ and S receives $d_{j,k,z}^S$.
- (8) For $j \in [m]$, $z \in [2^d]$, R computes $r_{j,z}^R = \sum_{k \in [d]} d_{j,k,z}^R$ and S computes $r_{j,z}^S = \sum_{k \in [d]} d_{j,k,z}^S$.
- (9) For $j \in [m]$, $z \in [2^d]$, R and S invoke functionality $\mathcal{F}_{\text{Interval}}^{\delta p}$ where R inputs $r_{j,z}^R$ and S inputs $r_{j,z}^S$. R receives $b_{j,z} = 1\{r_{j,z}^R + r_{j,z}^S \leq \delta^p\}$.
- (10) For $j \in [m]$, R computes $b_j = \bigvee_{z \in [2^d]} b_{j,z}$.
- (11) For $j \in [m]$, R and S invoke functionality $\mathcal{F}_{\text{OT}}$ where R inputs b_j and S inputs $(\perp, q_j)$. R receives u_j .
- (12) R outputs $I = \{u_j \mid b_j = 1 \text{ for } j \in [m]\}$.

Figure 26: Protocol of prefix-optimized fuzzy PSI for L_p distance under receiver-sided unique cell assumption.

Protocol Π_{FPSI}^{Lp}

Parameters: Distance threshold δ . Bit length $\ell = \lambda + \log m + p \log \delta$.

Input: S inputs $Q = \{q_j\}_{j \in [m]} \in \mathbb{U}^{d \times m}$ and R inputs $W = \{w_i\}_{i \in [n]} \in \mathbb{U}^{d \times n}$.

Protocol:

- (1) For $i \in [n]$, R computes $\{C_{i,z}^R\}_{z \in [2^d]} = \text{neigh}_{2\delta}(w_i)$. For $j \in [m]$, S computes $C_j^S = \text{cell}_{2\delta}(q_j)$.
- (2) R sets $\text{List} = \{(C_{i,z}^R \| k \| w_{i,k} + t, |t|^p)\}_{i \in [n], k \in [d], z \in [2^d], t \in [-\delta, \delta]}$
- (3) R and S invoke functionality $\mathcal{F}_{\text{SO-OPPRF}}$ where R inputs List and S inputs $\{C_j^S \| k \| q_{j,k}\}_{j \in [m], k \in [d]}$. R receives $\{e_{j,k}^R\}_{j \in [m], k \in [d]}$ and S receives $\{e_{j,k}^S\}_{j \in [m], k \in [d]}$.
- (4) For $j \in [m]$, $k \in [d]$, S and R invoke functionality $\mathcal{F}_{\text{B2A}}$, where S inputs $e_{j,k}^S$ and R inputs $e_{j,k}^R$. S receives $s_{j,k}^S$ and R receives $s_{j,k}^R$.
- (5) For $j \in [m]$, R computes $r_j^R = \sum_{k \in [d]} s_{j,k}^R$ and S computes $r_j^S = \sum_{k \in [d]} s_{j,k}^S$.
- (6) For $j \in [m]$, R and S invoke functionality $\mathcal{F}_{\text{Interval}}^{\delta p}$ where R inputs r_j^R and S inputs r_j^S . R receives $b_j = 1\{r_j^R + r_j^S \leq \delta^p\}$.
- (7) For $j \in [m]$, R and S invoke functionality $\mathcal{F}_{\text{OT}}$ where R inputs b_j and S inputs $(\perp, q_j)$. R receives u_j .
- (8) R outputs $I = \{u_j \mid b_j = 1 \text{ for } j \in [m]\}$.

Figure 27: Protocol of fuzzy PSI for L_p distance under receiver-sided unique block assumption.

Protocol $\Pi_{\text{FPSI-Px}}^{L_\infty}$

Parameters: Distance threshold δ . Bit length $\mu = \log \delta + 2$ and $\ell = \lambda + \log(md\mu)$

Input: $\mathcal{S}$ inputs $Q = \{\mathbf{q}_j\}_{j \in [m]} \in \mathbb{U}^{d \times m}$ and $\mathcal{R}$ inputs $W = \{\mathbf{w}_i\}_{i \in [n]} \in \mathbb{U}^{d \times n}$.

Protocol:

- (1) For $i \in [n]$, $\mathcal{R}$ computes $\{C_{i,z}^{\mathcal{R}}\}_{z \in [2^d]} = \text{neigh}_{2\delta}(\mathbf{w}_i)$. $\mathcal{S}$ shuffles Q and computes $C_j^{\mathcal{S}} = \text{cell}_{2\delta}(\mathbf{q}_j)$ for $j \in [m]$.
- (2) $\mathcal{R}$ sets $\text{List} = \{(C_{i,z}^{\mathcal{R}} \| k \| \tilde{\mathbf{w}}_{i,k,h}, 0^\ell)\}_{i \in [n], z \in [2^d], k \in [d], h \in [\mu]}$ where $\{\tilde{\mathbf{w}}_{i,k,h}\}_{h \in [\mu]} = \text{PxTrie}(\mathbf{w}_{i,k} - \delta, \mathbf{w}_{i,k} + \delta)$
- (3) $\mathcal{S}$ computes $\{C_j^{\mathcal{S}} \| k \| \tilde{q}_{j,k,h}\}_{j \in [m], k \in [d], h \in [\mu]}$ where $\{\tilde{q}_{j,k,h}\}_{h \in [\mu]} = \text{PxPath}(\mathbf{q}_{j,k}, \delta)$
- (4) $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{so-OPPRF}}$ where $\mathcal{R}$ inputs List and $\mathcal{S}$ inputs $\{C_j^{\mathcal{S}} \| k \| \tilde{q}_{j,k,h}\}_{j \in [m], k \in [d], h \in [\mu]}$. $\mathcal{R}$ receives $\{e_{j,k,h}^{\mathcal{R}}\}_{j \in [m], k \in [d], h \in [\mu]}$ and $\mathcal{S}$ receives $\{e_{j,k,h}^{\mathcal{S}}\}_{j \in [m], k \in [d], h \in [\mu]}$.
- (5) For $j \in [m], k \in [d]$, $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{ConSel}}^{\text{Eq}, \mu}$ where $\mathcal{R}$ inputs $\{e_{j,k,h}^{\mathcal{R}}\}_{h \in [\mu]}$ and $\mathcal{S}$ inputs $\{-e_{j,k,h}^{\mathcal{S}}\}_{h \in [\mu]}$. $\mathcal{R}$ receives $r_{j,k}^{\mathcal{R}}$ and $\mathcal{S}$ receives $r_{j,k}^{\mathcal{S}}$.
- (6) For $j \in [m]$, $\mathcal{R}$ computes $r_j^{\mathcal{R}} = \sum_{k \in [d]} r_{j,k}^{\mathcal{R}}$. $\mathcal{S}$ computes $r_j^{\mathcal{S}} = \sum_{k \in [d]} r_{j,k}^{\mathcal{S}}$.
- (7) For $j \in [m]$, $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{Eq}}$ where $\mathcal{R}$ inputs $r_j^{\mathcal{R}}$ and $\mathcal{S}$ inputs $-r_j^{\mathcal{S}}$. $\mathcal{R}$ receives $b_j = 1\{r_j^{\mathcal{R}} = -r_j^{\mathcal{S}}\}$.
- (8) For $j \in [m]$, $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{OT}}$ where $\mathcal{R}$ inputs b_j and $\mathcal{S}$ inputs $(\perp, \mathbf{q}_j)$. $\mathcal{R}$ receives $\mathbf{u}_j$.
- (9) $\mathcal{R}$ outputs $\{\mathbf{u}_j \mid b_j = 1 \text{ for } j \in [m]\}$.

Figure 28: Protocol of prefix-optimized fuzzy PSI for L_∞ distance under receiver-sided unique block assumption.

Protocol $\Pi_{\text{FPSI-Px}}^{L_p}$

Parameters: Distance threshold δ . Bit length $\ell = \lambda + \log m + p \log \delta$.

Input: $\mathcal{S}$ inputs $Q = \{\mathbf{q}_j\}_{j \in [m]} \in \mathbb{U}^{d \times m}$ and $\mathcal{R}$ inputs $W = \{\mathbf{w}_i\}_{i \in [n]} \in \mathbb{U}^{d \times n}$.

Protocol:

- (1) For $i \in [n]$, $\mathcal{R}$ computes $\{C_{i,z}^{\mathcal{R}}\}_{z \in [2^d]} = \text{neigh}_{2\delta}(\mathbf{w}_i)$. $\mathcal{S}$ shuffles Q and computes $C_j^{\mathcal{S}} = \text{cell}_{2\delta}(\mathbf{q}_j)$ for $j \in [m]$.
- (2) $\mathcal{R}$ sets $\text{List} = \{(C_{i,z}^{\mathcal{R}} \| k \| \sigma \| \tilde{\mathbf{w}}_{i,k,h,\sigma}, \mathbf{w}^* - \mathbf{w}_{i,k})\}_{i \in [n], z \in [2^d], k \in [d], h \in [\mu], \sigma \in [0,1]}$ where $\{\tilde{\mathbf{w}}_{i,k,h,0}\}_{h \in [\mu]} = \text{PxTrie}(\mathbf{w}_{i,k} - \delta, \mathbf{w}_{i,k})$ and $\{\tilde{\mathbf{w}}_{i,k,h,1}\}_{h \in [\mu]} = \text{PxTrie}(\mathbf{w}_{i,k} + 1, \mathbf{w}_{i,k} + \delta)$ and $\mathbf{w}^* = \text{UpBound}(\tilde{\mathbf{w}}_{i,k,h,\sigma})$ if $\sigma = 0$ otherwise $\mathbf{w}^* = \text{LowBound}(\tilde{\mathbf{w}}_{i,k,h,\sigma})$.
- (3) $\mathcal{S}$ computes $\{C_j^{\mathcal{S}} \| k \| \sigma \| \tilde{q}_{j,k,h,\sigma}\}_{j \in [m], k \in [d], h \in [\mu], \sigma \in [0,1]}$ where $\{\tilde{q}_{j,k,h,\sigma}\}_{h \in [\mu]} = \text{PxPath}(\mathbf{q}_{j,k}, \delta/2)$.
- (4) $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{so-OPPRF}}$ where $\mathcal{R}$ inputs List and $\mathcal{S}$ inputs $\{C_j^{\mathcal{S}} \| k \| \sigma \| \tilde{q}_{j,k,h,\sigma}\}_{j \in [m], k \in [d], h \in [\mu], \sigma \in [0,1]}$. $\mathcal{R}$ receives $\{v_{j,k,h,\sigma}^{\mathcal{R}}\}_{j \in [m], k \in [d], h \in [\mu], \sigma \in [0,1]}$ and $\mathcal{S}$ receives $\{v_{j,k,h,\sigma}^{\mathcal{S}}\}_{j \in [m], k \in [d], h \in [\mu], \sigma \in [0,1]}$.
- (5) For $j \in [m], k \in [d], h \in [\mu], \sigma \in [0,1]$, $\mathcal{S}$ and $\mathcal{R}$ invoke functionality $\mathcal{F}_{\text{B2A}}$, where $\mathcal{R}$ inputs $v_{j,k,h,\sigma}^{\mathcal{R}}$ and $\mathcal{S}$ inputs $v_{j,k,h,\sigma}^{\mathcal{S}}$. $\mathcal{R}$ receives $s_{j,k,h,\sigma}^{\mathcal{R}}$ and $\mathcal{S}$ receives $s_{j,k,h,\sigma}^{\mathcal{S}}$.
- (6) For $j \in [m], k \in [d], h \in [\mu], \sigma \in [0,1]$, $\mathcal{S}$ and $\mathcal{R}$ invoke Π_{getDist}^p , where $\mathcal{R}$ inputs $s_{j,k,h,\sigma}^{\mathcal{R}}$ and $\mathcal{S}$ inputs $(s_{j,k,h,\sigma}^{\mathcal{S}}, q_{j,k}, \tilde{q}_{j,k,h,\sigma})$. $\mathcal{R}$ receives $d_{j,k,h,\sigma}^{\mathcal{R}}$ and $\mathcal{S}$ receives $d_{j,k,h,\sigma}^{\mathcal{S}}$.
- (7) For $j \in [m], k \in [d]$, $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{ConSel}}^{\text{Cmp}, 2-\mu}$ where $\mathcal{R}$ inputs $\{d_{j,k,h,\sigma}^{\mathcal{R}}\}_{h \in [\mu], \sigma \in [0,1]}$ and $\mathcal{S}$ inputs $\{d_{j,k,h,\sigma}^{\mathcal{S}}\}_{h \in [\mu], \sigma \in [0,1]}$. $\mathcal{R}$ receives $r_{j,k}^{\mathcal{R}}$ and $\mathcal{S}$ receives $r_{j,k}^{\mathcal{S}}$.
- (8) For $j \in [m]$, $\mathcal{R}$ computes $r_j^{\mathcal{R}} = \sum_{k \in [d]} r_{j,k}^{\mathcal{R}}$ and $\mathcal{S}$ computes $r_j^{\mathcal{S}} = \sum_{k \in [d]} r_{j,k}^{\mathcal{S}}$.
- (9) For $j \in [m]$, $\mathcal{S}$ and $\mathcal{R}$ invoke functionality $\mathcal{F}_{\text{Interval}}^{\delta p}$, where $\mathcal{R}$ inputs $r_j^{\mathcal{R}}$ and $\mathcal{S}$ inputs $r_j^{\mathcal{S}}$. $\mathcal{R}$ receives $b_j := 1(r_j^{\mathcal{S}} + r_j^{\mathcal{R}} \leq \delta^p)$.
- (10) For $j \in [m]$, $\mathcal{R}$ and $\mathcal{S}$ invoke functionality $\mathcal{F}_{\text{OT}}$, where $\mathcal{R}$ inputs b_j and $\mathcal{S}$ inputs $(\perp, \mathbf{q}_j)$. $\mathcal{R}$ receives OT outputs $\mathbf{u}_j$.
- (11) $\mathcal{R}$ outputs $\{\mathbf{u}_j \mid b_j = 1 \text{ for } j \in [m]\}$.

Figure 29: Protocol of prefix-optimized fuzzy PSI for L_p distance under receiver-sided unique block assumption.

Table 4: The communication (MB) and running time (s) of our protocols for L_p distance under the receiver-sided unique cell assumption.

Size $m=n$	Dim. d	Metric	$\delta = 32$		$\delta = 64$		$\delta = 128$		$\delta = 256$		$\delta = 512$	
			Comm	Time	Comm	Time	Comm	Time	Comm	Time	Comm	Time
2^8	2	L_1	3.52	0.23	4.23	0.28	5.62	0.28	8.36	0.32	13.83	0.53
		L_2	3.66	0.27	4.41	0.30	5.82	0.28	8.94	0.44	14.39	0.58
	4	L_1	14.24	0.63	15.63	0.65	18.39	0.70	23.96	0.84	34.95	0.91
		L_2	14.68	0.58	16.53	0.64	19.32	0.78	25.00	0.90	36.12	1.20
	6	L_1	69.90	2.31	72.43	2.36	77.01	2.62	85.66	2.61	102.53	2.82
		L_2	72.38	2.37	75.42	2.46	80.48	2.58	89.64	2.70	107.01	3.61
2^{12}	2	L_1	36.73	1.26	48.10	1.40	70.37	1.86	114.44	2.73	202.10	4.24
		L_2	39.21	1.26	51.09	1.50	73.84	1.95	118.41	3.02	206.58	5.17
	4	L_1	204.80	6.43	228.55	6.67	274.09	7.31	363.26	9.08	539.68	12.76
		L_2	214.72	6.10	240.46	6.71	287.97	7.72	380.23	10.46	558.78	16.17
	6	L_1	1098.49	32.75	1139.36	33.70	1212.97	32.79	1352.21	38.21	1622.54	42.32
		L_2	1139.51	35.11	1188.59	36.70	1270.34	37.35	1417.80	40.73	1696.37	54.99
2^{16}	2	L_1	567.83	14.77	750.43	19.13	1107.61	26.11	1814.41	40.05	3220.26	70.40
		L_2	608.85	17.04	799.66	19.92	1164.98	27.85	1879.99	48.77	3294.09	79.57
	4	L_1	3261.14	112.59	3642.86	113.98	4374.04	120.13	5804.69	156.48	8634.00	211.10
		L_2	3425.32	100.17	3839.89	112.74	4603.68	129.88	6084.17	167.54	8948.50	283.50
	6	L_1	17578.29	622.81	18237.60	655.66	19421.77	606.16	21658.27	688.11	25996.49	799.07
		L_2	18255.49	621.42	19050.23	627.93	20368.92	628.95	22740.88	749.81	27215.29	1119.00

Table 5: The communication (MB) and running time (s) of our protocols under the sender-sided unique cell assumption.

Size $m=n$	Dim. d	Metric	$\delta = 32$		$\delta = 64$		$\delta = 128$		$\delta = 256$		$\delta = 512$	
			Comm	Time	Comm	Time	Comm	Time	Comm	Time	Comm	Time
2^8	2	L_∞	2.34	0.26	3.02	0.25	4.38	0.34	7.10	0.41	12.53	0.49
		L_1	3.20	0.37	3.89	0.37	5.24	0.40	7.96	0.55	13.40	0.64
		L_2	3.20	0.34	3.89	0.39	5.24	0.46	7.96	0.55	13.40	0.67
	4	L_∞	8.54	0.55	9.89	0.51	12.61	0.64	18.05	0.73	28.93	0.94
		L_1	13.67	0.84	15.03	0.88	17.74	0.87	23.18	0.99	34.06	1.16
		L_2	13.67	0.68	15.03	0.84	17.74	0.90	23.18	0.96	34.06	1.17
2^{12}	6	L_∞	40.03	1.83	42.07	1.80	46.15	1.98	54.30	2.24	70.67	2.30
		L_1	68.85	2.68	70.90	2.41	74.98	2.78	83.12	2.95	99.49	2.95
		L_2	68.85	2.70	70.90	2.76	74.98	2.76	83.12	2.87	99.49	3.22
	2	L_∞	25.33	1.11	36.21	1.29	57.98	1.77	101.55	2.52	188.72	4.19
		L_1	35.65	1.60	46.53	1.72	68.30	2.19	111.86	3.03	199.04	4.51
		L_2	35.65	1.58	46.53	1.76	68.30	2.24	111.86	3.11	199.04	4.58
2^{16}	4	L_∞	124.07	4.88	145.84	5.25	189.41	5.86	276.58	7.20	451.02	10.36
		L_1	201.03	7.34	222.80	7.74	266.36	8.72	353.54	10.18	527.98	14.20
		L_2	201.03	7.40	222.80	7.98	266.36	8.57	353.54	10.20	527.98	13.40
	6	L_∞	628.55	23.78	661.26	23.91	726.72	24.79	857.69	28.92	1119.79	32.37
		L_1	1082.43	38.90	1115.14	40.96	1180.59	40.51	1311.57	45.02	1573.67	49.11
		L_2	1082.43	40.07	1115.14	40.82	1180.59	41.64	1311.57	44.12	1573.67	52.33
2^{16}	2	L_∞	393.99	18.72	568.43	21.57	917.44	28.09	1615.98	41.79	3013.60	69.55
		L_1	551.86	25.32	726.30	31.87	1075.31	35.44	1773.85	47.82	3171.47	75.53
		L_2	551.86	25.20	726.30	28.25	1075.31	35.62	1773.85	53.31	3171.47	90.51
	4	L_∞	1976.03	93.68	2325.08	98.78	3023.61	113.03	4421.18	139.09	7217.55	192.08
		L_1	3196.50	169.05	3545.55	145.08	4244.08	163.49	5641.65	196.81	8438.03	249.97
		L_2	3196.50	148.73	3545.55	162.31	4244.08	156.41	5641.65	190.25	8438.03	260.12
2^{16}	6	L_∞	10056.07	492.81	10580.55	505.97	11630.04	522.84	13730.14	600.87	17932.58	669.67
		L_1	17297.40	875.75	17821.89	837.55	18871.38	832.43	20971.48	963.54	25173.92	1027.60
		L_2	17297.40	870.23	17821.89	835.03	18871.38	808.55	20971.48	987.70	25173.92	1015.14